

# MARSEA: MULTIVARIATE RELATIONS BY SELECTIVE EXCLUSION IN ATTENTION

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## ABSTRACT

Attention represents relations *pairwise*:  $A_{ij}$  is a statement about one query and one key. An important class is not of this form. In *competitive exclusion* a set of variables jointly bears on a target, its members compete for a finite influence budget, and which prevails is settled by comparison *within* the set — the winner taking the bulk, close rivals progressively less, everything past a content-decided cutoff exactly none. We prove that no *query-local* scheme — rows computed independently per query, a class containing softmax, every relative-position or geometric bias, and every elementwise gate — can enforce any non-product-form degree constraint, and that a second, independent obstruction stops any row-normalized scheme from bounding *every* key’s fan-out below the query-to-key ratio, which is why doubly stochastic attention does not help — an obstruction that is severe wherever a small key set serves many queries and weak in self-attention, where the separation theorem carries the argument alone. **MarSea** — *multivariate relations by selective exclusion in attention* — imposes *content-dependent degree constraints on both margins*, and on each margin it predicts *two* things rather than one — a **capacity**, how much a node may commit or absorb, and a **shape**, how that is distributed over its incident edges including which are exactly zero. Each margin takes two steps: one inherits the mass an unmodified layer would allocate, keeping an exclusive margin commensurate with a non-exclusive one, and one re-shapes that mass over the relation at a *predicted* exclusivity, producing exact zeros. **The bound we claim is therefore on degree, not on mass** — how many candidates a node reaches inside its relation, which is the quantity a query-local scheme provably cannot bound and a full-support one cannot represent. *Every predicted parameter reads its participant together with the field that participant faces* — the key with its score column, the query with its incoming allocation — so the coupling that defeats the separation theorem is in the parameters as well as in the programs. Both are closed-form; neither renormalizes rows; both run natively over the visible prefix in a decoder. Read as retrieval of a ground-truth relation, the recovery interval is exactly a *precision–recall* interval with *sharp* endpoints, one per error type, and it is non-empty on every column with a positive margin — so exact recovery is never obstructed, and what a hard column costs is the *width* of the target the head must hit. This yields a falsifiable prediction for multi-hop evidence selection: with  $m$  gold facts among  $n$  candidates, *any* full-support attention has evidence precision exactly  $m/n$  — an identity, not a tendency — so it degrades as candidates are added and no training arrests it, while MarSea’s support size does not involve  $n$ . Our load-bearing experiment sweeps  $m$  at fixed task difficulty, separating the expressivity claim from the effect of exclusion-shaped training, and we pre-commit to reporting either outcome. This concerns degradation with *candidate count* and not the positional U-shape, whose architectural origin is separately understood.

## 1 INTRODUCTION

An attention matrix is a weighted bipartite graph:  $A_{ij}$  is an edge from key  $j$  to query  $i$ . Softmax decides which edges carry mass *pairwise* — each weight from one query and one key — under a single constraint per row,  $\sum_j A_{ij} = 1$ .

Many relations are not pairwise. Our object is **multivariate competitive exclusion**: a set  $\mathcal{A}$  jointly bears on  $b$ , its members compete for a finite influence budget, and which prevails is settled by comparison *within*  $\mathcal{A}$ . Losing is graded near the top of the order and absolute below it — *little to nothing*, not uniformly nothing. Competition may be **ranked**, by a priority order over kinds, or **graded**, by a measured quantity. We say *exclusion* rather than *prioritization* because ordering is not the scarce capability — row-softmax already ranks — but *ending* the order at a cutoff the candidates themselves decide is.

**One scene, both margins.** At a traffic intersection many signals may be present at once — lights, hand-held signs, pedestrians, an emergency vehicle, an officer — under a priority order that differs by jurisdiction and so must be learned from content. Two statements follow, one per margin. **Preemption**: among the signals bearing on a lane the highest-priority one renders the rest irrelevant (*fan-in*). **Scope**: each signal governs a *set* of lanes, and how many varies by signal (*fan-out*). Neither is a property of any query–key pair; both concern a node’s degree. The same structure appears wherever a model must decide which of many candidates bear on a conclusion: which  $m$  of  $n$  retrieved passages support an answer, which instruction outranks which when they conflict.

**Orientation.** Semantic influence ( $\mathcal{A} \rightarrow b$ ) is not attention direction (key  $\rightarrow$  query), and nothing ties them: if the competing signals are keys, preemption is a statement about a *row*; if they are queries, the identical relation is a statement about a *column*. Which holds depends on how the representation is learned and may differ across heads, so a design constraining one margin silently assumes an orientation it cannot verify. MarSea constrains both.

**Two obstructions, and MarSea.** Sec. 3 shows this is not a modelling preference: any scheme computing each query’s row from that query’s own inputs — softmax, the relative-position and geometric-bias family, every elementwise gate — provably cannot enforce such a constraint however expressive its scoring function (Thm. 1), and a second, independent obstruction shows that no row-normalized scheme bounds every key’s fan-out by a common constant (Prop. 1). We therefore read  $\mathcal{A}$  as a relational graph and impose **content-dependent degree constraints on both margins** — *degree* in the weighted sense, the sum of incident edge weights, which is exactly what a budget constrains, with cardinality control following from the exclusivity and exact under Assumption 2: for each key, a joint program over its fan-out edges inherits the mass an unmodified layer allocates and redistributes it over the relation at a predicted exclusivity  $\tau_j$ ; for each query, a capped projection does the same on the other margin at a predicted exclusivity  $\tau_i$ , beneath a ceiling fixed at one unit. Every head reads its participant *and its field* —  $\tau_j$  and the gate see the score column,  $\tau_i$  and its gate the incoming allocation — so no parameter of the mechanism is a function of a key or query alone. The column step is sparsemax (Martins & Astudillo, 2016; Peters et al., 2019) — strictly concave, exactly solvable by sorting, non-iterative, unlike Sinkhorn attention (Sander et al., 2022) or capsule routing (Sabour et al., 2017) — and the row step is a threshold, not a normalization, which keeps Prop. 1 from applying to it.

**Contributions.** (i) A **change of represented object**, from the query–key pair to a multivariate relation carried by **both margins** of the attention graph — and on each margin by **two** predicted quantities rather than one. A **capacity** says how much influence a node may commit or absorb; a **shape** says how that influence is distributed over its incident edges, *including which edges are exactly zero*. Both are predicted from content, and both run over the whole margin or a declared part of it (Sec. 2.3, App. E). **Exclusivity is the pair**: a capacity alone bounds a total without deciding membership, and a shape alone redistributes a total the mechanism does not control — which is why one margin, or one of the two quantities, is not the object. Table 1 places every mechanism we know of on these four cells; **MarSea is the only entry that fills them**. The object is forced rather than chosen, by a **separation theorem** and an independent **conservation obstruction** that together require query coupling and a non-uniform column capacity.

**We claim the two shape cells and inherit the two capacities, and the disclaimer is deliberate.** Zhang et al. (2023) learn *both* marginals of a transport, each from its own participant (App. C); a predicted capacity is theirs, not ours, and MarSea takes its capacities from what an unmodified layer allocates — which is what keeps an exclusive margin commensurate with a non-exclusive one and makes  $\sum_j c_j = n_q$  a consequence rather than an anchor (Prop. 2). **What no prior mechanism does is predict the shape**: MESH’s transport is entropy-regularized and its output dense, with the regu-

Table 1: **The modelling object, and where every mechanism we know of sits in it.** A multivariate relation is carried by *both* margins, and on each margin it takes *two* predicted quantities: a **capacity** — how much influence the node may commit or absorb — and a **shape** — how that influence is distributed over its incident edges, including which edges are *exactly* zero. Capacity without shape bounds a total but does not decide membership; shape without capacity redistributes a total the mechanism does not control. *Exclusivity is the pair, and no prior mechanism holds both cells on either margin.* “learned( $k$ )” means a capacity predicted from the participant alone; “inherited” means taken from what an unmodified layer allocates, which is what makes an exclusive margin commensurate with a non-exclusive one (Prop. 2). **MarSea claims the two shape cells and inherits the two capacities**; MESH is its mirror image, learning both capacities and predicting neither shape — and only the shape cells produce the exact zeros that Prop. 6 keys on, which is why every “dense” row is pinned at  $m_j/n_q$  however well its capacities are learned.  $\dagger$  marks a mechanism that runs on a declared *part* of a margin rather than all of it.

| Mechanism                           | fan-out margin (per key)      |                                                  | fan-in margin (per query) |                                                     |
|-------------------------------------|-------------------------------|--------------------------------------------------|---------------------------|-----------------------------------------------------|
|                                     | capacity                      | shape                                            | capacity                  | shape                                               |
| Row-softmax                         | free                          | dense                                            | = 1 fixed                 | dense                                               |
| Geometric bias / pairwise gate      | free                          | dense                                            | $\leq 1$ fixed            | dense                                               |
| $\text{softmax}_1$ , learnable sink | free                          | dense                                            | $\leq 1$ fixed const.     | dense                                               |
| Differential attention              | free                          | dense, signed                                    | = $1 - \lambda$ fixed     | dense, signed                                       |
| Row-entmax, top- $k$ row            | free                          | dense                                            | free                      | zeros, $\alpha$ fixed                               |
| Adaptively sparse                   | free                          | dense                                            | free                      | zeros, $\alpha$ per head                            |
| Selective Attention                 | free                          | dense                                            | = 1 fixed                 | dense                                               |
| Sinkformer, doubly stochastic       | = $n_q/n_k$ uniform           | dense                                            | = 1 fixed                 | dense                                               |
| LOTFormer                           | = uniform                     | dense                                            | = 1 fixed                 | dense                                               |
| Slot Attention                      | = 1 uniform                   | dense                                            | = 1 fixed                 | dense                                               |
| MESH                                | = learned( $k$ )              | dense, global $\lambda$                          | = learned( $q$ )          | dense, global $\lambda$                             |
| Expert-choice routing               | = fixed $k$ , content-free    | zeros, hard top- $k$                             | —                         | —                                                   |
| <b>MarSea<math>^\dagger</math></b>  | inherited, on $\mathcal{E}_j$ | <b>zeros</b> , $\tau_j(k_j, s_{\cdot j}, \nu_j)$ | inherited, $\leq 1$       | <b>zeros</b> , $\theta_i(\tau_i(q_i, \tilde{A}_i))$ |

larization strength a global hyperparameter, so no edge is ever exactly zero and no per-key parameter decides where the support ends — and by Prop. 6 its column precision is therefore pinned at  $m_j/n_q$  exactly as row-softmax’s is. Conversely the sparse-attention family predicts a shape query-locally, which Thm. 1 shows cannot enforce a constraint on the other margin at all. The shape parameters here read their participant *together with the field it faces*, which Cor. 3 shows a participant-only parameter provably cannot substitute for. (ii) A closed-form **mechanism** of five components realizing all four cells (Sec. 2), with a native causal form in which prefix growth is provably edge-removing. (iii) A **fidelity analysis** (Sec. 3.6) recasting every guarantee as precision and recall on *both* margins — a ceiling binding every full-support mechanism irrespective of training, and a recovery interval with *sharp* endpoints, one per error type. (iv) A **prediction** for multi-hop evidence selection (Sec. 4) and a **protocol** (Sec. 5) whose load-bearing experiment separates the expressivity claim from exclusion-shaped training and reports either outcome.

**Related work.** App. C treats all of it in full. *Geometry in scores* (Hu et al., 2018; Wu et al., 2021; Gao et al., 2021; Liu et al., 2022; Hou et al., 2024) is query-local by Def. 1, so Thm. 1 covers it. So is *differential attention* (Ye et al., 2025), which subtracts two row-softmax maps to cancel over-allocation to irrelevant context; its weights are signed, so the theorem applies but Cor. 1 does not (App. C). *Transport-based attention* (Sander et al., 2022; Shahbazi et al., 2025) and Slot Attention (Locatello et al., 2020) identify the marginals as the design surface and fix the column marginal to *uniform*, equalizing fan-out at  $n_q/n_k$  rather than bounding it by a content-dependent constant; Tay et al. (2020) balances a block *sorting* matrix rather than the attention itself and is not in this group, while Zhang et al. (2023) *does* learn non-uniform marginals — both of them — and is the closest prior art to MarSea’s *capacities* on either margin. **What it does not predict is a shape**: its transport is entropy-regularized and dense at a global regularization strength, so it has no exact zeros and Prop. 6 pins its column precision at  $m_j/n_q$  exactly as it does row-softmax’s (Table 1, App. C). **Selective Attention** (Leviathan et al., 2025) is the one attention mechanism we know of that **escapes Thm. 1**: its mask accumulates along the prefix, so what query  $i$  may give key  $j$  depends on what *earlier queries* decided about  $j$ . Expert-choice routing (Zhou et al., 2022) makes the same move outside attention, each expert selecting its own top- $k$  tokens. Both decide on the receiving margin over a set of senders at once; neither constrains a marginal, and neither predicts from content

how much a key should emit — which is the step that turns coupling into a degree bound. *Sparse attention* (Martins & Astudillo, 2016; Peters et al., 2019; Correia et al., 2019) and the regularized-argmax framework (Niculae & Blondel, 2017; Blondel et al., 2020) supply our variational form, which is *not* our contribution; we depart from Correia et al. (2019) in predicting exclusivity per key *from the key and its column* rather than as a per-head constant, which Cor. 3 argues is necessary. Closest in motivation, Veličković et al. (2025) prove that softmax attention must disperse as the number of items grows and propose adaptive temperature as a remedy; Prop. 10 is that claim made quantitative on the fan-out margin, and Prop. 6 is why sharpening a full-support distribution cannot reach  $P_j = 1$ . *Position bias* (Liu et al., 2024) is a distinct phenomenon with its own architectural accounts (Chowdhury, 2026; Wu et al., 2025; Herasimchyk et al., 2026), which differ on how much of it survives without positional encodings; we make no claim about that axis (Sec. 4).

## 2 METHOD

With  $n_q$  queries,  $n_k$  keys, values  $V_j$  and relational descriptors  $\gamma_{ij}$ , scores combine content and relation:

$$s_{ij} = q_i \cdot k_j / \sqrt{d} + b(\gamma_{ij}). \quad (1)$$

MarSea replaces the normalization step of an attention layer and has *five* components, introduced together so that the object analyzed in Sec. 3 is the one we propose; App. A collects notation and the guarantee each carries.

### 2.1 STAGE 1: THE FAN-OUT PROGRAM

**The exclusive relation a key participates in is rarely its whole column.** A left-turn arrow governs the left-turn lanes and is exclusive over nothing else. We therefore fix, per key, an **exclusive set**  $\mathcal{E}_{\cdot j} \subseteq [n_q]$  — decided from the content and relation of the participants, and known before the program runs — and the fan-out stage takes *two* steps, one per cell of the object.

**Step 1 fixes how much, and it is the layer’s own attention.** With  $A^{\text{sm}}$  the ordinary row-softmax of the scores over the visible pairs — each *query’s* row normalized over the keys it sees,  $\sum_j A_{ij}^{\text{sm}} = 1$ ; no column is normalized here, the column stage only *reads* the column sums of this matrix —

$$A^{\text{sm}} := \text{softmax}_{\text{row}}(S), \quad c_j := \sum_i A_{ij}^{\text{sm}}, \quad \bar{c}_j := \sum_{i \in \mathcal{E}_{\cdot j}} A_{ij}^{\text{sm}}. \quad (2)$$

**The capacity is inherited, not predicted,** and it is inherited entrywise: every pair outside the relation carries exactly what an unmodified layer gives it, so with an empty relation the mechanism *is* standard attention, not an approximation of it. Three things follow at no cost. An exclusive column carries the same total as a non-exclusive one; every row’s supply is one unit before step 2 touches it; and  $\sum_j c_j = n_q$  holds *as a consequence* rather than as an imposed anchor. (An earlier draft placed step 1 at a column softmax rescaled to  $c_j$ ; that preserved the total but not the entries, and an empty relation then reproduced standard attention only in aggregate. We record the change because App. G checks the entrywise identity.)

**Step 2 fixes which, and yields the exact zeros.** The exclusive set’s quota  $\bar{c}_j$  is redistributed over that set alone,

$$p_{\mathcal{E}_{\cdot j}, j} = \arg \max_{p \in \Delta^{|\mathcal{E}_{\cdot j}|-1}} \{ \langle p, \tau_j s_{\mathcal{E}_{\cdot j}, j} \rangle + H_{\alpha}^{\top}(p) \}, \quad \tilde{A}_{\mathcal{E}_{\cdot j}, j} = \bar{c}_j p_{\mathcal{E}_{\cdot j}, j}, \quad \tilde{A}_{ij} = A_{ij}^{\text{sm}} \quad \forall i \notin \mathcal{E}_{\cdot j}. \quad (3)$$

The relation is re-shaped and written back into the column standard attention produced; every other entry of that column is untouched. The column total is therefore *exactly*  $c_j$  whatever the exclusive set is, and mass can never be relocated off it (Prop. 2). **What this buys and what it forgoes should be said together.** A key can never emit *more* into its relation than an unmodified layer would, and it can never emit *less* either: a distractor key keeps its whole softmax mass, and a large  $\tau_j$  concentrates that mass onto fewer members rather than shrinking it. Suppression of a key’s total is therefore not something this margin does; it decides membership, and what a loser loses is reach, not mass. The row stage is where a query declines a key (Sec. 2.2). Every result below is stated on the index set the program runs over, so the unrestricted case  $\mathcal{E}_{\cdot j} = [n_q]$  is the special case and nothing is reproved for it.

**Two limiting columns deserve a sentence each.** When exactly one member of the relation is true, Thm. 2 has  $W_j = 0$  and its interval is  $[1/\delta_j, \infty)$ : for every  $\tau_j \geq 1/\delta_j$  the program puts the *entire* quota  $\bar{c}_j$  on the largest-scoring member and an exact zero on every other, and because sparsemax is 1-Lipschitz in its argument the column moves *continuously* in  $\tau_j$  from spread to that allocation — a hard assignment reached through a continuous parameter, with the top gap of  $s_{\cdot j}$ , which  $\tau_\phi^K$  reads, as the signal that the column has one target. When *no* member is true the program still allocates, a simplex having no empty point: such a key reaches whichever member scores it highest, and the row stage is where that query declines it. The composition is what matters. A distractor whose highest-scoring member is not the answer query is removed from the answer row by Stage 1 alone, as an exact and permanent zero (Prop. 9), and only distractors that concentrate *onto* the answer query reach its row program at all; App. G checks both, and E8 reports the split. Here  $H_\alpha^T$  is the Tsallis  $\alpha$ -entropy, so equation 3 is  $\alpha$ -entmax( $\tau_j s_{\cdot j}$ ). We fix  $\alpha = 2$ , i.e. sparsemax, where the solution is unique, exactly computable by sorting in  $O(n_q \log n_q)$  with no fixed-point iteration, and differentiable almost everywhere (Martins & Astudillo, 2016; Peters et al., 2019):

$$p_{ij} = [\tau_j s_{ij} - \psi_j]_+, \quad (4)$$

affine in the score wherever positive and *exactly* zero past the threshold  $\psi_j$  — the precise sense of “little to nothing”. The exclusivity  $\tau_j$  *selects between the two regimes* rather than being tuned to one: support size is non-increasing in  $\tau_j$ , so a large  $\tau_j$  approaches ranked preemption and a small one graded competition, and one layer serves both because it is predicted from the key *together with the field it faces* — one head per margin, never one head used twice, the superscripts being load-bearing for reasons App. D gives:

$$\tau_j = \tau_\phi^K(k_j, s_{\cdot j}), \quad \tau_i = \tau_\phi^Q(q_i, \tilde{A}_i) \quad (\text{Sec. 2.2}). \quad (5)$$

**Two things now place MarSea outside the query-local class, not one:**  $\psi_j$  depends on the whole column, and so does the parameter that sets it. Nor is reading  $s_{\cdot j}$  a convenience — by Thm. 2 the temperatures that recover exactly are *exactly*  $[1/(W_j + m_j \delta_j), 1/W_j]$ , whose endpoints are *functions of that column* and whose width can be small, so a head seeing only  $k_j$  cannot target an interval it cannot observe.

$\tau_j$  **carries the whole fan-out claim, so it reads the statistic that answers it.** With the capacity inherited, the only thing the mechanism decides on this margin is where the support ends — and by Thm. 2 landing it means placing  $\tau_j$  so that  $k^*(\tau_j) = m_j$ . The exclusivity head therefore takes, alongside the key and its score column, the smooth *effective cardinality*  $\nu_j := \|p_{\cdot j}\|_2^{-2} \in [1, |\mathcal{E}_{\cdot j}|]$ , which equals  $k$  when  $p_{\cdot j}$  is uniform on  $k$  atoms and, unlike  $|\text{supp}(p_{\cdot j})|$ , carries a gradient. Nothing is circular: computation runs  $s_{\cdot j} \rightarrow \tau_j \rightarrow p_{\cdot j} \rightarrow \nu_j$ , and  $\nu_j$  enters the next block or the next layer’s head as an input, never as a definition of  $\tau_j$  within one solve. Whether  $\nu_j$  carries  $m_j$  at all is not an identity — it depends on  $\tau_j$  and on the column — and Fig. 2(b) measures it.

**What the two steps do and do not claim.** Every result in Sec. 3 uses only  $\bar{c}_j > 0$ , support recovery included. **We are explicit that the fan-out bound is now a bound on degree and not on mass.** Step 1 hands the exclusive set whatever an unmodified layer would have given it, so the mass a key emits into its relation is not a quantity this mechanism predicts; what it decides is how many members of that relation receive any of it. That is the cardinality half of Cor. 1, it is the half Thm. 1 forbids a query-local scheme from enforcing, and it is what Prop. 6 compares against, since a full-support mechanism reaches every candidate by construction. The step-1 total couples keys through  $A^{\text{sm}}$  and introduces no dependence of one row on another row’s inputs.

## 2.2 STAGE 2: THE FAN-IN PROGRAM

Stage 1 says nothing about what a *query* receives, and the fan-in stage mirrors it exactly — an exclusive set  $\mathcal{E}_i \subseteq [n_k]$ , a step that fixes how much, and a step that fixes which. **Step 1** caps the whole row at the unit an unmodified layer spends,

$$a_{i\cdot}^{(1)} = \arg \min_{a \geq 0} \frac{1}{2} \|a - \tilde{A}_i\|_2^2 \text{ s.t. } \sum_j a_j \leq 1, \quad \bar{c}_i := \sum_{j \in \mathcal{E}_i} a_{ij}^{(1)}. \quad (6)$$

This is the row as standard attention produced it, modified only where a column relation re-shaped it, held beneath the unit that row already carried: every row of  $A^{\text{sm}}$  sums to one, so the cap is the

identity on any row no key has re-shaped, and binds exactly where a fan-out concentration has *added* mass to the row — a condition on the relation’s own entries, which is what makes it well posed in finite precision (Prop. 2 and the remark after it). **It is not a second softmax.** The entries of  $\tilde{A}$  are already attention weights, and re-normalizing them would rescale the off-relation entries away from standard attention’s and — because a Stage-1 zero is a weight of 0, and a softmax of 0 is positive — would resurrect every zero Stage 1 produced. The harness checks that the zeros survive (Prop. 2(v)). **Step 2** re-shapes the relation’s share beneath its quota,

$$A_{i,\mathcal{E}_i} = \arg \min_{a \geq 0} \frac{1}{2} \|a/\tau_i - \tilde{A}_{i,\mathcal{E}_i}\|_2^2 \text{ s.t. } \sum_{j \in \mathcal{E}_i} a_j \leq \bar{c}_i \implies A_{i,\mathcal{E}_i} = \tau_i [\tilde{A}_{i,\mathcal{E}_i} - \theta_i]_+, \quad (7)$$

$$A_{ij} = a_{ij}^{(1)} \quad \forall j \notin \mathcal{E}_i,$$

with  $\theta_i \geq 0$  the *dual* of the quota,  $\theta_i = 0$  when the quota is slack and otherwise solving  $\sum_{j \in \mathcal{E}_i} \tau_i [\tilde{A}_{ij} - \theta_i]_+ = \bar{c}_i$ . As on the column, the relation is re-shaped and written back; the rest of the row is what step 1 left.

**The quota is a ceiling, not an equality, and the reason is a theorem we would otherwise break.** With an equality in place of equation 7’s inequality the target  $\bar{c}_i/\tau_i$  exceeds  $\sum_{j \in \mathcal{E}_i} \tilde{A}_{ij}$  whenever  $\tau_i$  is small enough, the dual goes negative, and every coordinate of the relation — including entries Stage 1 set to *exactly zero* — becomes positive. That contradicts Prop. 9(iv), which says a Stage-1 false negative is unrecoverable, and the harness confirms it on roughly half of random rows. The inequality keeps  $\theta_i \geq 0$  for every  $\tau_i > 0$  (Prop. 2(v)). **The solution has the same functional form as the unrestricted program** — one common floor and one common gain — so every result stated in terms of  $\theta_i$  transfers with nothing reproved; only the equation fixing  $\theta_i$  changes, from a unit to the quota.

Write  $\tilde{R}_i := \sum_{j \in \mathcal{E}_i} \tilde{A}_{ij}$  for the relation’s supply on row  $i$ . **The regime is set by  $\tau_i \tilde{R}_i$  against the quota, not by  $\tau_i$  alone.** For  $\tau_i \leq \bar{c}_i/\tilde{R}_i$  the ceiling is slack,  $\theta_i = 0$  and  $A_{i,\mathcal{E}_i} = \tau_i \tilde{A}_{i,\mathcal{E}_i}$ : the support is untouched and the query *spends less than its quota* — which softmax cannot do however irrelevant every key is, and which softmax<sub>1</sub> (Miller, 2023) supplies only content-independently, by a fixed additive constant and without exact zeros; Xiao et al. (2024) consider it and prefer a learnable sink. For  $\tau_i > \bar{c}_i/\tilde{R}_i$  the ceiling binds,  $\theta_i > 0$ , and the query spends exactly its quota concentrated on fewer keys, with  $\tau_i \rightarrow \infty$  admitting only  $\arg \max_{j \in \mathcal{E}_i} \tilde{A}_{ij}$ . **At  $\tau_i = 1$  step 2 returns exactly the row step 1 produced:** on a row the unit cap left alone,  $\bar{c}_i = \tilde{R}_i$  and the quota is met without a floor; on a row it bound, the quota is what step 1’s own floor left on the relation, and by uniqueness of the projection step 2 recovers that same floor (the harness confirms this on every cap-binding row). So  $\tau_i = 1$  is the identity on  $a^{(1)}$  — the standard-attention start — and the two regimes are read against it:  $\tau_i > 1$  raises the floor and sharpens,  $\tau_i < 1$  lowers it and spreads. (An earlier draft said  $\tau_i = 1$  was not an identity; that compared step 2 with  $\tilde{A}$  rather than with  $a^{(1)}$ , and it was wrong.) E8 reports the distribution of  $\tau_i$  about 1. This also scopes the second margin: the ceiling bounds fan-in *mass* unconditionally but fan-in *degree* only where it binds, a slack row leaving  $\text{supp}(A_{i,\mathcal{E}_i}) = \text{supp}(\tilde{A}_{i,\mathcal{E}_i})$  untouched. The floor is common to a row but its *value* is set by the whole relation — the row-wise analogue of the column coupling in equation 3, and the sense in which MarSea is *joint* on both sides. Crucially equation 7 is a constrained optimization but **not a normalization**: the constraint is an inequality, and a row supplied with less keeps less.

**Two honest notes on this margin.** First, the fan-in step is query-local by Def. 1 — it is a function of the row and the keys — so Thm. 1 does not distinguish it from a row-entmax with a learned temperature, and any row-margin comparison in Sec. 5 is a comparison of *which keys Stage 1 let reach the row*, not of the row program itself (App. D). Second, when step 1’s cap binds it truncates the whole row at one floor, so a key’s fan-out concentration on one column can remove an off-relation entry from another query’s row; the complement is fixed by step 1 in the sense of Prop. 2(iv), but it is fixed *after* that cap, not at the unmodified layer’s values.

### 2.3 CONDITIONAL ACTIVATION IS THE RELATION

Not every key exercises exclusivity and not every query is contested, and imposing a degree constraint on structure that has none is worse than wasteful. In an earlier draft this was a pair of gates,  $\omega_\phi^K$  on keys and  $\omega_\phi^Q$  on queries, separate from the programs. **Under the two-step form the gates are**

**not separate components: they are the empty-relation case.** A key with  $\mathcal{E}_{.j} = \emptyset$  runs no program and keeps its softmax column entrywise; a query with  $\mathcal{E}_{i.} = \emptyset$  runs no program and keeps its step-1 row. So **participation is a property of the relation, not of the participant alone**, exactly as the motivation demands — a signal preempts *when competing signals are present* — and it is decided by the same pairwise compatibility that decides the relation (App. E):  $e_{ij} = \mathbf{1}[\langle u_\phi(k_j), v_\phi(q_i) \rangle > 0]$ , which reads the pair and induces both slices of one object.

Two consequences. Because  $\mathcal{E}_{i.}$  is a slice of the same  $\mathcal{E}$  as every  $\mathcal{E}_{.j}$ , a query enters equation 7 *exactly when* some key’s relation contains it; there is no second trigger and no separate row gate to tie to the column one. And the degenerate optimum is now literal: **an empty relation everywhere is standard attention**, entrywise, which App. G checks to machine precision. That optimum is *easier* to reach than any collapse of  $\tau_j, \tau_i$ , so the  $|\mathcal{E}_{.j}|$  and  $|\mathcal{E}_{i.}|$  distributions are diagnostics from epoch one and E8 scores relation membership against annotated participation where it exists: a detector, not a scheme that avoids it. Cost is unchanged, since  $e_{ij}$  is a rank- $r$  inner product computed alongside the scores. One position is excluded from every relation by construction: the first token of a decoder is the attention sink, not a candidate for anything, and a query whose only visible key re-shapes its quota away from it would be left with an empty row — so  $e_{i0} = e_{0j} = 0$ , and the sink’s row and column stay at standard attention.

## 2.4 SCALING: BLOCKING FOR INCREMENTALITY, HIERARCHICAL SELECTION FOR SCALE

Two constructions extend equation 3. **Blocking** runs the program within tiles of  $B$  queries, making the per-step cost  $O(n_k B \log B)$  — linear in the key count, as attention already is, with the program size and sort factor held at the block rather than growing with the prefix. **Hierarchical top- $K$**  addresses scale: since the sparsemax support is a *prefix* of the sorted order (Lem. 1), retaining the top  $K$  per block and solving one exact program at the root on *original* scores reproduces the global solution whenever  $K \geq k^*$  (Prop. 11).  $K$  is a generous fixed constant set from the measured  $k^*$  distribution (E8) rather than assumed. App. B has both in full.

## 2.5 CAUSAL FORM: THE PROGRAM OVER A VISIBLE PREFIX

In a decoder, the queries that may attend to key  $j$  are those at positions  $t \geq j$ , and MarSea’s fan-out program runs over exactly that set — a *component*, not an adaptation, since in a causal model there is no counterfactual “full” column that the prefix approximates: which queries key  $j$  governs is a question about the queries that exist. Every result in Sec. 3 therefore applies verbatim, with  $m_j, \delta_j, W_j$  computed over the visible set  $Q_j(t)$ . What changes is that the column grows, and Prop. 12 records how:  $\psi_j^{(t)}$  is non-decreasing, so *exclusion is permanent* while admission is provisional, and the incremental *frozen-prefix* variant over-admits and never under-admits. That monotonicity is at fixed  $\tau_j$ , so membership and  $\tau_j$  are *sealed at the block boundary*, as the block’s inherited quota  $\bar{c}_j^{(b)}$  is; a head re-reading a growing column could move  $\psi_j$  down and forfeit permanence. So the streaming implementation need only ever *evict*, and the frozen-prefix error of the class Stage 2 can remove at all rather than the one it provably cannot (App. D). We ablate both emission policies.

# 3 THEORY

*Every claim in this section is proved in App. F, which restates each result before proving it; the few given here in prose are stated formally there.*

## 3.1 TWO OBSTRUCTIONS

**Definition 1** (Query-locality, and product form). Write query  $i$ ’s inputs as  $x_i := (q_i, \gamma_i) \in \mathbb{R}^d \times \Gamma^{n_k}$  with admissible set  $\mathcal{D}_i$ . A scheme is **query-local** if for every  $i$  the row  $A_i$  is a function of  $(x_i, \{k_j\}_j)$  alone, independent of  $x_{i'}$  for  $i' \neq i$ . A constraint  $\mathcal{C} \subseteq \mathbb{R}_{\geq 0}^{n_q}$  is **product-form** if  $\mathcal{C} = \prod_i \mathcal{C}_i$ ; a capacity bound  $\sum_i x_i \leq c$  is not, since whether one coordinate is admissible depends on the others. The first is independence in the mechanism, the second in the constraint.

Row-softmax, every additive relative-position or geometric bias and every elementwise gate  $A_{ij} = \alpha_{ij}g(q_i, k_j, \gamma_{ij})$  are query-local; MarSea is not.

**Theorem 1** (Query-local schemes attain exactly a product set). *Fix a key  $j$ , evaluate a query-local scheme over  $\mathcal{D} = \prod_i \mathcal{D}_i$  with  $\{k_j\}$  fixed, and put  $\mathcal{R}_i := \{A_{ij}(x_i) : x_i \in \mathcal{D}_i\}$ . Then the attainable set of fan-out columns is exactly the product,*

$$\{A_{\cdot j}(x) : x \in \mathcal{D}\} = \prod_{i=1}^{n_q} \mathcal{R}_i, \quad (8)$$

and the scheme satisfies  $A_{\cdot j} \in \mathcal{C}$  for all  $x \in \mathcal{D}$  iff  $\prod_i \mathcal{R}_i \subseteq \mathcal{C}$ . Hence no query-local scheme enforces a non-product-form  $\mathcal{C}$  except vacuously.

**Assumption 1** (Non-degenerate replication). *Fix key  $j$ . There exist  $a > 0$  and  $M \leq n_q$  such that some admissible input has at least  $M$  queries  $i$  with  $A_{ij}(x_i) \geq a$ . We call  $(a, M)$  a replication pair for key  $j$ .*

**Corollary 1** (Cardinality fails, and fails on real inputs). *Cardinality  $\mathcal{C}_{\text{card}} = \{x : |\text{supp}(x)| \leq m\}$  with  $1 \leq m < |\mathcal{E}_{\cdot j}|$  and capacity  $\mathcal{C}_{\text{cap}} = \{x : \sum_i x_i \leq c\}$  with  $c > 0$  are both non-product-form for  $|\mathcal{E}_{\cdot j}| \geq 2$ , so by Thm. 1 no query-local scheme enforces either. The failure is realized on constrained input spaces: given a replication pair  $(a, M)$  inside the set (Assumption 1), some admissible input has  $M$  coordinates at least  $a$ , so **cardinality fails whenever  $m < M$  and capacity fails whenever  $c < Ma$ .***

**MarSea claims the cardinality half and not the capacity half, and we say so here rather than let it be inferred.** By equation 2 the mass a key emits into its relation is inherited from an unmodified layer, so no *budget* is predicted and no bound on  $\sum_{i \in \mathcal{E}_{\cdot j}} \tilde{A}_{ij}$  is claimed. What the mechanism decides is  $|\text{supp}(\tilde{A}_{\cdot j})|$ , and that is the quantity the corollary shows a query-local scheme cannot bound, that Prop. 6 converts into a precision ceiling, and that Fig. 2(a) measures. One consequence is worth stating plainly: on each margin a single parameter now carries the whole claim —  $\tau_j$  on the column,  $\tau_i$  through  $\theta_i$  on the row — where a predicted budget would have supplied a second, independent constraint. That makes the interval-hit rate of Sec. 5 the diagnostic rather than one of several. It does *not* make  $\tau_j \rightarrow 0$  a route back to standard attention: by Lem. 1 that limit is the *uniform* allocation of the inherited quota over the relation, a column re-shaped to the wrong shape rather than left alone, and the only way back to standard attention is an empty relation (Prop. 2(vi); App. D).

The hypothesis is generic rather than adversarial: wherever many candidates are near-equivalent with respect to one key — topically plausible distractors retrieved for the same question — a query-local gate, knowing nothing of the others, must treat them alike. **S0 measures  $(a, M)$  on trained-backbone columns** rather than assuming it; the probe already run is synthetic (App. H).

**Proposition 1** (Conservation obstruction). *If  $\sum_j A_{ij} = 1$  for every query, then  $\max_j \sum_i A_{ij} \geq n_q/n_k$ : no row-normalized scheme can bound every key’s fan-out below the query-to-key ratio, and doubly stochastic attention attains it with equality for every key.*

Neither implies the other: Prop. 1 forbids bounding all columns by a common constant but not a single one, while Thm. 1 forbids a query-local scheme from bounding even one. Together they force query coupling *and* a column marginal that is learned and non-uniform — the total is conserved either way, and what MarSea changes is how it is allocated, which is why a *uniform* marginal does not address cardinality.

**How hard the second bites depends on the shape of the attention, and the mechanism is defined for all of them.** Both programs are stated for arbitrary  $n_q$  and  $n_k$ , so MarSea applies to encoder self-attention, to cross-attention, and to the decoder self-attention this paper deploys in; what varies with the shape is not the mechanism but the *strength* of Prop. 1. Where a small key set serves many queries — cross-attention, encoder-decoder attention, any layer with  $n_k \ll n_q$  — the bound  $n_q/n_k$  grows with the query count and the obstruction is severe: no row-normalized scheme keeps any key’s fan-out bounded as queries are added. **In self-attention  $n_q = n_k$  and the bound is the constant 1, and we think it is weak there.** Worse than weak: under causal masking the bound is attained at the first position, where the only visible key is the first and  $A_{11} = 1$  before anything

is trained, so in a decoder the witness is an artifact of the mask rather than a statement about fan-out control. The proposition remains true and it still forbids bounding *every* column below one, but we do not lean on it in the decoder, and a reader should not read the abstract’s “severe” as applying to Sec. 4’s setting. **Nor does conditional activation rescue it.** The same pigeonhole runs against MarSea’s own inherited capacities:  $\sum_j c_j = n_q$  over all keys gives  $\max_j c_j \geq n_q/n_k$ , exactly as for the unmodified layer it inherits from. MarSea does not escape conservation of the total, and never claimed to; what it changes is *which* members of a relation the total reaches, which a uniform column marginal cannot do. **The load therefore shifts between the two obstructions with the shape.** Thm. 1 is shape-independent, forbidding a query-local scheme from bounding even one column at any  $n_q$  and  $n_k$ , and it is what carries the argument in a decoder; Prop. 1 is what additionally rules out the transport-based remedy, and it is strongest where the key set is small.

### 3.2 WHAT THE TWO STEPS GUARANTEE, AND WHAT SHRINKING A RELATION DOES

**Proposition 2** (The two-step form conserves and cannot relocate). *For every exclusive set  $\mathcal{E}$  and every  $\tau_j, \tau_i > 0$ : (i)  $\sum_i \tilde{A}_{ij} = c_j$  exactly, whatever  $\mathcal{E}_j$  is — the column carries the mass an unmodified layer gives it; (ii) hence  $\sum_j c_j = n_q$ , as a consequence rather than an imposed anchor; (iii)  $\sum_j A_{ij} \leq \sum_j A_{ij}^{\text{sm}}$  for every query — the mechanism never gives a row more than the unmodified layer gave it, which is one unit in exact arithmetic — with equality only when step 1’s cap binds, so a query may still spend less; (iv) entries off  $\mathcal{E}$  are fixed by step 1 and unchanged by step 2 on both margins, so no margin claim can be met by relocating mass out of the relation; (v)  $\theta_i \geq 0$  for every  $\tau_i > 0$ , so a Stage-1 zero is never resurrected and the relation’s row mass never exceeds its quota  $\tilde{c}_i$ ; (vi) with  $\mathcal{E} = \emptyset$  the output equals  $A^{\text{sm}}$  entrywise.*

**Why (iii) is stated against the row’s own mass, and what that buys in finite precision.** In exact arithmetic  $\sum_j A_{ij}^{\text{sm}} = 1$  and (iii) is the unit bound. We write it against the row’s own mass because the two are not the same number in floating point, and the difference is a property of the *reduction kernel* rather than of the mechanism: an fp32 softmax row sums to  $1 + \delta$  with  $|\delta| \leq n_k \epsilon / 2$ , the worst case being a row with one dominant entry and a long tail lying just under half an ulp of it, so that a  $L$ -lane reduction discards  $1/L$  of that tail and  $\delta$  reaches  $n_k \epsilon / (2L)$ . That is  $\approx 10^{-6}$  for a pairwise reduction and  $6 \times 10^{-5}$  at  $n_k = 16,384$  for a 16-lane one. **Testing whether a row exceeds one therefore makes the cap’s binding set device-dependent, and no fixed tolerance is correct on every kernel.** Testing whether the *relation* added mass does not: by (iv),  $\tilde{A} = A^{\text{sm}}$  off  $\mathcal{E}$ , so

$$\sum_j \tilde{A}_{ij} - \sum_j A_{ij}^{\text{sm}} = \sum_{j \in \mathcal{E}_i} (\tilde{A}_{ij} - A_{ij}^{\text{sm}}) \quad (9)$$

exactly — a difference of in-relation terms that reads no row total. The implementation binds equation 6 on equation 9 and projects a binding row onto  $\sum_j A_{ij}^{\text{sm}}$ , so the binding *decision* is a function of in-relation terms alone on every kernel, (v) and (vi) hold *exactly* — an empty relation has excess bitwise 0.0 and so reproduces standard attention bitwise rather than to a tolerance. **What finite precision still costs is a window, not a constant.** A row whose excess is positive but smaller than the projection’s own arithmetic can see is left uncapped, so the *realized* row total may exceed  $\sum_j A_{ij}^{\text{sm}}$  by that much:  $10^{-7}$  on our reference CPU and  $2 \times 10^{-6}$  at  $n_k = 16,384$  on the deployment GPU, both measured. (iii) is a statement about the program and is exact; the implementation meets it up to that window, which App. G reports and which the release gates against the tolerance its invariants allow. App. G likewise reports the kernel’s own  $\delta$  as a device statistic beside the checks rather than inside them.

**Proposition 3** (Shrinking a relation only helps). *Let  $T_j \subseteq \mathcal{E}' \subseteq \mathcal{E}$ . Then  $W_j$  and  $m_j$  are unchanged, and (i)  $\delta_j(\mathcal{E}') \geq \delta_j(\mathcal{E})$ ; (ii) the relative interval width  $m_j \delta_j / W_j$  is non-decreasing; (iii) if Assumption 2 holds on  $\mathcal{E}$  it holds on  $\mathcal{E}'$ , and the recovery interval on  $\mathcal{E}$  is contained in the one on  $\mathcal{E}'$ ; (iv) the converse fails — Assumption 2 can hold on  $\mathcal{E}'$  where it fails on  $\mathcal{E}$ . Restricting to a relation therefore never narrows the admissible temperature set and can create one where none existed.*

Part (iv) is the reason the restriction matters asymptotically rather than cosmetically.  $\delta_j$  is a minimum over the target set minus a maximum over its complement, so over a complement that grows with the context it is non-increasing and converges to a limit that is negative whenever a non-target can outrank the weakest target; past that point Sec. 3.5 is vacuous on the full column. Over a relation whose size does not track the context it is context-free by construction. Table 5 measures both,

and the harness finds Assumption 2 failing on the full set but holding on the relation for 11.2% of columns drawn.

### 3.3 STRUCTURE OF THE FAN-OUT PROGRAM

The fan-out program has one structural property everything else rests on. With  $G(k) := \sum_{r \leq k} (z_{(r)} - z_{(k)})$  non-decreasing in  $k$ , the support of  $\text{sparsemax}(z)$  is exactly the  $k^*$  largest entries with  $k^* = \max\{k : G(k) < 1\}$ : a *prefix* of the sorted order (Lem. 1). Three things follow. *Monotone exclusivity*:  $k^*(\tau)$  is non-increasing, from  $n_q$  at  $\tau \rightarrow 0^+$  to  $|\arg \max_i s_i|$  at  $\tau \rightarrow \infty$ . And the two facts Sec. 2 relies on — exactness of hierarchical retention above  $k^*$  (Prop. 11) and permanence of exclusion as the causal prefix grows (Prop. 12).  $G$  also *computes* the endpoints of Thm. 2.

### 3.4 WHAT IS REALIZABLE, AND WHERE THE DIFFICULTY LIES

A bound on others is not a capability, so we give the positive counterpart. *Every competitive selection is realizable*: for any partition  $\{T_j\}$  there are scores, exclusivities and budgets making  $\text{supp}(A_{\cdot j}) = T_j$  for every  $j$  at once and at *every* admissible  $\tau_i$  — the floors are not free witnesses, being duals of the unit cap — robustly to sup-norm score perturbation below  $1/2$ , with cardinalities free apart from what a partition forces (Prop. 4).

The negative half must be stated carefully, because one half of what we want *is* available to a query-local scheme (Prop. 5). (i) Selection at a *single* target is: row-sparsemax at large temperature admits only  $\arg \max_j s_{ij}$ . (ii) Scope — a bound  $|T_j| \leq \kappa_j$  — is not: a replication pair  $(a, M)$  with  $\kappa_j < M$  drives  $|\text{supp}(A_{\cdot j})| \geq M > \kappa_j$  on some admissible input. (iii) The non-local content is therefore neither in isolation but their *conjunction*, which MarSea satisfies by construction. Query-locality restricts a row’s dependence on *other queries’* inputs, not on keys, so target-side exclusivity as such is not what is unavailable: what (ii) shows is that *independent* per-target selection over-commits a source (App. D). “The fastest lane” is answerable one lane at a time; “one vehicle per lane, each taking the fastest lane still open to it” is not. **This concession governs which experiments count**: a gain at  $m = 1$  is not evidence for Thm. 1.

### 3.5 EXACT RECOVERY

**Assumption 2** (Margin and spread).  $\delta_j := \min_{i \in T_j} s_{ij} - \max_{i \notin T_j} s_{ij} > 0$ ; and  $W_j := \sum_{i \in T_j} (s_{ij} - \min_{i' \in T_j} s_{i'j})$ , which is  $\geq 0$  by definition rather than by hypothesis.

**Theorem 2** (Exact support recovery, with sharp endpoints). *Let  $\alpha = 2$  and Assumption 2 hold. Then  $\text{supp}(\tilde{A}_{\cdot j}) = T_j$  if and only if*

$$\tau_j \in \left[ \frac{1}{W_j + m_j \delta_j}, \frac{1}{W_j} \right) \quad (1/W_j := \infty \text{ if } W_j = 0), \quad (10)$$

and this interval is non-empty for every column with a positive margin, since  $m_j \delta_j > 0$ .

Both endpoints are exact because  $G(m_j + 1) = \tau_j(W_j + m_j \delta_j)$  is computable rather than merely bounded below. Exact recovery is therefore never obstructed on the column — some temperature always attains it — and the obstruction is instead *width*, the interval’s relative width being  $m_j \delta_j / W_j$ . Keys whose intervals are disjoint,  $W_{j_2} + m_{j_2} \delta_{j_2} \leq W_{j_1}$ , admit no common global  $\tau$ , so **content-dependent exclusivity is necessary**, not merely convenient (Cor. 3). Because the interval is a function of the score column the same argument runs *within* a key across contexts, which is why equation 5 reads  $s_{\cdot j}$ . The two regimes fall out of the endpoints — for  $m_j = 1$ ,  $W_j = 0$  and any  $\tau_j \geq 1/\delta_j$  isolates the target; for large  $m_j$  the upper endpoint binds and a *small*  $\tau_j$  is needed — and both occur in one instance.

### 3.6 FIDELITY: PRECISION AND RECALL ON BOTH MARGINS

Reading the allocation as retrieval of the ground-truth relation makes the guarantees legible and locates where each error type enters. We carry *both* margins throughout.

**Definition 2** (Support precision and recall, both margins). For key  $j$  with target set  $T_j$ ,  $m_j = |T_j|$  and  $S_j := \text{supp}(p_{\cdot j})$ , and query  $i$  with true source set  $K_i$  and  $S_i := \text{supp}(A_{i, \mathcal{E}_i})$  — the support on the relation, since off it the row carries the unmodified layer’s dense entries by construction and precision there is not a quantity the mechanism decides — put  $P_j = |S_j \cap T_j|/|S_j|$ ,  $R_j = |S_j \cap T_j|/m_j$ ,  $\rho_j = \sum_{i \in T_j} p_{ij}$ ;  $P_i = |S_i \cap K_i|/|S_i|$ ,  $R_i = |S_i \cap K_i|/|K_i|$ , with the convention that precision is 1 on an empty support. Where the two stages must be distinguished we write  $S_j^{(1)} := \text{supp}(\tilde{A}_{\cdot j})$  and  $S_j^{(2)} := \text{supp}(A_{\cdot j})$ , and likewise  $S_i^{(1)}, S_i^{(2)}$  on the row, all taken on the relation;  $S_j^{(1)} = S_j$  since  $\bar{c}_j > 0$ . Where  $T_j \not\subseteq \mathcal{E}_j$  the excluded targets are recall misses before any program runs, and we report **relation recall**  $|T_j \cap \mathcal{E}_j|/m_j$  separately so that a mask error is never charged to a temperature.

Under any *normalized* allocation mass-precision and mass-recall coincide at  $\rho_j$ : the two error types become definable only once exact zeros exist, so this is a capability a fully supported mechanism lacks, not a vocabulary.

*Full support pins precision, on both margins* (Prop. 6). If a mechanism assigns  $p_{ij} > 0$  to every pair — row-softmax, any additive bias on it, any elementwise gate with positive factors — then for every key, every score configuration and every temperature,  $P_j = m_j/n_q$  and  $P_i = |K_i|/n_k$  *exactly*. Both are independent of the scoring function, so neither improves by training, by architecture within the class, or by any temperature; both tend to 0 as the candidate set grows, while recall is 1 trivially.

This is not a technicality about how support is defined: a mechanism whose output set is fixed in advance has surrendered precision as a *controllable* quantity, the scores being free to rank the targets perfectly while the precision is unchanged. Thm. 1 says a query-local mechanism cannot enforce the constraint and Prop. 6 that a full-support one cannot represent the answer; both must be cleared, which is why MarSea couples and zeroes.

The interval of Thm. 2 is itself a precision–recall interval (Prop. 7), and because both endpoints are exact so is the correspondence — Fig. 1, with  $\rho_j = 1$  inside as well. The interval is never empty, so the obstruction is not attainability but *width*: a column with large within-set spread leaves a narrow target, which is what makes a head reading that column necessary rather than merely helpful. Outside it the failure is graded rather than abrupt, each margin shedding true members in non-decreasing order of strength, ties together, at a rate exact in  $\tau_j$  on the column and in  $\theta_i$  on the row, so the two are separately diagnosable (Prop. 8).

The two stages compose in one direction only (Prop. 9): Stage 2 only shrinks supports, so it only loses recall and a Stage-1 false negative is **unrecoverable**, while an *exact* precision once achieved is preserved through any admission floors whatsoever. The composition is **ratcheted**, not cancelling —  $\theta_i$  is a row threshold and  $P_j$  a column quantity, so Stage 2 may also lower a column’s precision, which is why E7 reports both readings rather than attributing a deficit by subtraction. Recall is the fragile side on both margins — at least 3:1 on the column, 2:1 on the row (App. D) — so no single  $F$ -style aggregate should be reported. The stage *order* is settled by theorem, precision-preservation being available only in this order, which is what makes frozen-prefix safe.

**Corollary 2** (Both quantities, both margins, simultaneously). Let Assumption 2 hold for every key, and let the two ground truths be consistent,  $K_i = \{j : i \in T_j\}$ . If  $\tau_j \in [\frac{1}{W_j + m_j \delta_j}, \frac{1}{W_j})$  and  $K \geq m_j$  for every  $j$ , and  $\theta_i < \min_{j : i \in T_j} c_j(1 - \tau_j W_j)/m_j$  for every  $i$ , then  $P_j = R_j = 1$  on every column and  $P_i = R_i = 1$  on every row, all four at once. Consistency is what makes “both margins at once” well posed: without it a key may allocate to a query that does not count it as a source, and no floor common to the row removes that edge while keeping the true ones.

**This is not a point on a trade-off curve.** Under Assumption 2, sweeping  $\tau_j$  traces the **L** of Fig. 1 — corner at  $P_j = R_j = 1$ , no interior point,  $R_j = 1$  with  $P_j = m_j/k^*$  below the interval and  $P_j = 1$  with  $R_j = k^*/m_j$  above (Lem. 1 gives  $|S_j| = k^*$ ) — while a full-support mechanism does not move from  $(1, m_j/n_q)$  at any temperature. Recall is therefore not what precision is bought with,

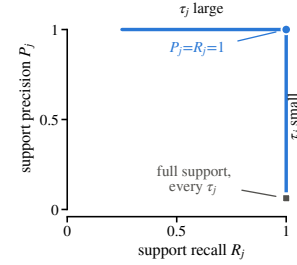


Figure 1: **The trace is predicted in full.** Stage 1 at 400 temperatures, one column,  $n_q = 64$ ,  $m_j = 4$ . Full support is the grey square, at *every*  $\tau_j$ .

and it is given exactly rather than bounded. The conditions are not prices paid, but nor are they free:  $\tau_j$  must be *found* in an interval never empty but possibly narrow,  $K \geq m_j$  is what E8 protects, and  $\theta_i$  is reachable only through  $\tau_i$ . E7 measures all three, and the curve.

Smoothing is not a way around a large candidate set either. Were the fan-out computed by column-*softmax*, holding the mass outside the target set relative to that inside at  $\varepsilon$  would take  $\tau_j \geq \frac{1}{\delta_j} \log \frac{n_q - m_j}{m_j \varepsilon}$ , in the worst case over configurations meeting Assumption 2 and with the bound tight there — growing like  $\log n_q$ , whereas exact recovery at  $\alpha = 2$  needs  $\tau_j \geq 1/(W_j + m_j \delta_j)$ , with no  $n_q$  displayed (Prop. 10; on what survives inside  $\delta_j$ , App. E).

**Smoothing does not relax a large candidate set — it is what makes it bite**, since every additional non-target contributes its own small share while a terminated tail removes the dependence outright. Sec. 4 turns that into a measurement.

## 4 APPLICATION: MULTIVARIATE EVIDENCE SELECTION

Deciding which  $m$  of  $n$  retrieved passages support an answer is a joint selection among topically plausible distractors:  $m > 1$  with coupling across targets, not the case Prop. 5(i) concedes. Three results meet there: by Prop. 6 any full-support attention has evidence precision exactly  $m/n$ , falling to zero as candidates are added; by Prop. 10 the temperature needed to hold leakage fixed grows like  $\log n$  in the worst case; and MarSea’s support size involves no  $n$  on the relation — a candidate outside it keeps its full-support weight, so the ceiling is removed only on the covered part of a row, and E3 reports the coverage rate  $|\mathcal{E}_j|/n$  alongside. **The theory therefore predicts that evidence selection degrades with candidate count for architectural reasons, and that MarSea’s does not.** The setting is a causal decoder, native by Sec. 2.5 rather than a reformulation.

**What we are not claiming:** long-context degradation also has a *positional* component with its own account (Chowdhury, 2026; Wu et al., 2025; Herasimchyk et al., 2026), which E3 controls for and App. D separates from ours in full.

## 5 EXPERIMENTS

Every claim is matched to a measurement that could falsify it, and **E3 is load-bearing**: it, not aggregate accuracy, decides whether the theory is doing the work.

**Setup.** We fine-tune a borrowed 1–3B decoder — Qwen2.5-1.5B (Apache 2.0) primary, Llama-3.2-3B as a cross-family check — with MarSea replacing *the softmax line of the attention step and nothing else* in the layers that carry the backbone’s retrieval heads (Wu et al., 2024),  $\tau_\phi^K, \tau_\phi^Q$  and the two relation heads as small modules reading a key or query together with a streaming summary of its column or row, and compare against *the same decoder, the same layers, the same adapters and the same schedule with unmodified attention* — a mechanism ablation, not a leaderboard entry. Every baseline is the same substitution at the same line. Since MarSea’s claim concerns the number of *candidates*, not raw token length, sweeps run at moderate context (2–16K tokens) with  $n$  up to a few hundred passages; App. H gives the reporting rules.

**Datasets, and what each is for.** **RULER** (Hsieh et al., 2024) is the *identifying* instrument, needle and distractor counts both being configuration parameters so that  $m$  moves *at constant task difficulty*; **MuSiQue** (Trivedi et al., 2022) is the *externalising* task, with **HotpotQA** (Yang et al., 2018) secondary and never alone; and **IHEval** (Zhang et al., 2025) supplies what evidence selection cannot, a *ranked* order over conflicting instruction tiers. App. H gives the reasons, including why neither instrument is reported alone.

**Baselines.** Six, each isolating one design decision and retuned under a matched schedule rather than quoted. Two are decisive and they ablate different things. **A learned-marginal transport in the MESH form** ablates *capacity against shape*: it predicts both marginals and no shape, which is the mirror image of what we do, and Prop. 6 predicts its column precision is pinned at  $m_j/|\mathcal{E}_j|$  regardless. **An exclusivity head reading  $k_j$  alone**,  $\tau_\phi^K(k_j)$  with the column statistic withheld, ablates

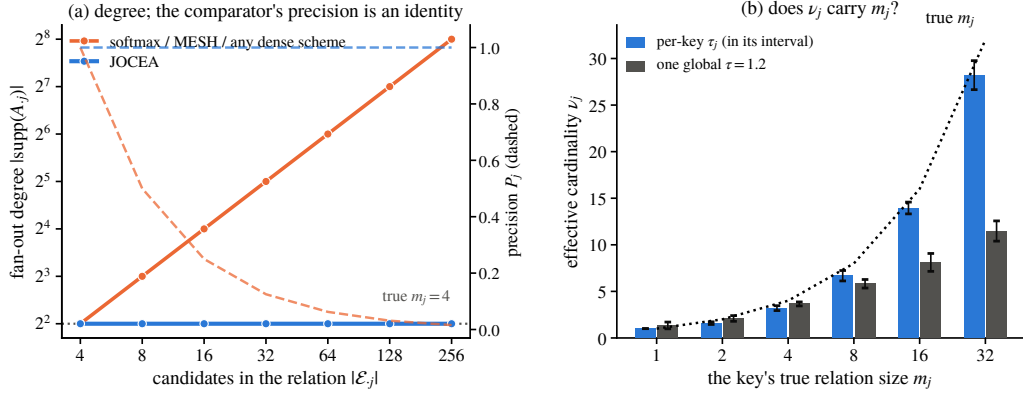


Figure 2: **E1, the replication probe.** (a) The true relation size is held at  $m_j = 4$  and the candidate set  $|\mathcal{E}_j|$  grows from 4 to 256; 12 seeds,  $\tau_j$  taken inside Thm. 2’s interval. **The measured quantity is degree, not mass:** by equation 2 the mass is inherited from an unmodified layer, so a mass sweep would show MarSea tracking softmax and would be the wrong instrument. Solid, left axis: fan-out degree. Any full-support scheme reaches every candidate (slope +1); MarSea holds 4.00 throughout (slope  $-0.0000$ ), which is Cor. 1’s cardinality half. Dashed, right axis: precision, which for the comparator is  $P_j = m_j/|\mathcal{E}_j|$  at every temperature and scoring function (Prop. 6) — an identity, not a tendency — and 1.000 for MarSea. (b) The one link that is not an identity. With the capacity inherited,  $\tau_j$  carries the whole fan-out claim and must place  $k^*$  at  $m_j$ ; its head reads  $\nu_j = \|p_{\cdot j}\|_2^{-2}$ , and whether  $\nu_j$  tracks  $m_j$  depends on  $\tau_j$  and the column. Six keys, true sizes 1–32 at  $n_q = 63$ , error bars one standard deviation. Under a per-key  $\tau_j$  inside its interval it tracks; under one global  $\tau$  it saturates, which is Cor. 3 in training-free form. Both panels are training-free possibility results. Reproduced by `v1_probe.py`.

the *field argument*; Cor. 3 predicts it fails wherever two keys need disjoint intervals, and Fig. 2(b) is the training-free form of the same test (App. H).

**E1 — the replication probe (run).** The probe measures **degree, not mass**, for the reason Sec. 2.1 gives: step 1 hands the relation whatever an unmodified layer would, so mass is not the bounded quantity and a mass sweep would show MarSea tracking softmax. Fig. 2(a) holds the true relation size at  $m_j = 4$  and grows the candidate set  $|\mathcal{E}_j|$  from 4 to 256. MarSea’s fan-out degree is 4.00 at every size — slope  $-0.0000$  — while any full-support scheme reaches all  $|\mathcal{E}_j|$  of them, slope  $+1.0000$ . Precision follows as an identity rather than a tendency: 1.000 against 1.000, 0.500, 0.250, 0.125, 0.062, 0.031, 0.016. **That gap is an identity on the comparator’s side:** by Prop. 6 a mechanism with  $p_{ij} > 0$  everywhere has  $P_j = m_j/|\mathcal{E}_j|$  at every temperature and every scoring function, so no training, no additive bias and no elementwise gate moves it. It is Cor. 1’s cardinality half on MarSea’s side:  $\tau_j$  from Thm. 2’s interval terminates the tail, and the interval is non-empty on every column with a positive margin.

**What panel (b) adds is the one link that is not an identity.** With the capacity inherited,  $\tau_j$  carries the whole fan-out claim, and landing it means placing  $k^*(\tau_j)$  at  $m_j$ ; the head reads  $\nu_j = \|p_{\cdot j}\|_2^{-2}$ , and whether  $\nu_j$  tracks the true  $m_j$  depends on  $\tau_j$  and on the score column rather than following from the definitions. It does under a per-key  $\tau_j$  inside its interval — 1.00, 1.59, 3.20, 6.68, 13.95, 28.21 against true sizes 1 to 32 — and saturates under one global  $\tau$ , at 1.34, 2.09, 3.66, 5.82, 8.11, 11.48, compressing a factor of 32 into a factor of 8.6. That is Cor. 3 in training-free form, and the reason a uniform-exclusivity null cannot supply the head with a usable input at all. *Both panels remain possibility results:*  $\mathcal{E}_j$  and  $\tau_j$  are set by hand and Stage 2 is inert, so they show what a column-coupled mechanism *can* express where query-local schemes provably cannot, not that a trained  $\tau^K$  will find it. S0 is the measurement (App. H).

**E2 — the  $m$ -sweep at fixed  $n$  (a Thm. 2 test).** On RULER, hold  $n$  and sequence length fixed and sweep the needle count  $m \in \{1, 2, 4, 8, 16\}$ . **An earlier draft called this load-bearing and read a**

**precision gap growing with  $m$  as evidence for Thm. 1. That reading is arithmetically inverted:** by Prop. 6 the comparator’s evidence precision is the identity  $m/n$ , which *rises* with  $m$  at fixed  $n$ , so MarSea’s precision gap  $1 - m/n$  *shrinks* with  $m$  if MarSea is at ceiling, and could only grow if MarSea failed at small  $m$ . What growing  $m$  stresses is the recovery interval, whose endpoints both scale like  $1/m$ : the head must land a narrower absolute target. E2 therefore reports MarSea’s own precision, exact-set accuracy and the *interval-hit rate stratified by  $m$* , and its falsification is a hit rate falling with  $m$ . The comparator arm carries no information here beyond confirming the identity.

**E3 — the  $n$ -sweep at fixed  $m$  (load-bearing).** Hold  $m$  fixed and sweep the candidate count  $n$ . *This is the experiment the paper lives on*, and it is Fig. 2(a) on real data: the comparator’s degree grows as  $n$  and its precision falls as  $m/n$  (Cor. 1’s cardinality half and Prop. 6), while MarSea’s degree stays at  $k^* = m$  *on the columns where Assumption 2 holds*. That qualifier is the test. If the relation is the whole candidate set, App. E predicts the assumption fails on a growing fraction of columns as  $n$  grows (Table 5: 63.6%, 14.9%, 1.2% recovery at 32, 128, 512), so E3 reports, per  $n$ , the fraction of active columns with  $\delta_j > 0$ , the interval-hit rate on those, and MarSea’s precision on those against on all; **a flat precision on the  $\delta_j > 0$  columns beside a falling fraction of such columns is the theory’s own prediction**, and a flat precision on *all* columns would mean the relation is doing the selection rather than the program (App. E, cost (a)). *Both outcomes are reportable and we pre-commit to reporting whichever we observe.*

**E3, continued — controls and what is reported.** E3 reports supporting-factor precision and recall and answer F1 against  $n$ , alongside the measured relation-coverage rate  $|\mathcal{E}_j|/n$ . At the *support* level the comparator’s precision is  $m/|\mathcal{E}_j|$  with no seed variance, an identity; at the *task* level a softmax decoder can emit exactly  $m$  items, so no such identity holds and we make no no-variance prediction there. Two controls are mandatory, since without both an  $n$ -effect is indistinguishable from a length- or position-effect: every point runs twice against inert padding of equal token count (Du et al., 2025), and gold sits at a fixed relative position with the position-marginal profile reported alongside (Chowdhury, 2026). **We pre-register what the generator does and does not hold fixed, measured before any accuracy was seen.** Sequence length is held to 96.4–98.8% of the requested 16,384 tokens across the whole sweep and on the padding control, a spread of 2.4% that is *not* monotone in  $n$ . The gold’s position, requested at the midpoint, is held to 0.470 of the token sequence at  $n = 8$  and 0.510 at  $n = 128$ : a drift of 4 points which *is* monotone in  $n$ , because distractor needles are placed at uniform random depths and a needle is not the length of an average haystack sentence, so the two do not cancel exactly. The gold therefore stays within  $\pm 2$  points of mid-sequence at every  $n$ , where the positional profile is flattest. **We do not ask the reader to take that on trust: E3depth measures accuracy against depth on the same trained model, and we report the drift’s worst-case contribution as its measured slope between 0.3 and 0.5 times 0.04**, beside the  $n$ -effect it could otherwise be confused with. App. H gives the grids and the aggregation: which layers carry MarSea, which head defines the row, and that a passage’s support is the union over its token keys.

**E4–E9 — controls, externalisation, fidelity, diagnostics, ablations.** E4 repeats E2 at  $m = 1$  (RULER S-NIAH), the regime Prop. 5(i) concedes, where a *large* gap is the interesting negative; E5 externalises to MuSiQue and HotpotQA-distractor, corroborating E2, never replacing it; E6 tests the *ranked* half on IHEval, reporting accuracy against conflicting-tier count as a slope, not a pooled mean. Where gold evidence is annotated, E7 reports all five quantities of Def. 2 — never a single aggregate — recall *twice*, the measured trace against Fig. 1, and the **interval-hit rate**: the fraction of active columns whose realized  $\tau_j$  satisfies equation 10. **That last is the sharpest single test of the theory**, being the one quantity the theory does not already determine, and it is available only because  $\tau_\phi^K$  reads that column. E8 logs the relation’s coverage on both margins, the  $|\mathcal{E}_j|$  and  $|\mathcal{E}_i|$  distributions and the realized  $k^*$ ; E9 ablates every design decision, **including each head’s field argument and the tying of  $\tau_\phi^K$  to  $\tau_\phi^Q$** ; the field-argument ablation is contribution (i)’s load-bearing test, and its capacity-against-shape arm is a published mechanism (Zhang et al., 2023), not a construction of ours. All three are in App. H, and E1–E9 together are staged cheapest-first at  $\approx 336$  H100-hours with a *pre-committed pivot* per stage (Table 8).

**Limitations.** The guarantees are single-layer, concern *support* not the layer output, hold only on the relation, and fidelity needs an annotated relation; App. D has these and the rest in full; App. E frees Assumption 2 from the candidate count. **Apart from E1, Sec. 5 has not been run.**

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## REPRODUCIBILITY STATEMENT

All analytical claims are stated with explicit hypotheses and proved in App. F; each is additionally checked by a self-contained numpy program supplied as supplementary material, which seeds each block separately, runs in under 25 seconds on one CPU core, and reproduces every figure in Tables 7 and 6, rows marked “each” being printed one per case. **Every claim verifies at zero violations**, with the count reported *per claim* in Tables 7 and 6 rather than as one headline number, because the rows are not commensurable and not all of them could have failed: a dagger marks every row whose conclusion follows from how the generator builds its input, a double dagger every row that re-expresses evaluations already counted elsewhere, and App. G says what each does and does not test. **What the harness catches is algebra errors, off-by-one errors in sorted-order arguments, and hypotheses that are accidentally stronger than intended** — which is how the non-sharp form of Thm. 2 was found. It is not evidence about real data, and the interval width on a trained network is E7. No block discards a column for failing a claim: the sharp interval is non-empty on every column drawn, so the two blocks that previously reported *accepted* counts now sample the interval directly; three rows testing the regime above  $1/W_j$  are defined only where  $W_j > 0$  and count accordingly. E1 is synthetic and training-free; its construction is in App. H. All datasets named in Sec. 5 are public.

## AI USE STATEMENT

Large language model assistance was used for literature search, LaTeX editing, and consistency checking. All theorem statements, proofs, and experimental design decisions are the authors’; every claim retained was verified by the authors, and the numerical verification was executed and its output inspected rather than reported from a model’s description of it.

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## A NOTATION AND COMPONENT SUMMARY

Table 2 lists the five components against the guarantee each carries; Table 3 collects every symbol used in the paper.

Table 2: The components. Every guarantee is scoped to the relation.

|                    | acts on                                               | parameters                                                      | guarantee                             |
|--------------------|-------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------|
| Stage 1 (fan-out)  | col. $\tilde{A}_{\cdot j}$ on $\mathcal{E}_{\cdot j}$ | $\bar{c}_j$ inherited; $\tau_j(k_j, s_{\cdot j}, \nu_j)$        | Thm. 2, Lem. 1, Prop. 2               |
| Stage 2 (fan-in)   | row $A_{i\cdot}$ on $\mathcal{E}_i$                   | cap 1; $\bar{c}_i$ inherited; $\tau_i(q_i, \tilde{A}_{i\cdot})$ | Prop. 8(ii), 9                        |
| Relation (§2.3)    | which pairs enter                                     | $e_{ij} = \mathbf{1}[(u_\phi(k_j), v_\phi(q_i)) > 0]$           | Prop. 3; $\emptyset = \text{softmax}$ |
| Hierarchy (§2.4)   | how Stage 1 is solved                                 | block $B$ , retention $K$                                       | Prop. 11                              |
| Causal form (§2.5) | the set each program sees                             | visible prefix $Q_j(t)$                                         | Prop. 12                              |

Table 3: Notation. Read each row as a left pair, then a right pair. In the fidelity group the left column is the fan-out (column) side and the right column the fan-in (row) side.

| Symbol                                                      | Meaning                                                                                                                               | Symbol                                 | Meaning                                                                                                                                                                        |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Inputs and scores</i>                                    |                                                                                                                                       |                                        |                                                                                                                                                                                |
| $n_q, n_k$                                                  | numbers of queries, keys                                                                                                              | $\gamma_{ij}$                          | relational descriptor for $(i, j)$                                                                                                                                             |
| $q_i, k_j \in \mathbb{R}^d$                                 | query and key vectors                                                                                                                 | $b(\gamma_{ij})$                       | relational bias in the score                                                                                                                                                   |
| $V_j$                                                       | value vector of key $j$                                                                                                               | $\mathcal{A}, b$                       | competing set; what it bears on                                                                                                                                                |
| $s_{ij}$                                                    | score, Eq. equation 1                                                                                                                 | $O_i$                                  | $= \sum_j A_{ij} V_j$ ; output for query $i$                                                                                                                                   |
| <i>Stage 1: the fan-out program (column), Sec. 2.1</i>      |                                                                                                                                       |                                        |                                                                                                                                                                                |
| $\Delta^{n-1}$                                              | probability simplex in $\mathbb{R}^n$                                                                                                 | $c_j$                                  | key $j$ 's <i>inherited</i> column mass, $\sum_i A_{ij}^{\text{sm}}$ ; $\bar{c}_j$ its relation's quota                                                                        |
| $\alpha$                                                    | entmax order, fixed at 2                                                                                                              | $p_{\cdot j}$                          | allocation, $\alpha$ -entmax( $\tau_j s_{\cdot j}$ )                                                                                                                           |
| $H_\alpha^T(p)$                                             | Tsallis $\alpha$ -entropy of $p$                                                                                                      | $\psi_j$                               | column threshold, Eq. equation 4                                                                                                                                               |
| $\tau_j > 0$                                                | exclusivity of key $j$ , $= \tau_\phi^K(k_j, s_{\cdot j})$ , Eq. equation 5                                                           | $\nu_j$                                | column effective cardinality $\ p_{\cdot j}\ _2^{-2}$                                                                                                                          |
| $\tilde{A}_{ij}$                                            | $= \bar{c}_j p_{ij}$ on $\mathcal{E}_{\cdot j}$ , $= A_{ij}^{\text{sm}}$ off it; the Stage-1 output                                   |                                        |                                                                                                                                                                                |
| <i>Stage 2: the fan-in program (row), Sec. 2.2</i>          |                                                                                                                                       |                                        |                                                                                                                                                                                |
| $\tau_i > 0$                                                | exclusivity of query $i$ , $= \tau_\phi^Q(q_i, \tilde{A}_{i\cdot})$ ; identity at 1 on every row                                      | $\theta_i \geq 0$                      | <i>admission floor</i> : the dual of the quota $\bar{c}_i$ , a threshold on $\tilde{A}_{i\cdot}$ set by the whole row; attains $[0, \max_j \tilde{A}_{ij}]$ as $\tau_i$ varies |
| $\tilde{R}_i$                                               | the relation's supply $\sum_{j \in \mathcal{E}_i} \tilde{A}_{ij}$ ( <i>not</i> row recall $R_i$ )                                     | $A_{ij}$                               | $= \tau_i[\tilde{A}_{ij} - \theta_i]_+$ on $\mathcal{E}_{i\cdot}$ , $= a_{ij}^{(1)}$ off it                                                                                    |
| $\tau_\phi^K, \tau_\phi^Q, u_\phi, v_\phi$                  | the content-predicting heads                                                                                                          | $\mathcal{E}_{\cdot j}, \mathcal{E}_i$ | the exclusive sets, slices of one relation                                                                                                                                     |
| <i>Expressivity separation, Sec. 3</i>                      |                                                                                                                                       |                                        |                                                                                                                                                                                |
| $x_i$                                                       | $= (q_i, \gamma_{i\cdot}) \in \mathbb{R}^d \times \Gamma^{n_k}$ , Def. 1                                                              | $\mathcal{R}_i$                        | values of $A_{ij}$ as $x_i$ varies, at fixed $j$ ( <i>not</i> row recall $R_i$ )                                                                                               |
| $\mathcal{D}_i, \mathcal{D}$                                | admissible inputs; $\mathcal{D} = \prod_i \mathcal{D}_i$                                                                              | $(a, M)$                               | replication pair, Assumption 1                                                                                                                                                 |
| $\mathcal{C}$                                               | constraint set on a column                                                                                                            | $\kappa_j$                             | scope bound asked of key $j$ : the requirement $ T_j  \leq \kappa_j$                                                                                                           |
| <i>Sorted order and recovery, Sec. 3.5</i>                  |                                                                                                                                       |                                        |                                                                                                                                                                                |
| $z_{(r)}$                                                   | $r$ -th largest entry of $z$                                                                                                          | $G(k), k^*$                            | prefix statistic; support size, Lem. 1                                                                                                                                         |
| $\text{supp}(\cdot)$                                        | the set of nonzero coordinates                                                                                                        | $T_j, m_j$                             | target set of key $j$ and its size                                                                                                                                             |
| $\delta_j$                                                  | <i>inter-set margin</i> $\min_{i \in T_j} s_{ij} - \max_{i \notin T_j} s_{ij}$ ; positive iff every target outscores every non-target | $W_j$                                  | <i>within-set spread</i> $\sum_{i \in T_j} (s_{ij} - \min_{i' \in T_j} s_{i'j}) \geq 0$ ; zero iff the targets tie, automatic at $m_j = 1$                                     |
| <i>Fidelity, on both margins, Def. 2</i>                    |                                                                                                                                       |                                        |                                                                                                                                                                                |
| $S_j$                                                       | column support $\text{supp}(p_{\cdot j})$                                                                                             | $K_i$                                  | true <i>source</i> set of query $i$                                                                                                                                            |
| $P_j, R_j$                                                  | column precision, recall                                                                                                              | $S_i$                                  | row support $\text{supp}(A_{i\cdot})$                                                                                                                                          |
| $\rho_j$                                                    | target mass $\sum_{i \in T_j} p_{ij}$                                                                                                 | $P_i, R_i$                             | row precision, recall                                                                                                                                                          |
| $S^{(1)}, S^{(2)}$                                          | support before, after Stage 2                                                                                                         |                                        |                                                                                                                                                                                |
| <i>Conditional activation and scaling, Secs. 2.3–2.4</i>    |                                                                                                                                       |                                        |                                                                                                                                                                                |
| $e_{ij}$                                                    | relation membership, Sec. 2.3; $\mathcal{E} = \{(i, j) : e_{ij} = 1\}$                                                                | $B$                                    | query block size                                                                                                                                                               |
|                                                             |                                                                                                                                       | $K$                                    | scores retained per block; $K \geq k^*$ suffices                                                                                                                               |
| <i>Selective exclusivity, App. E; stated, not evaluated</i> |                                                                                                                                       |                                        |                                                                                                                                                                                |
| $\mathcal{E}$                                               | <i>exclusive subgraph</i> : the pairs the two programs range over                                                                     | $\mathcal{E}_{\cdot j}, \mathcal{E}_i$ | its column and row slices; $ \mathcal{E}_{\cdot j} $ is key $j$ 's <i>scope</i>                                                                                                |

## B BLOCKING AND HIERARCHICAL SELECTION

Two constructions extend equation 3; they address different obstacles and conflating them has misled us once. **Blocking** partitions queries into blocks of size  $B$  and runs the program within the tile, with key  $j$ 's per-block quota  $\bar{c}_j^{(b)}$  inherited from the block's own softmax over its visible queries, so that  $\sum_j \bar{c}_j^{(b)} = |Q_b|$  holds by the same consequence as equation 2; admitting a query perturbs only its own block, so the per-step cost is  $O(n_k B \log B)$  against  $O(n_k t \log t)$  unblocked. Note that “perturbs

only its own block” is a statement about blocks, not about keys: admitting a query changes the row-softmax of that query and hence every  $c_j^{(b)}$  in the block, which is what a streaming implementation recomputes at each boundary. Both are linear in the key count — in a decoder  $n_k$  is the context length, so blocking does not make the step cost independent of it; what blocking removes is the growth of the program size and of the sort factor with the prefix, replacing  $t \log t$  by  $B \log B$ . What remains is emission, not compute. **Hierarchical top- $K$**  addresses scale: because the sparsemax support is a *prefix* of the sorted order (Lem. 1), retaining the top  $K$  scores per block, merging survivors recursively and solving one exact program at the root on *original* scores reproduces the global solution exactly whenever  $K \geq k^*$  (Prop. 11), in  $O(n_q)$  time and  $O(K)$  memory per column. Intermediate entmax is unnecessary: lower levels perform selection only, and running entmax at each computes a normalization the next discards.

$K$  should be a generous fixed constant, *not* a learned per-block budget: a learned count makes a discrete commitment no higher level can undo, adding a recall failure path that no higher level can repair, and buys nothing  $\tau_j$  does not already give. How generous is regime-dependent, and the qualification is doing work: over 4000 random columns with  $n_q \leq 2048$  (App. G),  $k^*$  has 99th percentile 5 and maximum 8 when the column is exclusive but reaches 74 when columns are deliberately broad. A fixed  $K$  in the tens is therefore safe *on exclusive columns*, not universally. Because membership is learned (Sec. 2.3) the relation *can* come to admit only well-separated columns, which are exactly the ones where  $k^*$  is small, so the hierarchy would be asked to be exact only where exactness is cheap. We do not assume it does: the relation is learned, not constructed, so the interaction is available rather than guaranteed. E8 reports the realized  $k^*$  stratified by  $|\mathcal{E}_j|$ , and its falsification clause is what protects the hierarchy in the meantime.

## C RELATED WORK IN FULL

**Geometry in scores, and other query-local remedies.** Relation modules over box geometry, directional and contextual relative-position encodings, content-predicted spatial modulation, anchor-modulated positional attention and explicit position-relation priors (Hu et al., 2018; Wu et al., 2021; Gao et al., 2021; Liu et al., 2022; Hou et al., 2024) all inject relational information into the *score*. Every one is query-local by Def. 1, so Thm. 1 applies to all of them, and MarSea is complementary rather than competing: relational information still enters the score through  $b(\gamma_{ij})$ , and what changes is that the allocation is then decided jointly per key. The same applies to a more recent remedy aimed at our own motivating symptom. Ye et al. (2025) take the diagnosis that attention over-allocates to irrelevant context and subtract two row-softmax maps so that the common-mode part cancels. Each map is a function of its own row, so the difference is query-local and the attainable-set identity of Thm. 1 covers it. **Our other two instruments do not transfer to it**, and it is worth being exact about why: the weights are *signed*, so Cor. 1’s realization argument, which lower-bounds a column sum by  $Ma$  from a replication pair with  $a > 0$  over the non-negative orthant, has no force where a column sum can be small by cancellation; and each row sums to  $1 - \lambda$  rather than 1, so Prop. 1 applies only with that constant in place of one. It is not an allocation in the sense used here, and the fidelity quantities of Sec. 3.6 are not defined on it without modification. It is a good demonstration that the phenomenon is real and worth a mechanism, and it is a natural addition to the matched-sparsity baseline group of App. H.

**Transport-based attention, and Slot Attention as its closest relative to Stage 1.** Sinkformers replace row-softmax with Sinkhorn normalization to obtain doubly stochastic attention (Sander et al., 2022), following earlier use of Sinkhorn balancing for sparse attention (Tay et al., 2020), and recent work makes it efficient (Shahbazi et al., 2025). Tay et al. (2020) is in fact a different construction — Sinkhorn is applied to a block *sorting* matrix and attention over the reordered blocks remains an ordinary row-softmax — so the group that actually constrains the attention matrix is Sander et al. (2022) and Shahbazi et al. (2025). That group identifies the marginals as the surface governing mass allocation and then sets the column marginal to *uniform*: every key emits identically. Slot Attention (Locatello et al., 2020) belongs here more squarely than its usual framing suggests, and it repays being read in this paper’s orientation: its slots emit the queries and its inputs supply the keys, so slot = query and input = key. It normalizes twice. The *first* normalization is a softmax over slots, which in these indices is  $\sum_i \text{attn}_{ij} = 1$  for every input — **a column marginal in exactly the sense of Sec. 2, fixed uniformly at one**, and the closest published relative of Stage 1. The second divides by

the sum over inputs, giving  $\sum_j A_{ij} = 1$  per slot, which is an ordinary row normalization. Zhang et al. (2023) identify the pair as a *single* Sinkhorn iteration, and one iteration does not satisfy both marginals: the second step destroys the first, so the matrix that multiplies the values is row-stochastic and its columns are unconstrained. Two things follow, and they cut in opposite directions. The construction reaches for a column marginal, which is the right object — but it is uniform and equal to one, where  $c_j$  is learned and free to differ across keys, and the competing set is a bank of slots drawn each forward pass from one shared learned Gaussian — deliberately interchangeable, so that no slot specializes — rather than the layer’s own keys, so what competes is not the data whose relation is being decided. And because what survives is the row normalization, Prop. 1 applies to the result.

**MESH is the closest prior art to MarSea’s capacities, on both margins, and the difference has to be stated precisely rather than dismissed.** Zhang et al. (2023) improve the construction by changing the transport *cost*, which is the headline; but their Appendix A also *learns* the marginals — and learns both of them: “we learn both  $a = m \cdot \text{softmax}(h_a(Z))$  and  $b = m \cdot \text{softmax}(h_b(X))$  with neural networks  $h_a, h_b$  shared across the  $m$  slots or  $n$  input elements respectively”, so that  $\sum_i a_i = \sum_j b_j = m$ . Read in the orientation above, that is a learned, non-uniform, content-predicted capacity on the key margin *and* on the query margin, each normalized to a layer total — the same objects as  $c_j$  with  $\sum_j c_j = n_q$  and the row’s unit cap. We had written that both leave the marginals fixed; that was wrong, and it was wrong on both margins, not one.

**What is absent is the other half of the object.** A capacity says how much a node may emit or absorb; it does not say *which* edges carry it, and MESH does not predict that. Their coupling is an entropy-regularized transport whose solution is *dense* — the attention matrix has no exact zeros — and the strength of the entropy term is a fixed global hyperparameter, tuned once for the model, not a quantity any head predicts per slot or per input. So there is no analogue of  $\tau_j$  reading a score column, no support to recover, no interval of Thm. 2 to land in, and nothing that could make Prop. 6 inapplicable: MESH is a full-support mechanism, so its column precision is pinned at  $m_j/n_q$  by that proposition exactly as row-softmax’s is, however well its capacities are learned. **That is the sizable difference, and it is a difference in the modelling object rather than in the parameterization** (Table 1).

Two smaller distinctions follow from the same place. **(i) The field argument.** Their  $h_a$  reads the input element *alone* and  $h_b$  the slot alone; the capacities are coupled across participants only through a softmax normalization, and neither reads a shape. Our  $\tau_\phi^K$  reads the key *together with* its score column and  $\nu_j = \|p_{\cdot j}\|_2^{-2}$ , a statistic of the realized column — the criterion Sec. 2 states for every head, which Cor. 3 shows is necessary rather than convenient, and whose consequence Fig. 2(b) measures. **(ii) An equality inside a transport, against a cap outside one.** Their marginals are equality constraints of a doubly stochastic problem, so both margins are pinned to prescribed totals and Prop. 1 applies with equality; our fan-in side is an *inequality* cap that never renormalizes, which is what lets a query spend less than a unit when nothing available deserves it.

**Query coupling without a constraint.** Selective Attention (Leviathan et al., 2025) subtracts an accumulated mask from the attention logits, with  $F_{ij} = \sum_{k \leq i-1} S_{kj}$  built from a reused head, so what query  $i$  may give key  $j$  depends on what *earlier queries* decided about  $j$ . **It is therefore not query-local by Def. 1, and Thm. 1 does not cover it** — the only attention mechanism we are aware of for which that is true, and the closest existing relative of the coupling MarSea introduces. What differs is what the coupling is *for*.  $F$  is a monotone, unbounded penalty accumulated along the prefix: it makes a key progressively less available, but it enforces nothing. No quantity is constrained, no cardinality is decided, and the row remains a softmax, so  $\sum_j A_{ij} = 1$  and Prop. 1 applies verbatim — some key still retains a full unit of fan-out however aggressively the mask accumulates. Our claim is not that queries should be coupled but that the coupling should *solve a program with a marginal*, which is what turns coupling into a degree bound and a recovery guarantee. Selective Attention is evidence that the field is reaching for the first half, and we would rather say so than let a reviewer find it.

**Capacity in routing, and the correction we owe expert choice.** Token-choice mixture-of-experts caps how many tokens an expert accepts and drops the overflow (Lepikhin et al., 2021; Fedus et al., 2022). Expert-choice routing (Zhou et al., 2022) is the entry that deserves a mechanism-level answer rather than a motivational one, and an earlier version of this paper gave the wrong kind. There, each

expert independently selects its own top- $k$  tokens with  $k = nc/e$  set by a capacity factor; the selection is over tokens, it therefore *couples* them, it is a cardinality constraint on the receiving margin, and load balance holds by construction, so there is no expert overflow to discard — unlike token choice, where overflow is dropped. (A token chosen by *no* expert simply skips the layer; that is a different matter and Zhou et al. (2022) do not claim otherwise.) So the coupling and the cardinality constraint both exist there, and rejecting the comparison on the grounds that capacity is “a systems device carrying no claim about the data” was a category error. What is absent is the part the theory is about.  $k$  is a fixed hyperparameter, uniform across experts and independent of content, where  $\tau_j$  is predicted per key from the key and its column; the selection is a hard top- $k$  rather than the solution of a constrained program, so there is no allocation to recover, no temperature to place, and no interval of Thm. 2 to hit; and the adaptive dimension in this literature is the *token*’s number of experts (Yue et al., 2025), not the expert’s capacity. Note also what this means for our own framing: Cor. 1 is a statement about *query-local* schemes, and expert choice is not one, so it is not a counterexample to anything we prove. Nor does it *escape* the theorem: expert choice is not an attention mechanism, so it was never inside the theorem’s universe. What it shares with Selective Attention is the *shape* of the move — decide on the receiving margin, over a set of senders at once — and neither turns that move into a guarantee.

**Sparse attention, and why we need no relaxation.** Sparsemax (Martins & Astudillo, 2016) and  $\alpha$ -entmax (Peters et al., 2019) yield exact zeros, and adaptively sparse transformers learn  $\alpha$  as a static per-head scalar (Correia et al., 2019); writing attention as a regularized argmax over a constrained polytope is the framework of Niculae & Blondel (2017), generalized by Fenchel–Young losses (Blondel et al., 2020). MarSea is an instantiation of that framework on the fan-out axis with content-predicted parameters; **the variational form is not our contribution.** Our departure from Correia et al. (2019) is that exclusivity is predicted per key from the key *together with its score column*, which Cor. 3 argues is necessary and Thm. 2 argues must read the column, since the admissible interval is a function of it. The usual route to a differentiable selection is instead to relax a discrete operator — a transport relaxation of top- $k$  (Xie et al., 2020), or perturbation of an arbitrary optimizer (Berthet et al., 2020). We need neither: at  $\alpha = 2$  the program’s solution is a closed-form truncation (Lem. 1), differentiable almost everywhere, with no iteration and no sampling.

**Softmax disperses, which is our motivating claim by a cleaner route.** Veličković et al. (2025) prove that softmax attention cannot remain sharp as the number of items grows: even for a task as simple as finding the maximum key, any learned circuit must disperse at test time, and they propose adaptive temperature as an acknowledged ad-hoc remedy. This is support rather than competition, and it reaches the motivation more cleanly than we do — but the remedy is exactly where the two part, and the difference is this paper. Their temperature is applied to a *row* at inference to sharpen a distribution that keeps full support;  $\tau_j$  is predicted per *key* from that key’s column and trained, and the program it parameterizes produces exact zeros, so what it delivers is a *support* rather than a sharper point on the simplex. Prop. 10 is the quantitative form of their qualitative claim on the fan-out margin, and Prop. 6 says why sharpening alone cannot reach  $P_j = 1$ : precision is pinned by the support, and a temperature does not change the support.

**Representational separations for attention.** Sanford et al. (2023) is the reference class for Thm. 1: a positive separation on sparse averaging, where attention scales logarithmically in the input size and recurrent and feedforward models polynomially, together with a negative result showing attention itself scales linearly on triple detection. Their limitations are lower bounds obtained by communication complexity. Ours separates *within* attention rather than between architectures — query-local against query-coupled, at fixed depth and width — and both the separating object (a constraint rather than a function) and the method (an exact characterization of an attainable set, with an explicit witness supplied by Assumption 1) differ. What we take from that line is the standard it sets for what a separation claim has to exhibit, not a proof technique.

**Assignment, agreement, and efficiency.** DETR enforces one-to-one assignment at the *loss* via Hungarian matching (Carion et al., 2020); capsule routing gates information flow by agreement between predicted votes and poses (Sabour et al., 2017). MarSea differs from the capsule family in being non-iterative, requiring no capsule-structured layers, and not renormalizing the allocation into a distribution over receivers —  $\sum_i \hat{A}_{ij} = c_j$  is inherited from the layer, not fixed at one. *Efficiency-*

*motivated sparsifiers* for long context (Xiao et al., 2024) reduce the support to save memory; we constrain *degree* for a correctness guarantee, and Sec. 5 includes matched-sparsity baselines so that support *size* and *membership* are not confused.

**Position bias.** Liu et al. (2024) is a distinct phenomenon with its own now-established architectural account: causal masking and residual connections produce a U-shaped influence profile over *positions* already at initialization, independently of positional encodings (Chowdhury, 2026; Wu et al., 2025; Herasimchyk et al., 2026). We make no claim about that axis; see Sec. 4 and App. D.

## D LIMITATIONS, CONTINUED

**The four stated in Sec. 5, in full. Single-layer scope:** Thm. 1 bounds what one attention operation can guarantee, not the asymptotic expressivity of a deep query-local stack. **No output-error guarantee:** Thm. 2 concerns the *support* of the allocation and does not bound  $\|O_i - O_i^*\|$  without further assumptions on the values, and the margin-conditioned output bounds available for pairwise gating do not transfer, since under column coupling  $A_{ij}$  is not a function of the pair alone. **Everything is scoped to the active set:** conditional activation is part of the mechanism, so every result above concerns the relation, while pairs outside it inherit standard attention’s precision ceiling; worse, an empty relation everywhere is a degenerate optimum *easier* to reach than collapse in  $\tau_j$ , and we have a diagnostic for it rather than a training scheme that avoids it. **Fidelity needs ground truth:** precision against a relation nobody annotated is not a number, and we do not report one.

**The positional account, and why it is a different claim.** Causal masking and residual connections produce a U-shaped influence profile over *positions*, and the literature agrees it is architectural rather than learned while differing on what else is required. Chowdhury (2026) derives it at initialization independently of positional encodings; Wu et al. (2025) obtain it from the causal mask *competing with* relative positional encodings such as decay masks and RoPE, so encodings are part of the mechanism rather than factored out; Herasimchyk et al. (2026) treat causal masking, the residual stream, positional encodings and content as four interacting forces. We do not need to adjudicate that, and we do not: what all three supply is a statement about *where* a gold item sits and is orthogonal to ours, which is about *how many* candidates accompany it: the two would predict the same aggregate degradation for different reasons, which is why E3 holds gold position fixed while sweeping  $n$  and reports the position-marginal profile alongside. A gap that vanishes once position is controlled would mean we are measuring the U-shape and not the candidate count, and we would report that. We do not claim to explain the U-shape and MarSea is not a remedy for it.

**The failure-path tally, and what it omits.** On the column, recall can fail from  $\tau_j$  above the recovery interval, from  $\theta_i$  too high, and from  $K < m_j$  at a leaf, against a single precision path ( $\tau_j$  below the interval); on the row, recall can fail from  $\theta_i$  too high and by inheriting a Stage-1 false negative, against one precision path ( $\theta_i$  too low). That 3 : 1 and 2 : 1 counts only the two programs. Two further *precision* paths enter once the rest of the mechanism is in scope: a key outside a row’s relation contributes full-support attention to that row, and the frozen-prefix variant over-admits (Prop. 12(iii)). Both are precision-side, so they do not disturb the conclusion that recall is the fragile side; they do mean the ratio should be read as a lower bound on the asymmetry rather than an exact count.

**Why independent per-target selection over-commits a source.** This is the content of Prop. 5(ii), and it is worth stating in words. If  $M$  targets are alike enough with respect to one source that the same source wins at each of them, then a query-local scheme assigns it to all  $M$  — no row can know that another row has already claimed it. The failure is not that the scheme cannot select; it is that it cannot decline to be selected.

**Thm. 1’s hypotheses, in self-attention.** The theorem fixes  $\{k_j\}$  while each  $x_i$  ranges over  $\mathcal{D}_i$ , and it is fair to ask whether the two are jointly satisfiable when  $q_i$  and  $k_i$  are projections of one token. **Fixing the keys does not by itself collapse  $\mathcal{D}_i$ .** Write  $q_i = W_q u_i$ ,  $k_i = W_k u_i$  with  $W_q, W_k : \mathbb{R}^d \rightarrow \mathbb{R}^{d_h}$  and  $u_i$  the vector the projections read. For  $W_k$  of full rank, fixing every key confines  $u_i$  to  $u_i + \ker W_k$ , of dimension  $d - d_h$ ; and unless  $\text{row}(W_q) \subseteq \text{row}(W_k)$ , in which case  $W_q$  vanishes

on  $\ker W_k$  and queries are fixed outright, the restriction  $W_q|_{\ker W_k}$  has rank  $\min(d_h, d - d_h)$ . So  $q_i$  still moves in a  $d_h$ -dimensional set once  $d \geq 2d_h$ , and under pre-LN, where  $u_i$  additionally lies on a  $(d-2)$ -dimensional manifold, once  $d \geq 2d_h + 2$  — three or more heads at  $d = Hd_h$ . Two honest caveats: the rank statements are generic rather than universal, and trained weights are not a generic draw, so the row-space degeneracy is measurable and unmeasured; and rank is not conditioning — a small  $\sigma_{\min}(W_q|_{\ker W_k})$  makes the required displacement large, which the pre-LN constraint bounds. **What this establishes is the negative claim, not a positive one:** the key constraint alone does not make Thm. 1 vacuous. Whether the tuple  $(u_1, \dots, u_{n_q})$  ranges over a *product* is a separate question, and it is the one Assumption 1 answers — so the division of labour is that Thm. 1 supplies the *shape* of what a query-local scheme attains, on whatever product domain it is given, and Assumption 1 with Cor. 1 supplies the *bite*. **S0 measures the second**, which is the half a shrunken domain can actually threaten.

**Why the margins do not share a head.** A single exclusivity head would have to accept  $(k_j, s.j) \in \mathbb{R}^d \times \mathbb{R}^{n_q}$  and  $(q_i, \tilde{A}_{i.}) \in \mathbb{R}^d \times \mathbb{R}^{n_k}$  — spaces of different dimension that coincide only in self-attention with  $n_q = n_k$ , and never in cross-attention or in the causal form of Sec. 2.5, where the column runs over  $Q_j(t)$  and the row over the visible keys. Even where a shared head is definable it would be aimed at two targets at once:  $\tau_\phi^K$  must land in Thm. 2’s interval, whose endpoints are functions of the score column, while  $\tau_\phi^Q$  must place the dual  $\theta_i$  in Prop. 8(ii)’s, whose endpoints are order statistics of the incoming allocation. The relation is one object by construction, so there is no pair of gates to tie or untie; what E9 ablates on that side is whether  $e_{ij}$  reads the pair or the key alone. E9 ablates the tying of  $\tau_\phi^K$  to  $\tau_\phi^Q$  in self-attention, where it is definable, and we predict it costs; the cost is what makes the separation mechanism rather than bookkeeping. **Why the cap follows the column program, and why the margins are not fused.** Applied beforehand the cap would merely shift scores that the column program would normalize away; and coupling both margins into one equality-budgeted program would be a capacitated transport problem needing Sinkhorn-style alternation, where two one-sided constraints in turn need a single sort each. **Stage 2 stands alone, and that bounds what it can do.** equation 7 is defined for any non-negative row, which is what makes the relation of Sec. 2.3 coherent: a row generally mixes entries re-shaped by column relations with entries left to standard attention, and the cap applies to the mixture. **What the cap does there is bound the total, not re-commensurate the two scales.** Eq. equation 7 applies one offset and one gain, both uniform across  $j$ , and a group-blind affine map cannot correct a group-dependent scale mismatch: it thresholds both groups at the same absolute level, so whichever group carries the smaller magnitudes is eliminated first irrespective of relevance. We treat this as an approximation the mixture forces rather than a repair, and it is why Cor. 2 is stated for consistent ground truths on rows whose relation is the whole row. Every row-margin guarantee inherits the standalone property, their proofs using only that truncation is coordinatewise, monotone and fixes zero — properties the gain  $\tau_i > 0$  preserves, so each result is stated in terms of  $\theta_i$ , a threshold on  $\tilde{A}_{i.}$ . The converse is the limitation: applied to an ordinary attention row, equation 7 is a function of that query’s own inputs and the keys, hence query-local by Def. 1, so it bounds that row’s fan-in — which needs only the cap — while by Thm. 1 leaving fan-out entirely untouched. A row-side deployment of MarSea is therefore a real mechanism with a real guarantee, but it is not the paper’s central claim, which needs Stage 1. **Row mass is bounded above by the row’s own softmax mass, and not below.**  $\sum_j A_{ij} \leq \sum_j A_{ij}^{\text{sm}}$  (Prop. 2(iii)), which is one unit in exact arithmetic, so no query ever absorbs more than the unit softmax spends — but a query whose relation is slack, or whose  $\tau_i < 1$  scales its relation down, spends *less*. That is deliberate (it is what lets a query decline to be explained by the keys available) and it is a real difference from softmax, which spends exactly one unit always: output magnitudes now vary across queries within a layer, and downstream normalization sees a wider range than it would otherwise. **What the cap bounds unconditionally is fan-in mass.** It bounds fan-in *degree* only where it binds: on a slack row  $\theta_i = 0$  and  $\text{supp}(A_{i.}) = \text{supp}(\tilde{A}_{i.})$  exactly, so the row’s degree is whatever Stage 1 and the relation left it, and Stage 2 is a positive scalar gain. The degree constraint on this margin is therefore conditional on  $\tau_i > 1$  (a floor above step 1’s) in a way the column’s is not, and E8 reports the fraction of rows on which the cap binds beside the coverage rates. **The admission floor is a dual, not a knob, and its range is bounded.**  $\theta_i$  is determined by  $(\tilde{A}_{i.}, \tau_i)$ : as  $\tau_i$  ranges over  $(0, \infty)$  it traverses  $[0, \max_j \tilde{A}_{ij}]$  continuously and strictly increasingly, and  $\theta_i = 0$  is *forced* on any row arriving within budget. Two consequences. Every  $\theta_i$ -conditioned statement above should be read as quantifying over that attainable range rather than

over an arbitrary floor; and Prop. 8(ii)’s row interval can be *unreachable* at every admissible  $\tau_i$  — in particular a row whose weakest true source is its largest entry can never shed it, so  $R_i \geq 1/|K_i|$  unconditionally. Where a predicted cap would have made row exactness a matter of parameter choice, the unit cap makes it partly a matter of reachability, and E7 reports attainability rather than only the sign of the margin. **Truncation rescales the survivors.** The soft form  $\tau_i[\tilde{A}_{ij} - \theta_i]_+$  subtracts  $\theta_i$  from every edge it keeps and then applies a common gain, so relative magnitudes within a row are not preserved even where the support is exact; a hard variant preserves them at the cost of differentiability at the threshold, and we ablate both. **The two margins are constrained in different currencies,** a mass budget and an admission floor. The asymmetry is what preserves a closed-form, non-iterative solution, but a genuinely two-sided budgeted formulation — a capacitated transport problem — remains open.  $\alpha = 2$  **specificity.** Monotone exclusivity and Thm. 2 are proved for sparsemax, which is the setting we deploy; monotonicity is numerically consistent at  $\alpha \in \{1.2, 1.5, 1.8, 3\}$  but unproven there. **Not invariant to the query set.** The program couples queries, so MarSea is permutation-equivariant but not invariant to insertion or deletion of queries. **The causal form interacts with the key–value cache.**  $\psi_j$  is a function of the visible column, so admitting a new query alters previously computed entries, whereas ordinary causal attention fixes  $A_{ij}$  once computed. Prop. 12 bounds the damage in the direction that matters and blocking holds the cost at  $O(n_k B \log B)$  per step, but the two emission policies are a real choice — seal-at-boundary is exact at up to  $B$  tokens of latency, frozen-prefix is fully incremental and over-admits — and we report both rather than claim one dominates. Relatedly,  $\tau_j$  reads  $\nu_j$ , which is a statistic of  $p_{\cdot j}$  and so is unavailable from the block being solved; it is taken from the previous block of the same key (or the previous layer in an encoder), and is initialised at 1, the value of a singleton support, on the first. **Degeneracy.** If  $\tau_j$  collapses to a constant across keys, or  $\tau_i$  to a constant across queries, the corresponding margin loses all content dependence — the column reducing to a fixed-temperature entmax on the relation. **With the capacity inherited this is cheaper than it was:** one parameter per margin carries the whole claim. Two collapses must be told apart. An empty relation is standard attention entrywise (Prop. 2(vi)) and is the degenerate optimum easiest to reach;  $\tau_j \rightarrow 0$  on a non-empty relation is *not* — it spreads the inherited quota uniformly over the relation (Lem. 1: at  $\tau_j \rightarrow 0$  every prefix has  $G < 1$ , so the support is the whole relation and the entries tend to  $\bar{c}_j/|\mathcal{E}_{\cdot j}|$ ), a column of the wrong shape, and shows up as a fan-out degree of  $|\mathcal{E}_{\cdot j}|$  with precision  $m_j/|\mathcal{E}_{\cdot j}|$  on a column the mechanism did touch. All three are directly measurable as across-key and across-query variances and are reported. A subtler variant is a head that ignores its field argument, behaving as  $\tau_\phi^K(k_j, \cdot)$ ; it is invisible to those variances, which is why E9 ablates the field argument directly rather than inferring it.

## E SELECTIVE EXCLUSIVITY: SCOPING BOTH PROGRAMS TO A RELATION

Both programs of Sec. 2 are defined on an *index set*, and nothing in either requires that set to be a whole column or a whole row. This appendix states the generalization that follows, says which results survive it and which do not, and gives our reason for thinking it is the direction this mechanism goes. **Since the fifth revision it is the mechanism:** Sec. 2 is stated on the relation, and this appendix keeps the derivation and the costs, and governs how Assumption 2 should be read as the candidate set grows.

**The generalization.** A key is rarely exclusive over everything. A left-turn arrow governs the left-turn lanes in its direction and nothing else: it is not exclusive over the through lanes, and not exclusive over how many vehicles are waiting in them. The same holds on the query margin — a lane is exclusive over the signals that govern it, not over every signal at the intersection. Let  $\mathcal{E} \subseteq [n_q] \times [n_k]$  be an **exclusive subgraph**, with slices written in the paper’s own convention,

$$\mathcal{E}_{\cdot j} := \{i : (i, j) \in \mathcal{E}\}, \quad \mathcal{E}_i := \{j : (i, j) \in \mathcal{E}\}, \quad (11)$$

and run Stage 1 equation 3 on  $\mathcal{E}_{\cdot j}$  and Stage 2 equation 7 on  $\mathcal{E}_i$ , pairs outside  $\mathcal{E}$  keeping ordinary attention. We call this **selective exclusivity**. Two structural remarks. First, conditional activation (Sec. 2.3) is the special case  $\mathcal{E}_{\cdot j} \in \{\emptyset, [n_q]\}$ , and the query gate the case  $\mathcal{E}_i \in \{\emptyset, [n_k]\}$ : both gates become degenerate masks rather than separate components, and the five heads of App. A become three, at the cost of whatever parameterizes the mask. Second, both margins must be slices of *one* object. Independent per-margin masks would let Stage 2 act on entries of  $\tilde{A}_i$ , that Stage 1 never

constrained, which puts Prop. 9 and the consistency hypothesis of Cor. 2 back in question; we do not consider them.

**What transfers.** Table 4 is the accounting. The pattern is that results which are statements about a sparsemax and a truncation *over an index set* carry over with the index set replaced, while results that count the set do not.

Table 4: Restriction of both programs to an exclusive subgraph, statement by statement. Rows in the last group are the ones a development of this idea would owe a proof.

| Result                                                                                                                  | Under restriction                                                        | Because                                     |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------|
| <i>Verbatim, with <math>[n_q] \rightarrow \mathcal{E}_{.j}</math> and <math>[n_k] \rightarrow \mathcal{E}_i</math>.</i> |                                                                          |                                             |
| Lem. 1                                                                                                                  | unchanged                                                                | a property of sparsemax on a set            |
| Monotone exclusivity                                                                                                    | $\tau \rightarrow 0^+$ endpoint reads $ \mathcal{E}_{.j} $               | it names the set size                       |
| Thm. 1, separation                                                                                                      | unchanged                                                                | the product is over the index set           |
| Thm. 2 and its endpoints                                                                                                | unchanged                                                                | $W_j, m_j, \delta_j$ carry no set size      |
| Prop. 7, 8, 11                                                                                                          | unchanged                                                                | same                                        |
| Prop. 9, composition                                                                                                    | unchanged                                                                | one mask $\Rightarrow$ both stages agree    |
| Cor. 3, necessity                                                                                                       | unchanged                                                                | a comparison of intervals                   |
| Prop. 12, causal form                                                                                                   | unchanged if $\mathcal{E}$ is arrival-sealed                             | growth must add, never revoke               |
| <i>Hypothesis changes</i>                                                                                               |                                                                          |                                             |
| Assumption 2                                                                                                            | max taken over $\mathcal{E}_{.j} \setminus T_j$                          | the only place $n_q$ entered                |
| Cor. 1; Prop. 5(ii)                                                                                                     | need $ \mathcal{E}_{.j}  \geq 2, M \leq  \mathcal{E}_{.j} $              | product form at a singleton                 |
| Prop. 6, ceiling                                                                                                        | $m_j/ \mathcal{E}_{.j} $ for a <i>masked</i> comparator                  | support is $\mathcal{E}_{.j}$ , still fixed |
| Prop. 10, dilution                                                                                                      | $n_q \rightarrow  \mathcal{E}_{.j} $ at <i>matched domain</i>            | it describes a comparator                   |
| Cor. 2, joint attainment                                                                                                | consistency induced by $\mathcal{E}$ if $T_j \subseteq \mathcal{E}_{.j}$ | one object, not two ground truths           |
| <i>Genuinely different, and owed a proof</i>                                                                            |                                                                          |                                             |
| Prop. 1, conservation                                                                                                   | $n_q/n_k \rightarrow \mu/n_k, \mu$ the in-relation mass                  | off- $\mathcal{E}$ mass escapes it          |
| The unit row cap                                                                                                        | bounds in-relation mass only                                             | the cap sees only $\mathcal{E}_i$ .         |
| Budget normalization $\sum_j c_j = n_q$                                                                                 | <b>derived, not imposed</b>                                              | step 1 matches standard attention           |
| Prop. 4, realizability                                                                                                  | partition of the covered set only                                        | construction must be re-run                 |

**Why:  $n_q$  enters the theory in exactly one place.** Of the three quantities fixing Thm. 2’s interval,  $W_j$  and  $m_j$  are functions of the target set alone; only  $\delta_j$  looks outside it, and it looks by taking a *max over the complement*. A max over a growing pool is non-decreasing in the pool size, so  $\delta_j$  is *non-increasing in  $n_q$  whatever the score law*, and it converges to  $\min_{i \in T_j} s_{ij} - \text{ess sup}(\text{non-target law})$  — which is negative whenever a non-target can outrank the weakest target with positive probability. Past that point Assumption 2 fails outright, not because the scoring function is poor but because the assumption is a statement about a set that keeps growing.

**We state it this way rather than as a divergence, and the reason is our own model of the representation.** If the non-target law has unbounded support above —  $\mathcal{N}(\mu_0, \sigma^2)$ , say — then  $\delta_j \rightarrow -\infty$  at rate  $\sqrt{\ln n_q}$ , since the maximum is  $\mu_0 + \sigma \sqrt{2 \ln(n_q - m_j)}(1 + o(1))$ . But under pre-LN the vector each projection reads lies on a bounded manifold (App. D), so  $q_i \cdot k_j / \sqrt{d}$  is bounded and the scores are not Gaussian in the tail: the max converges *up to* an essential supremum instead of diverging. The monotone-convergence form above is what survives that, it is all the argument needs, and it is not removed by training, which moves the law but not the monotonicity. Past that point every result in Sec. 3.5 is vacuous rather than merely weak. Restricted to  $\mathcal{E}_{.j}$ , the max is over a set whose size does not grow with the context, and  $\delta_j$  becomes  $n_q$ -free by construction. Table 5 measures both, with a scoring function that already separates non-targets by two standard deviations.

**What this settles about the comparison with softmax, and what it does not.** Prop. 10 sets a  $\log n_q$  requirement against one with “no  $n_q$  in it at all”. On the unrestricted mechanism that contrast is stated more strongly than it should be:  $n_q$  survives inside  $\delta_j$  on *both* sides, so the honest residual is the logarithmic factor and not the presence or absence of  $n$ . Under restriction there are two gains and they deserve separate credit. (i) The  $\log |\mathcal{E}_{.j}|$  factor belongs to the *program* — a terminated tail rather than an exponential one — and is unaffected by the mask. (ii) The reduction of the domain from  $n_q$  to  $|\mathcal{E}_{.j}|$  belongs to the *mask*, and a masked softmax would enjoy it equally. What restriction contributes beyond either is that Assumption 2 stops being asymptotically vacuous, so (i) remains a claim about something. We regard this as the strongest argument for the generalization and the correct reading of Prop. 10 without it.

Table 5: Margin and exact recovery against candidate count, at  $|\mathcal{E}_j| = 8$ ,  $m_j = 3$ , non-targets outside the scope drawn  $\mathcal{N}(-2, 1)$ ; 3000 columns per row,  $\tau_j$  taken inside the scope’s interval. The scope margin is identical across rows by construction; the column margin is not. **Both  $\delta_j$  columns are means, and the recovery columns are not a function of them**, which is the honest way to read the table: recovery is per column, and Assumption 2 fails on a growing *fraction* of columns rather than switching off at a threshold. Already at  $n_q = 32$ , where the mean margin is still positive, 26.5% of columns have  $\delta_j \leq 0$  and can be recovered at no temperature; at  $n_q = 128$ , where the mean has gone negative, 24.5% still satisfy the assumption, which is what the residual 14.9% is made of. The scope column is 100% at every  $n_q$  because the max defining  $\delta_j$  is taken over a set that does not grow.

| $n_q$ | $\delta_j$ , full column | $\delta_j$ , on $\mathcal{E}_j$ | recovery, full column | recovery, on $\mathcal{E}_j$ |
|-------|--------------------------|---------------------------------|-----------------------|------------------------------|
| 8     | 1.127                    | 1.127                           | 100.0%                | 100.0%                       |
| 32    | 0.296                    | 1.127                           | 63.6%                 | 100.0%                       |
| 128   | -0.321                   | 1.127                           | 14.9%                 | 100.0%                       |
| 512   | -0.791                   | 1.127                           | 1.2%                  | 100.0%                       |
| 2048  | -1.184                   | 1.127                           | 0.0%                  | 100.0%                       |
| 8192  | -1.555                   | 1.127                           | 0.0%                  | 100.0%                       |

**What it costs, and what it no longer costs.** An earlier version of this appendix listed four costs of restriction. **The two-step form of Sec. 2 retires two of them and improves on a third**, because step 1 fixes the complement rather than leaving it free. We set out the accounting rather than quietly dropping it.

**(a) The separation theorem’s force is parameterized by  $|\mathcal{E}_j|$ , and vanishes at 1 — this one survives.** Cardinality on a singleton is product-form, so a query-local scheme enforces it by clipping and Thm. 1 says nothing; the replication pair of Assumption 1 must also be found inside the set, giving  $M \leq |\mathcal{E}_j|$ . A learned relation can therefore shrink its way out of the theorem. Worse, the degeneracy is invisible to the *collapse variances* of App. D: at  $|\mathcal{E}_j| = 1$  the mechanism is still masking and  $\tau_j$  still varies across keys while every constraint has quietly become product-form. It is not undetectable — every column then has  $k^* = 1$  and  $\nu_j = 1$ , and  $T_j \subseteq \mathcal{E}_j$  forces  $m_j \leq 1$ , so E8’s realized- $k^*$  report shows it — and any deployment must report the  $|\mathcal{E}_j|$  *distribution*, not its mean, from the first epoch, and carry  $|\mathcal{E}_j| \geq 2$  wherever Thm. 1 is invoked. It also adds an experimental axis: the E2 gap should shrink as  $|\mathcal{E}_j| \rightarrow 1$ , a pre-committed falsification of the same shape as E4.

**(b) The conservation obstruction was said to weaken — retired.** The worry was that a scheme could evade Prop. 1 by parking mass off  $\mathcal{E}$ . Under equation 2 the off-relation entries are whatever an unmodified layer allocates: they are fixed by step 1 and untouched by step 2 (Prop. 2(iv)), so they are not a free parameter and there is nothing to park. Prop. 1 is in any case a statement about row-normalized *baselines*, and restriction changes nothing any baseline does.

**(c) The unit row cap was said to stop covering the row — retired.** equation 6 caps the *whole* row at the unit it already carries, before any restriction is applied, and step 2 conserves the relation’s quota within it, so the row total never exceeds its own softmax mass and the complement keeps its step-1 value. The earlier version left the complement free and the row total could approach two; that is no longer the construction.

**(d) The capacity anchor was said to have no mask-independent analogue — improved on.**  $\sum_j c_j = n_q$  is no longer imposed at all:  $c_j$  is the mass an unmodified layer gives the key, so the identity is a consequence (Prop. 2(ii)) and holds at every relation. What this costs instead is the *capacity half* of the modelling object — the mechanism no longer predicts how much a key emits into its relation, only how many members receive it. That is stated in Sec. 2.1 and in Cor. 1, and it is why E1 measures degree.

**(e) A new cost, and the one we would press on if we were reviewing.** With the capacity inherited, each margin has a single parameter carrying its whole claim, where a predicted budget would have been a second and independent constraint. Collapse is correspondingly cheaper: where previously  $c_j$ ,  $\tau_j$  and  $\tau_i$  had to degenerate together, now either the relation emptying (standard attention, Prop. 2(vi)) or  $\tau_j \rightarrow 0$  on a non-empty relation (the inherited quota spread uniformly over it — *not* standard attention, and a column re-shaped to the wrong shape) removes the fan-out claim on its own. E8 reports the  $|\mathcal{E}_j|$  distribution, the across-key variance of  $\tau_j$  and the realized  $k^*$  distribution

from epoch one for this reason, and the implementation initialises the relation near-empty so that training starts at standard attention rather than at a uniformly re-shaped one.

**How the mask must be parameterized, and why that is not a retreat.** A free per-edge binary mask is  $n_q n_k$  discrete decisions, is unidentifiable, and is the strong form of exactly what App. B declines for a learned per-block  $K$ . The right object is a typed or low-rank compatibility that *induces* the mask,  $e_{ij} = \mathbf{1}[\langle u_\phi(k_j), v_\phi(q_i) \rangle > 0]$  at rank  $r$ , costing  $O((n_q + n_k)r)$  and reading the traffic example literally: a class of key against a class of query. This has a consequence for Sec. 2.5. Because  $e_{ij}$  depends on the pair alone, membership decided when a query arrives is never revoked,  $\mathcal{E}_j(t)$  is nested increasing, and Prop. 12 goes through unchanged; a mask that could reconsider does not, and in simulation breaks permanence of exclusion on 90.7% of prefixes. A pairwise mask is thus *required* by the causal form rather than merely tolerated.

That deserves saying plainly, because it looks at first like a concession: the paper’s argument is that a pairwise decision cannot govern a multivariate relation, and here membership is decided pairwise. The two are compatible, and the distinction is the point. *Whether this key and this query stand in the relation* is a question about the pair. *Which of the competing keys wins this query, given the others* is not, and that is what Thm. 1 is about. Pairwise decides who is in the room; the joint program decides who wins; the mask restricts a *domain* while the program does the *constraining*. This also clarifies rather than weakens the case for field-reading heads: the field argument was always needed by  $\tau_\phi^K$  and  $\tau_\phi^Q$ , whose job is to hit the interval of Thm. 2, and never by the membership decision.

**Where it helps least.** In the evidence-selection setting of Sec. 4,  $\mathcal{E}_j$  for an answer query is plausibly the entire retrieved candidate set, since the passages are candidates by construction, and a learned relation that admits them all changes nothing in E2 or E3 relative to the unrestricted form. Its natural settings are those with typed structure — a pronoun is exclusive over *mentions* and not over every token; a variable binding over the *references* to that variable. Fidelity measurement on the column needs  $T_j$  with several members, which typed structure supplies directly and multi-hop evidence supplies through the bridge entity’s later mentions (App. H, E7).

**Numerical check of the transfer.** The claims of the first group are checked directly, by running both programs on a scope rather than a margin, in the same harness that produces Table 7; Table 6 gives the counts. The margin block is Table 5. Two of these are checks that something *fails*: the  $n_q/n_k$  bound of Prop. 1 is escaped on every trial once mass may sit off  $\mathcal{E}$ , and permanence of exclusion breaks on 90.7% of prefixes if the mask is allowed to reconsider. The first is weaker than it looks and we mark it so: the generator draws the mask density from  $[0.02, 0.5]$ , so the covered mass cannot reach  $n_q/n_k$  and the escape is arranged rather than discovered — what the row shows is that the bound is not automatic under restriction, not how far it falls. As everywhere in App. G, these establish that the statements are true of the objects the program constructs; they are not evidence about trained networks, and no mask here is learned.

## F PROOFS

Every numbered claim in the paper is proved here, in the order it is used in Sec. 3. Results stated in the main text are **restated** before their proofs so this appendix can be read on its own; results the main text gives in prose are stated formally here for the first time.

Throughout,  $\Delta^{n-1} = \{p \in \mathbb{R}_{\geq 0}^n : \sum_i p_i = 1\}$ , and for a vector  $z$  we write  $z_{(1)} \geq z_{(2)} \geq \dots$  for its entries sorted descending. At  $\alpha = 2$  the solution of equation 3 is  $p_{ij} = [\tau_j s_{ij} - \psi_j]_+$  with  $\psi_j$  the unique threshold making the entries sum to one; existence, uniqueness, the sort-based computation and almost-everywhere differentiability are standard for sparsemax and  $\alpha$ -entmax (Martins & Astudillo, 2016; Peters et al., 2019), and we use them without reproof.

**Lemma 1** (Prefix property). *For  $z \in \mathbb{R}^n$  sorted descending let  $G(k) := \sum_{r \leq k} (z_{(r)} - z_{(k)})$ . Then  $G(k+1) - G(k) = k(z_{(k)} - z_{(k+1)}) \geq 0$ , so  $G$  is non-decreasing; consequently  $\text{supp}(\text{sparsemax}(z))$  is the set of the  $k^*$  largest entries with  $k^* = \max\{k : G(k) < 1\}$ , and  $\{k : G(k) < 1\}$  is a prefix.*

Table 6: Restriction to a scope, checked. Zero violations throughout, with  $\dagger$  as in Table 7. The fourth row is run once rather than at each  $n_q$ : the scope recovery check does not depend on  $n_q$  — the pool outside the scope never enters it — so repeating it six times would report one result as six. **That  $n_q$ -invariance is the appendix’s point and it is structural, not empirical**; what varies with  $n_q$  is the *full-column* comparison, which is Table 5. The last row confirms a *negative* — that the  $n_q/n_k$  bound is escaped once mass may sit off  $\mathcal{E}$ ; that it is escaped at all is arranged by the mask density the generator draws, so the row shows the bound is not automatic, not how far it falls. The harness separately prints the rate at which a revocable scope breaks permanence of exclusion, which the fifth row’s hypothesis excludes.

| Claim on a scope                          | Test                                                    | Violations    |
|-------------------------------------------|---------------------------------------------------------|---------------|
| Thm. 2, interval on $\mathcal{E}_j$       | $\text{supp} = T_j$ inside, $\delta_j$ over the scope   | 0 / 24000     |
| Thm. 2, lower endpoint                    | below it: $\text{supp} \supseteq T_j$                   | 0 / 12000     |
| Thm. 2, upper endpoint                    | at or above it: $\text{supp} \subseteq T_j$             | 0 / 10604     |
| Assumption 2 on $\mathcal{E}_j$           | exact recovery, $n_q$ -free by construction             | 0 / 3000      |
| Prop. 12, arrival-sealed scope            | threshold monotone, exclusion permanent                 | 0 / 131180    |
| Prop. 6 under a mask $^\dagger$           | masked softmax $P_j$ vs. $m_j/ \mathcal{E}_j $          | 0 / 4000      |
| Prop. 9(i),(ii),(iv), one mask $^\dagger$ | Stage 2 on $\mathcal{E}_i$ , Stage 1 on $\mathcal{E}_j$ | 0 / 12953 ea. |
| Prop. 1, corrected bound $^\dagger$       | $\max_j \sum_{i \in \mathcal{E}_j} A_{ij} \geq \mu/n_k$ | 0 / 3000      |
| Prop. 1, $n_q/n_k$ lost                   | the original bound is escaped off $\mathcal{E}$         | 0 / 3000      |

*Proof of Lem. 1.* Write  $G(k) = \sum_{r \leq k} z(r) - kz(k)$ . Then

$$\begin{aligned} G(k+1) - G(k) &= \left( \sum_{r \leq k+1} z(r) - (k+1)z(k+1) \right) - \left( \sum_{r \leq k} z(r) - kz(k) \right) \\ &= z(k+1) - (k+1)z(k+1) + kz(k) = k(z(k) - z(k+1)) \geq 0, \end{aligned}$$

since the entries are sorted descending. Hence  $G$  is non-decreasing, and  $\{k : G(k) < 1\}$  is a prefix of  $\{1, \dots, n\}$ .

For sparsemax,  $p = [z - \psi]_+$  with  $\sum_i [z_i - \psi]_+ = 1$ . If the support consists of the  $k$  largest entries then  $\psi = (\sum_{r \leq k} z(r) - 1)/k$ , and  $z(k) > \psi$  is equivalent to  $kz(k) > \sum_{r \leq k} z(r) - 1$ , i.e.  $G(k) < 1$ . The support size is therefore the largest such  $k$ , and by the prefix property this is well defined.  $\square$

**Monotone exclusivity.** Apply Lem. 1 to  $z = \tau s$ . Since  $G$  is positively homogeneous of degree one,  $G_{\tau s}(k) = \tau G_s(k)$  and  $k^*(\tau) = \max\{k : \tau G_s(k) < 1\}$ . As  $G_s(k) \geq 0$  is non-decreasing in  $k$ , the set  $\{k : \tau G_s(k) < 1\}$  is nested decreasing in  $\tau$ , so  $k^*(\tau)$  is non-increasing.  $G_s(1) = 0$  always, so  $k^* \geq 1$ ; as  $\tau \rightarrow 0^+$  every  $k$  qualifies and  $k^* = n_q$ ; as  $\tau \rightarrow \infty$  only  $k$  with  $G_s(k) = 0$  survive, which holds iff the top  $k$  scores are tied, giving  $k^* \rightarrow |\arg \max_i s_i|$ .

**Theorem 1 (restated).** Fix a key  $j$ , evaluate a query-local scheme over  $\mathcal{D} = \prod_i \mathcal{D}_i$  with  $\{k_j\}$  fixed, and put  $\mathcal{R}_i := \{A_{ij}(x_i) : x_i \in \mathcal{D}_i\}$ . Then the attainable set of fan-out columns is exactly the product,  $\{A_{\cdot j}(x) : x \in \mathcal{D}\} = \prod_{i=1}^{n_q} \mathcal{R}_i$ , and the scheme satisfies  $A_{\cdot j} \in \mathcal{C}$  for all  $x \in \mathcal{D}$  iff  $\prod_i \mathcal{R}_i \subseteq \mathcal{C}$ . Hence no query-local scheme enforces a non-product-form  $\mathcal{C}$  except vacuously.

*Proof of Thm. 1.* ( $\subseteq$ ) Fix  $x = (x_1, \dots, x_{n_q}) \in \mathcal{D}$ . By Def. 1,  $A_{ij}(x) = A_{ij}(x_i)$  for each  $i$ , so the column  $A_{\cdot j}(x) = (A_{ij}(x_i))_i$  has  $i$ -th entry in  $\mathcal{R}_i$ ; hence  $A_{\cdot j}(x) \in \prod_i \mathcal{R}_i$ .

( $\supseteq$ ) Let  $(a_1, \dots, a_{n_q}) \in \prod_i \mathcal{R}_i$ . For each  $i$  pick  $x_i \in \mathcal{D}_i$  with  $A_{ij}(x_i) = a_i$ , which exists by definition of  $\mathcal{R}_i$ . Because  $\mathcal{D}$  is the product  $\prod_i \mathcal{D}_i$ , the tuple  $x = (x_1, \dots, x_{n_q})$  is admissible, and by query-locality  $A_{\cdot j}(x) = (a_1, \dots, a_{n_q})$ .

This proves equation 8. The stated equivalence is immediate: a universally quantified guarantee over  $\mathcal{D}$  holds iff it holds on the attained set, which is exactly  $\prod_i \mathcal{R}_i$ . If  $\mathcal{C}$  is non-product-form and  $\prod_i \mathcal{R}_i \subseteq \mathcal{C}$ , then  $\prod_i \mathcal{R}_i$  is a product subset of  $\mathcal{C}$ ; the guarantee is then achieved not by coupling the entries but by restricting each entry’s range independently, which is what we mean by vacuous.  $\square$

**Corollary 1 (restated).** Capacity  $\mathcal{C}_{\text{cap}} = \{x : \sum_i x_i \leq c\}$  and cardinality  $\mathcal{C}_{\text{card}} = \{x : |\text{supp}(x)| \leq m\}$  are non-product-form for  $n_q \geq 2$ , so by Thm. 1 no query-local scheme enforces

either, for  $c > 0$  and  $1 \leq m < n_q$ . The failure is realized on constrained input spaces: given a replication pair  $(a, M)$ , some admissible input has  $\sum_i A_{ij} \geq Ma$ , so capacity fails whenever  $c_j < Ma$  and cardinality whenever  $m < M$ ; with  $M = n_q$ , no constant budget survives a query set larger than  $c_j/a$ .

*Proof of Cor. 1.* Not of product form, for  $c > 0$  and  $1 \leq m < n_q$ ; at  $c = 0$  or  $m \geq n_q$  both sets are products and the statement is vacuous, which is why those cases are excluded. Suppose  $\mathcal{C}_{\text{cap}} = \prod_i \mathcal{C}_i$  with  $c > 0$  and  $n_q \geq 2$ . The point  $ce_1$  (all mass on entry 1) lies in  $\mathcal{C}_{\text{cap}}$ , so  $c \in \mathcal{C}_1$ ; likewise  $c \in \mathcal{C}_2$ . Then  $c(e_1 + e_2) \in \prod_i \mathcal{C}_i$  but has sum  $2c > c$ , a contradiction. The same argument applied to  $\mathcal{C}_{\text{card}}$  with  $m < n_q$ , using two points whose supports are distinct singletons, gives a product element of support size  $2 > m$  when  $m = 1$ , and generalizes to any  $m < n_q$  by taking  $m + 1$  singletons.

*Violation.* Let  $I$  index  $M$  coordinates with  $\sup \mathcal{R}_i \geq a$ . For any  $\epsilon > 0$  choose  $x_i \in \mathcal{D}_i$  with  $A_{ij}(x_i) > a - \epsilon$  for  $i \in I$ , and arbitrary  $x_i$  otherwise; admissible because  $\mathcal{D}$  is a product. Then  $\sum_i A_{ij} > M(a - \epsilon)$ , and letting  $\epsilon \rightarrow 0$  gives the bound  $\geq Ma$ . Thus  $\sum_i A_{ij} > c$  once  $M > c/a$ . For cardinality, the same construction yields at least  $M$  strictly positive entries, so  $|\text{supp}(A_{\cdot j})| \geq M > m$ .

*Realization on constrained domains.* By Assumption 1 there is an admissible input with a set  $S \subseteq \{1, \dots, n_q\}$ ,  $|S| \geq M$ , on which  $A_{ij} \geq a$ . Since all entries are non-negative,

$$\sum_i A_{ij} \geq \sum_{i \in S} A_{ij} \geq |S|a \geq Ma.$$

Hence  $\sum_i A_{ij} > c_j$  whenever  $c_j < Ma$ . The same input has at least  $M$  strictly positive entries in column  $j$ , so  $|\text{supp}(A_{\cdot j})| \geq M > m$  whenever  $m < M$ . Taking  $M = n_q$  gives  $\sum_i A_{ij} \geq n_q a$ , which exceeds  $c_j$  once  $n_q > c_j/a$ .  $\square$

**Proposition 1 (restated).** If  $\sum_j A_{ij} = 1$  for every query, then  $\max_j \sum_i A_{ij} \geq n_q/n_k$ . No row-normalized scheme can bound every key's fan-out below  $n_q/n_k$ ; doubly stochastic attention attains this with equality for every key.

*Proof of Prop. 1.* Summing the row constraint over queries gives  $\sum_{i,j} A_{ij} = \sum_i 1 = n_q$ . A maximum is at least an average, so  $\max_j \sum_i A_{ij} \geq \frac{1}{n_k} \sum_j \sum_i A_{ij} = n_q/n_k$ . For a doubly stochastic scheme with uniform column marginal every column sums to exactly  $n_q/n_k$  by definition, attaining the bound. Since  $n_q/n_k \rightarrow \infty$  as  $n_q$  grows with  $n_k$  fixed, no constant bound on all columns is possible.

The restriction to *all* columns is essential. A row-normalized scheme may bound one column — e.g. by driving  $s_{ij} \rightarrow -\infty$  for a single  $j$  — whereupon the remaining columns absorb the mass. Cor. 1 excludes this for query-local schemes on different grounds, namely that the bound must hold uniformly over admissible inputs.  $\square$

**Proposition 4** (Every competitive selection is realizable). Let  $\{T_j\}$  partition  $\{1, \dots, n_q\}$  with  $m_j = |T_j| \geq 1$ . Then scores, exclusivities, a relation  $\mathcal{E}$  exist such that, with capacities inherited by equation 2,  $\text{supp}(A_{\cdot j}) = T_j$  at every admissible  $\tau_i$ , for every  $j$  simultaneously; the construction survives any sup-norm score perturbation below  $1/2$ . The cardinalities may differ arbitrarily subject only to what a partition forces,  $\sum_j m_j = n_q$ : scope 1 and scope  $n_q - n_k + 1$  coexist in one realization.

*Proof of Prop. 2.* (i) By equation 2,  $\tilde{A}_{ij} = A_{ij}^{\text{sm}}$  off  $\mathcal{E}_{\cdot j}$  and  $\sum_{i \in \mathcal{E}_{\cdot j}} \tilde{A}_{ij} = \bar{c}_j \sum_i p_{ij} = \bar{c}_j$  since  $p_{\cdot j}$  lies on the simplex; so  $\sum_i \tilde{A}_{ij} = \bar{c}_j + \sum_{i \notin \mathcal{E}_{\cdot j}} A_{ij}^{\text{sm}} = \sum_i A_{ij}^{\text{sm}} = c_j$ . (ii) Summing (i) over  $j$  and exchanging sums,  $\sum_j c_j = \sum_i \sum_j A_{ij}^{\text{sm}} = \sum_i 1 = n_q$ . (iii) Step 1 of Stage 2 is a projection onto  $\{a \geq 0 : \sum_j a_j \leq \sum_j A_{ij}^{\text{sm}}\}$ , which is the unit cap since the row-softmax row sums to one, so  $\sum_j a_{ij}^{(1)} \leq \sum_j A_{ij}^{\text{sm}}$ ; step 2 replaces the relation's entries by  $\tau_i u$  with  $\sum_{j \in \mathcal{E}_i} \tau_i u_j \leq \bar{c}_i = \sum_{j \in \mathcal{E}_i} a_{ij}^{(1)}$  and leaves the rest, so the row total does not increase. (iv) Off  $\mathcal{E}_{\cdot j}$  step 2 of Stage 1

does not write; off  $\mathcal{E}_i$ . step 2 of Stage 2 does not write. (v) The step-2 program is the projection of  $\tilde{A}_{i\mathcal{E}_i} \geq 0$  onto  $\{u \geq 0 : \sum u \leq \bar{c}_i/\tau_i\}$ . If the constraint is slack the projection is the identity on the non-negative orthant and  $\theta_i = 0$ ; if it binds, the KKT conditions give  $u_j = [\tilde{A}_{ij} - \theta_i]_+$  with  $\theta_i$  the multiplier of an *inequality*, hence  $\theta_i \geq 0$ . In either case  $\tilde{A}_{ij} = 0 \Rightarrow u_j = 0$ , and  $\sum_{j \in \mathcal{E}_i} A_{ij} \leq \bar{c}_i$ . (vi) With  $\mathcal{E} = \emptyset$  no step 2 runs on either margin,  $\tilde{A} = A^{\text{sm}}$ , and step 1 of Stage 2 projects a row already summing to one onto the unit cap, which fixes it.  $\square$

*Proof of Prop. 3.*  $W_j$  and  $m_j$  are functions of  $\{s_{ij} : i \in T_j\}$  alone and  $T_j$  is common to both sets. (i)  $\delta_j(\mathcal{E}) = \min_{T_j} s - \max_{\mathcal{E} \setminus T_j} s$  and  $\mathcal{E}' \setminus T_j \subseteq \mathcal{E} \setminus T_j$ , so the maximum over the smaller set is no larger. (ii) Immediate from (i) with  $W_j, m_j$  fixed. (iii) If  $\delta_j(\mathcal{E}) > 0$  then  $\delta_j(\mathcal{E}') \geq \delta_j(\mathcal{E}) > 0$ , so Assumption 2 holds on  $\mathcal{E}'$ ; the lower endpoint  $1/(W_j + m_j \delta_j)$  is decreasing in  $\delta_j$  and the upper endpoint  $1/W_j$  does not depend on the set, giving  $[1/(W_j + m_j \delta_j(\mathcal{E})), 1/W_j] \subseteq [1/(W_j + m_j \delta_j(\mathcal{E}')), 1/W_j]$ . (iv) Take  $T_j = \{1\}$ ,  $\mathcal{E}' = \{1, 2\}$ ,  $\mathcal{E} = \{1, 2, 3\}$  with  $s_1 = 1, s_2 = 0, s_3 = 2$ : then  $\delta_j(\mathcal{E}') = 1 > 0 > -1 = \delta_j(\mathcal{E})$ .  $\square$

*Proof of Prop. 4.* Take  $s_{ij} = \mathbb{1}[i \in T_j]$ . Fix  $j$ . Since the  $T_j$  are non-empty, every target in  $T_j$  scores 1 and every other query scores 0, so  $\delta_j = 1 - 0 = 1 > 0$  and, all scores within  $T_j$  being equal,  $W_j = \sum_{i \in T_j} (1 - 1) = 0$ . By Thm. 2 the recovery interval is  $[1/(W_j + m_j \delta_j), 1/W_j] = [1/m_j, \infty)$ , which is non-empty; choose any  $\tau_j$  in it. Then  $\text{supp}(p_{\cdot j}) = T_j$ , and because the program equation 3 is symmetric under permutation of equally scored coordinates and its solution is unique (Martins & Astudillo, 2016),  $p_{\cdot j}$  is uniform on  $T_j$ :  $p_{ij} = 1/m_j$  for  $i \in T_j$  and 0 otherwise.

Setting  $c_j = m_j$  gives  $\tilde{A}_{ij} = m_j \cdot (1/m_j) = 1$  for  $i \in T_j$  and 0 otherwise. Because  $\{T_j\}$  partitions  $\{1, \dots, n_q\}$  we have  $\sum_j c_j = \sum_j m_j = n_q$ , so the construction also honours the budget normalization of Sec. 2.1; the conclusion does not depend on it, since only  $c_j > 0$  is used. Each row of  $\tilde{A}$  then has exactly one non-zero entry, equal to 1, so equation 7 gives  $A_{ij} = \min(\tau_i, 1) > 0$  for the unique  $j$  with  $i \in T_j$  — the cap slack at  $\tau_i \leq 1$ , binding at  $\tau_i > 1$  with  $\theta_i = 1 - 1/\tau_i$  — and  $A_{ij} = 0$  otherwise. This is the claimed realization, at every admissible  $\tau_i$ ; note  $\theta_i$  is not a free witness, being the dual of a cap fixed at one.

For robustness, let  $s'$  satisfy  $\|s' - s\|_\infty < 1/2$ . Then for each  $j$  every target still scores above  $1/2$  and every non-target below  $1/2$ , so  $\delta'_j > 0$  and the top  $m_j$  entries of  $s'_{\cdot j}$  remain  $T_j$ . Assumption 2 therefore still holds and Thm. 2 applies with the perturbed constants, so  $\text{supp}(\tilde{A}_{\cdot j}) = T_j$  persists for every  $\tau_j$  in the corresponding interval.

Finally, on cardinalities: the partition is used twice above — to give  $\sum_j c_j = n_q$ , and to make each row of  $\tilde{A}$  carry exactly one non-zero — so the  $m_j$  are constrained by  $\sum_j m_j = n_q$  and by nothing else. Since each  $m_j \geq 1$ , one key may take  $n_q - n_k + 1$  targets while the rest take one apiece, which is the widest spread a partition permits. Dropping the partition to arbitrary non-empty  $\{T_j\}$  removes even that constraint at the cost of a longer Stage-2 step: a row with  $r > 1$  unit-valued entries gives  $\theta_i = 1 - 1/(r\tau_i)$  when the cap binds and all  $r$  survive, so the conclusion is unchanged.  $\square$

**Proposition 5** (Selection is local; scope-respecting selection is not). *Fix a query-local scheme. (i) Selection at a single target is available to it: row-sparsemax at large temperature admits only  $\arg \max_j s_{ij}$  at every target. (ii) Scope is not: for a scope bound  $\kappa_j$  (the requirement being  $|T_j| \leq \kappa_j$ ), a replication pair  $(a, M)$  with  $\kappa_j < M$  drives  $|\text{supp}(A_{\cdot j})| \geq M > \kappa_j$  on some admissible input. (iii) Hence the non-local content is neither in isolation but their conjunction, which MarSea satisfies by construction.*

*Proof of Prop. 5.* (i) Let  $A_{i\cdot} := \text{sparsemax}(\lambda s_{i\cdot})$  for  $\lambda > 0$ . This is query-local: the row depends on  $q_i, \gamma_i$  and  $\{k_j\}_j$  only, and on no  $x_{i'}$  with  $i' \neq i$ . By monotone exclusivity applied along the row,  $|\text{supp}(\text{sparsemax}(\lambda s_{i\cdot}))| \rightarrow |\arg \max_j s_{ij}|$  as  $\lambda \rightarrow \infty$ , which is 1 for scores in general position; and by Lem. 1 the surviving index is the largest, namely  $\arg \max_j s_{ij}$ . Choosing  $\lambda$  beyond the finite threshold at which the support size reaches its limit gives the claim.

(ii) By Assumption 1 there is an admissible input on which at least  $M$  queries satisfy  $A_{ij} \geq a > 0$ ; each such query contributes a strictly positive entry to column  $j$ , so  $|\text{supp}(A_{\cdot j})| \geq M$ . If  $\kappa_j < M$

this violates  $|T_j| \leq \kappa_j$ . The argument is the cardinality half of Cor. 1 and holds for every query-local scheme admitting a replication pair.

(iii) Immediate from (i) and (ii): the class attains target-side exclusivity, fails scope, and therefore fails their conjunction. MarSea attains the conjunction. Realizability is Prop. 4. For scope, the relevant quantity is a *cardinality*, and Lem. 1 supplies it directly:  $|\text{supp}(\tilde{A}_{\cdot j})| = k^*(\tau_j)$ , which by monotone exclusivity is non-increasing in  $\tau_j$  and equals  $m_j$  under Assumption 2 with  $\tau_j$  in equation 10. Since  $\tau_i > 0$  and  $\theta_i \geq 0$  give  $A_{ij} > 0 \Rightarrow \tilde{A}_{ij} > \theta_i$ , we have  $\text{supp}(A_{\cdot j}) \subseteq \text{supp}(\tilde{A}_{\cdot j})$  and hence  $|\text{supp}(A_{\cdot j})| \leq k^*(\tau_j)$ , with no appeal to  $c_j$  and no bound on  $\tau_i$  required. (The *mass* statement  $\sum_i \tilde{A}_{ij} = c_j \sum_i p_{ij} = c_j$  also holds by equation 3, giving  $|\{i : \tilde{A}_{ij} \geq t\}| \leq c_j/t$  by Markov; it is stated on the Stage-1 output because  $A_{ij} = \tau_i[\tilde{A}_{ij} - \theta_i]_+$  need not satisfy  $A_{ij} \leq \tilde{A}_{ij}$  when  $\tau_i > 1$ . That statement is not what (iii) needs.)  $\square$

**Theorem 2 (restated).** *Let  $\alpha = 2$  and Assumption 2 hold. Then  $\text{supp}(\tilde{A}_{\cdot j}) = T_j$  if and only if  $\tau_j \in [1/(W_j + m_j\delta_j), 1/W_j)$ , with  $1/W_j := \infty$  when  $W_j = 0$ ; this interval is non-empty for every column with a positive margin.*

*Proof of Thm. 2.* Let  $z := \tau_j s_{\cdot j}$  with  $\tau_j > 0$ . By Assumption 2,  $\min_{i \in T_j} s_{ij} > \max_{i \notin T_j} s_{ij}$ , so the  $m_j$  largest entries of  $z$  are exactly the indices in  $T_j$ . Write  $G$  for the quantity in Lem. 1 applied to  $z$ , and recall from that lemma that the support of  $\text{sparsemax}(z)$  is the top  $k^*$  entries with  $k^* = \max\{k : G(k) < 1\}$ . It therefore suffices to show  $k^* = m_j$ .

*Step 1:  $k^* \geq m_j$ , i.e.  $G(m_j) < 1$ .* By definition,

$$G(m_j) = \sum_{r \leq m_j} (z_{(r)} - z_{(m_j)}) = \tau_j \sum_{i \in T_j} \left( s_{ij} - \min_{i' \in T_j} s_{i'j} \right) = \tau_j W_j,$$

using that the top  $m_j$  entries are  $T_j$  and that  $z_{(m_j)}$  corresponds to the smallest score within  $T_j$ . Hence  $G(m_j) < 1$  iff  $\tau_j W_j < 1$  iff  $\tau_j < 1/W_j$ : an *equivalence*, not merely a sufficient condition.

*Step 2:  $k^* \leq m_j$ , i.e.  $G(m_j+1) \geq 1$ .* The top  $m_j$  entries are  $T_j$  and  $z_{(m_j+1)}$  corresponds to the largest score outside  $T_j$ , which by Assumption 2 is  $\min_{i \in T_j} s_{ij} - \delta_j$ . So  $G(m_j+1)$  is not merely bounded below but *available in closed form*: writing  $s_{\min} := \min_{i \in T_j} s_{ij}$ ,

$$G(m_j+1) = \sum_{r \leq m_j} (z_{(r)} - z_{(m_j+1)}) = \tau_j \left( \sum_{i \in T_j} (s_{ij} - s_{\min}) + m_j \delta_j \right) = \tau_j (W_j + m_j \delta_j),$$

the  $r = m_j+1$  term vanishing. Hence  $G(m_j+1) \geq 1$  iff  $\tau_j \geq 1/(W_j + m_j \delta_j)$ , again an equivalence.

By Lem. 1 the feasible set  $\{k : G(k) < 1\}$  is a prefix, so  $G(m_j+1) \geq 1$  implies  $G(k) \geq 1$  for all  $k \geq m_j+1$  and hence  $k^* \leq m_j$ . Combining the two steps,  $k^* = m_j$  — and the support is  $T_j$  — exactly when  $\tau_j$  lies in equation 10. Finally that interval is non-empty iff  $W_j < W_j + m_j \delta_j$ , i.e. iff  $m_j \delta_j > 0$ , which Assumption 2 and  $m_j \geq 1$  give (interpreting  $1/0 = \infty$ ).

*Remark.* Bounding  $G(m_j+1) \geq m_j \tau_j \delta_j$  instead yields the sufficient interval  $[1/(m_j \delta_j), 1/W_j)$ , which is non-empty only when  $m_j \delta_j > W_j$ ; that condition is an artifact of the bound and not a property of the problem.  $\square$

**Corollary 3** (Content-dependent exclusivity is necessary). *If two keys satisfy  $W_{j_2} + m_{j_2} \delta_{j_2} \leq W_{j_1}$  their recovery intervals are disjoint, so no single global  $\tau$  achieves exact recovery for both; a content-predicted  $\tau_j$  does.*

*Proof of Cor. 3.* Let  $I_j := [1/(W_j + m_j \delta_j), 1/W_j)$ , which by Thm. 2 is *exactly* the set of temperatures recovering key  $j$ . The hypothesis  $W_{j_2} + m_{j_2} \delta_{j_2} \leq W_{j_1}$  gives  $\sup I_{j_1} = 1/W_{j_1} \leq 1/(W_{j_2} + m_{j_2} \delta_{j_2}) = \inf I_{j_2}$ , and since  $I_{j_1}$  is open on the right,  $I_{j_1} \cap I_{j_2} = \emptyset$ . A single global temperature lies in at most one of the two sets, so — because Thm. 2 is an equivalence and not merely a sufficient condition — exact recovery genuinely fails for at least one of the two keys. A per-key  $\tau_j$  can be chosen in each interval separately.  $\square$

**Proposition 6** (The precision ceiling of full support). *Let a mechanism assign  $p_{ij} > 0$  to every pair — row-softmax, any additive bias on it, any elementwise gate with positive factors. Then for every key, every score configuration and every temperature,  $P_j = m_j/n_q$  on every column and  $P_i = |K_i|/n_k$  on every row. Both are independent of the scoring function, so neither improves by training, by architecture within the class, or by any temperature; both tend to 0 as the candidate set grows. Recall is 1 trivially.*

*Proof of Prop. 6.* If  $p_{ij} > 0$  for every  $i$  then  $S_j = \text{supp}(p_{\cdot j}) = \{1, \dots, n_q\}$ , so  $|S_j| = n_q$  and  $|S_j \cap T_j| = |T_j| = m_j$ , giving  $P_j = m_j/n_q$  by Def. 2. Neither quantity refers to the scores, so the value is unchanged by any reparameterization of the scoring function or any positive temperature; row-softmax has  $p_{ij} = e^{s_{ij}} / \sum_{i'} e^{s_{i'j}} > 0$  for all finite scores, and an elementwise gate with positive factors preserves positivity. The row statement is identical with the margins exchanged. Finally  $m_j/n_q \rightarrow 0$  as  $n_q \rightarrow \infty$  at fixed  $m_j$ .  $\square$

**Proposition 7** (The recovery interval is a precision–recall interval). *The endpoints of equation 10 govern the two error types separately, and each is an equivalence:  $\tau_j \geq 1/(W_j + m_j\delta_j)$  is exactly the **precision** requirement ( $P_j = 1$  iff it holds, and below it  $P_j < 1$  with  $R_j = 1$ ) and  $\tau_j < 1/W_j$  exactly the **recall** requirement ( $R_j = 1$  iff it holds, and above it  $R_j < 1$  with  $P_j = 1$ ), with  $P_j = R_j = \rho_j = 1$  inside. Since the interval is always non-empty, both are attainable at a single temperature on every column with a positive margin; what a hard column costs is not attainability but the width  $m_j\delta_j/W_j$  of the target the head must hit.*

*Proof of Prop. 7.* Write  $G$  for the prefix statistic of Lem. 1 applied to  $z = \tau_j s_{\cdot j}$  and recall  $k^* = \max\{k : G(k) < 1\}$ , with  $\text{supp} =$  the top  $k^*$  entries.

(i) Step 2 of the proof of Thm. 2 computes  $G(m_j+1) = \tau_j(W_j + m_j\delta_j)$  exactly, so  $k^* \leq m_j$  iff  $\tau_j \geq 1/(W_j + m_j\delta_j)$ . If instead  $\tau_j < 1/(W_j + m_j\delta_j)$  then  $G(m_j+1) < 1$ , so  $k^* > m_j$  and the support strictly contains  $T_j$ ; since by Assumption 2 the entries ranked beyond  $m_j$  lie outside  $T_j$ , we get  $|S_j| > m_j = |S_j \cap T_j|$  and hence  $P_j < 1$ , while  $R_j = 1$ . No extremal configuration is needed: the equivalence holds for every column.

(ii) Step 1 establishes  $k^* \geq m_j$  from  $G(m_j) = \tau_j W_j < 1$ , i.e.  $\tau_j < 1/W_j$ . If  $\tau_j \geq 1/W_j$  then  $G(m_j) \geq 1$ , so  $k^* < m_j$  and the support is a proper subset of the top  $m_j$  entries, which are exactly  $T_j$ ; hence  $|S_j \cap T_j| = |S_j| < m_j$ , giving  $R_j < 1$  with  $P_j = 1$ .

Combining, both equal one simultaneously iff  $\tau_j$  lies in  $[1/(W_j + m_j\delta_j), 1/W_j]$ , which is non-empty whenever  $\delta_j > 0$ .

*Exactness inside the interval.* By Thm. 2,  $S_j = \text{supp}(p_{\cdot j}) = T_j$  for every  $\tau_j$  in equation 10. Then  $|S_j \cap T_j| = |T_j| = m_j = |S_j|$ , so  $P_j = R_j = 1$  by Def. 2. Every coordinate outside  $T_j$  is exactly zero, so  $\rho_j = \sum_{i \in T_j} p_{ij} = \sum_i p_{ij} = 1$ .  $\square$

**Proposition 8** (Graded degradation, both margins, weakest-first). *Past the upper endpoint of its interval each margin sheds true members in non-decreasing order of strength — one at a time except that tied members shed together — with precision pinned at 1 throughout. (i) For  $\tau_j > 1/W_j$ ,  $P_j = 1$  and  $R_j = k(\tau_j)/m_j$  with  $k(\tau_j) = \max\{k : W_j(k) < 1/\tau_j\}$ ,  $W_j(k) := \sum_{r \leq k} (s_{(r)} - s_{(k)})$ . (ii) If the row separates, any  $\theta_i \in [\max_{j \notin K_i} \tilde{A}_{ij}, \min_{j \in K_i} \tilde{A}_{ij}]$  gives  $P_i = R_i = 1$ , and above it  $P_i = 1$  with  $R_i = k(\theta_i)/|K_i|$ , where  $k(\theta_i) := |\{j \in K_i : \tilde{A}_{ij} > \theta_i\}|$  counts the true sources still above the floor. The two margins are driven by different heads,  $\tau_\phi^K$  directly and  $\tau_\phi^Q$  through the dual  $\theta_i$ , reading different fields and aimed at these two different intervals, and are separately diagnosable. The row interval is reachable only through  $\tau_i$ ,  $\theta_i$  being a dual and not a knob (App. D).*

*Proof of Prop. 8. (i) Column.* Let  $\tau_j > 1/W_j$ . By part (ii) of the previous proof  $k^* < m_j$ , so the support is contained in the top  $m_j$  entries, which by Assumption 2 are exactly  $T_j$ ; hence  $S_j \subseteq T_j$ , giving  $P_j = 1$  and  $R_j = |S_j|/m_j = k^*/m_j$ . For  $k \leq m_j$  the top  $k$  entries all lie in  $T_j$ , so  $G(k) = \tau_j \sum_{r \leq k} (s_{(r)} - s_{(k)}) = \tau_j W_j(k)$ , and  $k^* = \max\{k : \tau_j W_j(k) < 1\} = \max\{k : W_j(k) < 1/\tau_j\}$ , which is the claimed  $k(\tau_j)$ . Monotonicity of  $W_j(k)$  in  $k$  is Lem. 1, so the surviving set is a prefix in score order and the lowest-scoring targets are shed first.

(ii) Row. Write  $\mu := \max_{j \notin K_i} \tilde{A}_{ij}$  and  $\lambda := \min_{j \in K_i} \tilde{A}_{ij}$ , with  $\mu < \lambda$  by hypothesis. For  $\theta_i \in [\mu, \lambda)$  and  $j \in K_i$  we have  $\tilde{A}_{ij} \geq \lambda > \theta_i$ , so  $A_{ij} > 0$ ; for  $j \notin K_i$  we have  $\tilde{A}_{ij} \leq \mu \leq \theta_i$ , so  $A_{ij} = \tau_i[\tilde{A}_{ij} - \theta_i]_+ = 0$ . Hence  $\text{supp}(A_i) = K_i$  and both row precision and row recall equal 1.

For monotonicity,  $\theta \mapsto \{j : \tilde{A}_{ij} > \theta\}$  is nested decreasing, so recall is non-increasing. For precision, the surviving set is always a top set in the order induced by  $\tilde{A}_i$ ; because that order places every element of  $K_i$  above every element of its complement, each element removed as  $\theta_i$  increases is drawn from the complement until the complement is exhausted, and only then from  $K_i$ . Precision is therefore non-decreasing until it reaches 1 and constant at 1 thereafter.

For the rate, let  $\theta_i \geq \lambda$ . Every surviving coordinate satisfies  $\tilde{A}_{ij} > \theta_i \geq \lambda = \min_{j \in K_i} \tilde{A}_{ij}$ , and by separation every coordinate outside  $K_i$  is at most  $\mu \leq \lambda$ , so  $S_i \subseteq K_i$  and  $P_i = 1$ ; hence  $|S_i \cap K_i| = |S_i| = |\{j \in K_i : \tilde{A}_{ij} > \theta_i\}| = k(\theta_i)$  and  $R_i = k(\theta_i)/|K_i|$ , which is the claim. Coordinates leave in non-decreasing order of  $\tilde{A}_{ij}$  and tied coordinates leave together, since the surviving set is a superlevel set of  $\tilde{A}_i$ ; the same remark applies to  $k(\tau_j)$  in (i), where a tie at rank  $k$  makes  $W_j(k) = W_j(k+1)$  and  $k^*$  falls by more than one.

Note finally that  $\theta_i$  is not free: it is the dual of the unit cap, so as  $\tau_i$  ranges over  $(0, \infty)$  the floor traverses  $[0, \max_j \tilde{A}_{ij}]$  continuously and strictly increasingly, and the interval above is reachable iff it meets that range (App. D).  $\square$

**Proposition 9** (Composition of the two stages). *For every  $\theta_i \geq 0$ : (i)  $S_j^{(2)} \subseteq S_j^{(1)}$ , so Stage 2 can only lose recall. (ii) If  $P_j^{(1)} = 1$  then  $P_j^{(2)} = 1$  unconditionally, for any admission floors whatsoever. (iii) If  $P_j^{(1)} < 1$ , Stage 2 may raise precision — and may also lower it, since  $\theta_i$  is indexed by the row while  $P_j$  is a column quantity. (iv) A Stage-1 false negative is **unrecoverable**:  $\tilde{A}_{ij} = 0$  implies  $A_{ij} = 0$  for every  $\theta_i \geq 0$ . Statements (i), (ii) and (iv) hold verbatim on the row margin. So does (iii)’s conclusion, but not its reason: on the row  $\theta_i$  and  $P_i$  are both row quantities, and precision falls there when a floor common to the row removes a weak true source before a strong false one.*

*Proof of Prop. 9.* (i)  $A_{ij} = \tau_i[\tilde{A}_{ij} - \theta_i]_+$  with  $\tau_i > 0$  and  $\theta_i \geq 0$ , so  $A_{ij} > 0$  requires  $\tilde{A}_{ij} > \theta_i \geq 0$ ; hence  $S_j^{(2)} \subseteq S_j^{(1)}$ . Intersecting with  $T_j$  and dividing by the constant  $m_j$ ,  $|S_j^{(2)} \cap T_j| \leq |S_j^{(1)} \cap T_j|$  gives  $R_j^{(2)} \leq R_j^{(1)}$ .

(ii)  $P_j^{(1)} = 1$  means  $S_j^{(1)} \subseteq T_j$ . By (i),  $S_j^{(2)} \subseteq S_j^{(1)} \subseteq T_j$ , so  $|S_j^{(2)} \cap T_j| = |S_j^{(2)}|$  and  $P_j^{(2)} = 1$ , vacuously so on an empty support by the convention of Def. 2. Nothing in the argument constrains  $\theta_i$ , which is the content of the “unconditionally”.

(iii) For  $i \notin T_j$  with  $\tilde{A}_{ij} > 0$ , the edge survives iff  $\tilde{A}_{ij} > \theta_i$ , so it is removed exactly when  $\tilde{A}_{ij} \leq \theta_i$ . Whether  $P_j^{(2)}$  then exceeds  $P_j^{(1)}$  depends on how many true edges are removed alongside; shrinking a support raises precision only when the removals are disproportionately false, which is why the statement is a “may” and not an inequality.

(iv) If  $\tilde{A}_{ij} = 0$  then  $A_{ij} = \tau_i[0 - \theta_i]_+ = 0$  for every  $\theta_i \geq 0$ : the map  $x \mapsto \tau_i[x - \theta]_+$  is non-decreasing and fixes 0, so no admission floor can raise a zero.

The row statements are the same computation with the roles of the indices exchanged; (i)–(iv) used only that truncation is coordinatewise, monotone, and fixes 0. In particular the gain  $\tau_i > 0$  enters none of them: it rescales surviving entries and moves no support.  $\square$

**Proposition 10** (A smooth column dilutes with the candidate set). *Suppose the fan-out were computed by column-softmax and let Assumption 2 hold. Then the mass outside the target set relative to that inside obeys  $\sum_{i \notin T_j} p_{ij} / \sum_{i \in T_j} p_{ij} \leq \frac{n_q - m_j}{m_j} e^{-\tau_j \delta_j}$ , so  $\tau_j \geq \frac{1}{\delta_j} \log \frac{n_q - m_j}{m_j \varepsilon}$  suffices to hold the ratio at  $\varepsilon$ ; and the bound is attained with equality — so that temperature is also required — whenever the non-targets tie at  $\max_{i \notin T_j} s_{ij}$  and the targets at  $\min_{i \in T_j} s_{ij}$ . The requirement therefore grows like  $\log n_q$  in the worst case over configurations meeting Assumption 2, whereas exact recovery at  $\alpha = 2$  needs only  $\tau_j \geq 1/(W_j + m_j \delta_j)$ , in which  $n_q$  does not appear. App. E reads the comparison honestly:  $n_q$  survives inside  $\delta_j$  on both sides, so the residual difference is the logarithmic factor.*

*Proof of Prop. 10.* By Assumption 2,  $s_{ij} \leq \max_{i' \notin T_j} s_{i'j}$  for every  $i \notin T_j$  and  $s_{ij} \geq \min_{i' \in T_j} s_{i'j}$  for every  $i \in T_j$ , and these two quantities differ by  $\delta_j$ . The normalizing constant cancels in the ratio, so

$$\frac{\sum_{i \notin T_j} e^{\tau_j s_{ij}}}{\sum_{i \in T_j} e^{\tau_j s_{ij}}} \leq \frac{(n_q - m_j) \exp(\tau_j \max_{i \notin T_j} s_{ij})}{m_j \exp(\tau_j \min_{i \in T_j} s_{ij})} = \frac{n_q - m_j}{m_j} e^{-\tau_j \delta_j},$$

using  $\tau_j > 0$  and the definition of  $\delta_j$ , which is the stated bound. Setting the right-hand side to  $\varepsilon$  and solving for  $\tau_j$  gives a *sufficient* temperature, since the criterion was reached from an upper bound. It is also necessary in the worst case: if every non-target sits at  $\max_{i \notin T_j} s_{ij}$  and every target at  $\min_{i \in T_j} s_{ij}$ , both inequalities above hold with equality, the ratio is exactly  $\frac{n_q - m_j}{m_j} e^{-\tau_j \delta_j}$ , and no smaller  $\tau_j$  holds it at  $\varepsilon$ . This is the configuration the  $\log n_q$  claim is about; on configurations with the non-targets spread well below their maximum a smaller  $\tau_j$  suffices. The comparison with Thm. 2 is immediate either way, that bound being  $\tau_j \geq 1/(W_j + m_j \delta_j)$  with no dependence on  $n_q$ .  $\square$

**Proposition 11** (Exactness of hierarchical selection, and fidelity under retention). *Let  $k^* = |\text{supp}(\text{sparsemax}(\tau_j s_{\cdot j}))|$ . (i) If  $K \geq k^*$ , hierarchical top- $K$  retention followed by one exact program on the survivors yields  $p_{\cdot j}$  identical to the global solution, for any block size and merge arity. (ii) If Assumption 2 holds with  $\tau_j$  in equation 10 and  $K < m_j$ , the surviving support is the  $K$  highest-scoring members of  $T_j$ , whence  $P_j = 1$ ,  $R_j = K/m_j$  and  $\rho_j = 1$ : under-retention is a pure recall failure, exact in rate — and invisible to  $\rho_j$ .*

*Proof of Prop. 11.* (i) *Exactness.* By Lem. 1 the support of  $\text{sparsemax}(z)$  is the set of the  $k^*$  largest entries of  $z$ , and the solution on that set is  $p_i = z_i - \psi$  with  $\psi$  determined by those entries alone. It therefore suffices that no member of the global top- $k^*$  is discarded at any level, since the root program then sees a superset of the global support and reproduces both  $k^*$  and  $\psi$ .

Retention is by score order within each group, so a member of the global top- $k^*$  is discarded at some level only if more than  $K$  entries of that group outrank it; every such entry is then also globally ranked above it, hence itself a member of the global top- $k^*$ , giving more than  $K$  members of a set of size  $k^*$  and so  $K < k^*$ . Contrapositively,  $K \geq k^*$  implies no member is ever discarded, at any level and for any block size or merge arity. The root group receives every survivor, and solving the program there over a superset of the global support returns the global  $p_{\cdot j}$  exactly, the coordinates outside the support being zero in both.

(ii) *Under-retention.* Here  $k^* = m_j$ , since Assumption 2 with  $\tau_j$  in equation 10 gives  $\text{supp}(p_{\cdot j}) = T_j$  by Thm. 2; so  $K < m_j$  means  $K < k^*$  and part (i) does not apply. What does still apply is the *retention* argument used inside (i): retention is by score order at every level, and a globally top- $K$  entry is discarded only if more than  $K$  entries of its group outrank it, each of which is then itself globally top- $K$  — impossible. So the survivor set is exactly the  $K$  globally highest-scoring entries.

By Assumption 2 the top  $m_j$  entries are  $T_j$ , so for  $K < m_j$  the survivor set is the  $K$  highest-scoring members of  $T_j$  (ties at rank  $K$  broken arbitrarily, the conclusion being unaffected). Solving the program on that set: writing  $G$  for the statistic of Lem. 1,  $k^* = m_j$  means  $G(k) < 1$  for every  $k \leq m_j$ , hence in particular for every  $k \leq K$ ; and  $G$  restricted to the survivors agrees with the global  $G$  for  $k \leq K$ , since  $G(k)$  depends only on the top  $k$  entries, which are the same in both. So the support is all  $K$  survivors. Therefore  $S_j$  is a subset of  $T_j$  of size  $K$ , giving  $|S_j \cap T_j| = |S_j| = K$  and so  $P_j = 1$  and  $R_j = K/m_j$  by Def. 2. Finally the allocation is normalized over the survivors, all of which lie in  $T_j$ , so  $\rho_j = \sum_{i \in T_j} p_{ij} = 1$ .  $\square$

**Proposition 12** (Prefix monotonicity: growth is edge-removing). *Let  $\psi_j^{(t)}$  be the Stage-I threshold on the visible set  $Q_j(t)$ , at fixed  $\tau_j$ . Then (i)  $\psi_j^{(t)}$  is non-decreasing in  $t$ ; (ii) exclusion is permanent ( $p_{ij}^{(t)} = 0$  implies  $p_{ij}^{(t')} = 0$  for  $t' \geq t$ ) while admission is provisional; (iii) the frozen-prefix variant over-admits and never under-admits, so its error is one-sided.*

*Proof of Prop. 12.* (i) At  $\alpha = 2$  the threshold  $\psi$  is the unique solution of  $\sum_{i \in Q} [z_i - \psi]_+ = 1$ , where  $z = \tau_j s_{\cdot j}$  restricted to the visible set  $Q$ . Write  $F_Q(\psi) := \sum_{i \in Q} [z_i - \psi]_+$ . For fixed  $Q$ ,  $F_Q$  is continuous, non-increasing, and strictly decreasing wherever positive, so the solution of  $F_Q(\psi) = 1$  is unique. Enlarging the set to  $Q' = Q \cup \{i^\dagger\}$  adds a non-negative term pointwise, so

$F_{Q'}(\psi) \geq F_Q(\psi)$  for every  $\psi$ . Evaluating at the old solution gives  $F_{Q'}(\psi_j^{(t)}) \geq 1$ , and since  $F_{Q'}$  is non-increasing the solution of  $F_{Q'}(\psi) = 1$  is at least  $\psi_j^{(t)}$ . Hence  $\psi_j^{(t+1)} \geq \psi_j^{(t)}$ , and by induction  $\psi_j$  is non-decreasing along the prefix.

(ii)  $p_{ij}^{(t)} = [z_i - \psi_j^{(t)}]_+$ , so  $p_{ij}^{(t)} = 0$  means  $z_i \leq \psi_j^{(t)}$ . By (i),  $\psi_j^{(t')} \geq \psi_j^{(t)} \geq z_i$  for  $t' \geq t$ , whence  $p_{ij}^{(t')} = 0$ . The converse fails:  $z_i > \psi_j^{(t)}$  does not imply  $z_i > \psi_j^{(t')}$ , so an admitted query may be evicted.

(iii) The frozen variant assigns index  $i$  the value it had when  $i$  first became visible, i.e. at  $t = i$ , giving  $p_{ij}^{\text{frozen}} = [z_i - \psi_j^{(i)}]_+$ . Since  $\psi_j^{(i)} \leq \psi_j^{(T)}$  by (i), we have  $p_{ij}^{\text{frozen}} \geq p_{ij}^{(T)}$  pointwise, so  $p_{ij}^{(T)} > 0 \Rightarrow p_{ij}^{\text{frozen}} > 0$  and the frozen support contains the exact one. The inclusion may be strict, so relative to the exact prefix column the frozen variant admits a superset and never drops a member: its error is one-sided in *support*, which is what (iii) asserts and what the composition argument of Prop. 9 needs. We do not claim it is one-sided in *fidelity*:  $P_j$  and  $R_j$  are measured against  $T_j$ , not against the exact support, and this proposition assumes no margin condition, so an extra admitted index may be a target and precision may rise. Under Assumption 2 with  $\tau_j$  in equation 10, where the exact support is  $T_j$ , the two coincide and the frozen variant's extra admissions are exactly false positives.  $\square$

**Corollary 2 (restated).** *Let Assumption 2 hold for every key and let the ground truths be consistent,  $K_i = \{j : i \in T_j\}$ . If  $\tau_j \in [\frac{1}{W_j + m_j \delta_j}, \frac{1}{W_j})$  and  $K \geq m_j$  for every  $j$ , and  $\theta_i < \min_{j: i \in T_j} c_j(1 - \tau_j W_j)/m_j$  for every  $i$ , then  $P_j = R_j = 1$  on every column and  $P_i = R_i = 1$  on every row, all four at once.*

*Proof of Cor. 2.* We first record the weakest true edge. By Thm. 2 the support of  $\tilde{A}_{\cdot j}$  is exactly  $T_j$ , of size  $m_j$ , so with  $z = \tau_j s_{\cdot j}$  the sparsemax threshold is  $\psi_j = (\sum_{i \in T_j} z_i - 1)/m_j$  and  $p_{ij} = z_i - \psi_j$  for  $i \in T_j$ . The minimum over  $T_j$  is attained at the smallest score, so writing  $s_{\min} := \min_{i \in T_j} s_{ij}$ ,

$$\min_{i \in T_j} p_{ij} = \tau_j s_{\min} - \frac{\tau_j \sum_{i \in T_j} s_{ij} - 1}{m_j} = \frac{1}{m_j} \left( 1 - \tau_j \left( \sum_{i \in T_j} s_{ij} - m_j s_{\min} \right) \right) = \frac{1 - \tau_j W_j}{m_j},$$

using  $W_j = \sum_{i \in T_j} (s_{ij} - s_{\min})$  from Assumption 2. This is strictly positive because  $\tau_j < 1/W_j$  by equation 10. Multiplying by  $c_j$ , the weakest true edge is  $\min_{i \in T_j} \tilde{A}_{ij} = c_j(1 - \tau_j W_j)/m_j$ .

Since  $A_{ij} = \tau_i [\tilde{A}_{ij} - \theta_i]_+$  with  $\tau_i > 0$  and  $\theta_i \geq 0$ ,  $A_{ij} > 0$  requires  $\tilde{A}_{ij} > \theta_i \geq 0$ , so  $\text{supp}(A_{\cdot j}) \subseteq \text{supp}(\tilde{A}_{\cdot j}) = T_j$ . Equality holds iff  $A_{ij} > 0$  for every  $i \in T_j$ , i.e. iff  $\theta_i < \tilde{A}_{ij}$  for every such  $i$  — for which the hypothesis  $\theta_i < \min_{j: i \in T_j} c_j(1 - \tau_j W_j)/m_j$  is sufficient, being a bound against the weakest true edge of every column that claims  $i$ . (It is not necessary: a row could sit below its own true edges without being below the minimum over columns.)

*Joint attainment.*  $K \geq m_j = k^*$  gives, by Prop. 11, that the hierarchical construction returns the global solution, so it suffices to argue for the latter. With  $\tau_j$  in equation 10, Thm. 2 gives  $\text{supp}(\tilde{A}_{\cdot j}) = T_j$ , hence  $P_j = R_j = 1$  on the column before truncation (Prop. 7). By the computation above the weakest true edge of column  $j$  is  $c_j(1 - \tau_j W_j)/m_j$ , so  $\theta_i$  strictly below the minimum of these over the columns claiming  $i$  leaves every  $i \in T_j$  with  $A_{ij} > 0$ , while every entry already zero stays zero; the column support is therefore unchanged and  $P_j = R_j = 1$  after truncation as well. For the row, consistency gives  $K_i = \{j : i \in T_j\}$ , so for  $j \notin K_i$  we have  $i \notin T_j$  and hence  $\tilde{A}_{ij} = 0$  by the column recovery just established; thus  $\max_{j \notin K_i} \tilde{A}_{ij} = 0 \leq \theta_i$ , while  $\theta_i < \min_{j \in K_i} \tilde{A}_{ij}$  by hypothesis. The row therefore separates — it is not a further assumption — and Prop. 8(ii) gives  $P_i = R_i = 1$ . All four hold under the single parameter choice in the hypotheses. *Consistency is used exactly once, here, and cannot be dropped:* an inconsistent pair admits a key with  $\tilde{A}_{ij} > 0$  and  $j \notin K_i$ , which no  $\theta_i$  below the true edges can remove.  $\square$

## G NUMERICAL VERIFICATION OF THE ANALYTICAL CLAIMS

**Protocol.** Unless a claim requires otherwise, a column is drawn by choosing  $n_q$  uniformly from a stated range, choosing a target-set size  $m_j$ , and drawing scores so that Assumption 2 holds by construction: targets receive scores in  $[0, 1)$  and non-targets scores below  $-\delta$  for a margin  $\delta$  drawn uniformly from  $[0.2, 1.2)$ . This plants a known ground truth with a known positive margin, which is what makes support recovery checkable rather than merely plausible. Claims that must hold for *arbitrary* inputs (Prop. 9) instead draw unstructured Gaussian scores for half the trials and planted-margin scores for the other half, so that both the  $P^{(1)} = 1$  and  $P^{(1)} < 1$  branches are exercised. **No block discards a column because the claim failed on it**, which is a consequence of the sharp form of Thm. 2: two blocks previously sampled  $\tau_j$  from a merely sufficient set and discarded the columns on which that set was empty, and once the interval is known to be non-empty whenever  $\delta_j > 0$  both sample the interval itself. **Three counts are nonetheless smaller than the number of columns drawn, and the tables show it** (10,098, 10,604 and 20,472 against 12,000, 12,000 and 24,000): those rows test behaviour *above* the upper endpoint  $1/W_j$ , which is  $+\infty$  when  $W_j = 0$ , so a column with  $m_j = 1$  has no such regime and contributes nothing to that row while contributing normally to every other. That is the test being undefined, not a draw being rejected, and we report the counts that were actually run rather than the counts attempted. Exact-arithmetic claims are checked exactly on integers and sets; floating-point claims use  $10^{-9}$ . Each block seeds its own generator, so the program is deterministic and the table is reproduced exactly by running it; total runtime is under 25 seconds on one CPU core and the only dependency is `numpy`. The last block checks App. E and is reported there.

**One check is about the device rather than the mechanism, and we separate it.** The unit cap of equation 6 is the one place where a floating-point reduction could have entered a *decision* rather than a value. As the remark after Prop. 2 sets out, the implementation binds on the relation’s own mass excess equation 9 and projects onto  $\sum_j A_{ij}^{\text{sm}}$ , so the binding set is a function of in-relation terms alone and the softmax reduction’s error never reaches it; the excess is accumulated in double precision from single-precision entries cast before subtracting, giving it no error of its own, and a row with an empty relation has excess exactly 0.0, so an empty relation cannot bind the cap on any kernel. **We therefore report the kernel’s own row-sum error  $\delta$  as a statistic of the machine, beside the checks and not inside them**, and a released implementation should do the same: the guarantee is device-independent and the measurement of  $\delta$  is what demonstrates that it had to be made so. On the reference CPU,  $\delta$  reaches  $3.0 \times 10^{-5}$  at  $n_k = 8192$  and  $6.0 \times 10^{-5}$  at 16,384, against  $\approx 10^{-6}$  for the pairwise reduction of the deployment GPU — a spread no fixed tolerance spans, which is why the criterion reads no row total.

**A second device statistic, and it is the one that bounds what the implementation delivers.** The projection that enforces the cap is itself computed in single precision, so it cannot act on an excess smaller than its own arithmetic resolves; a row inside that *window* is left uncapped and keeps its excess. The window is a property of the device’s prefix reduction exactly as  $\delta$  is a property of its softmax reduction:  $1 \times 10^{-7}$  on the reference CPU,  $1.1 \times 10^{-6}$  at  $n_k = 8192$  and  $2.0 \times 10^{-6}$  at 16,384 on the deployment GPU, and as much as  $n_k \varepsilon / 4$  on a strictly sequential fp32 scan. We measure it by bisection on the target device before any run and refuse to start if it exceeds what the invariants allow, so the statement the release makes is  $\sum_j A_{ij} \leq \sum_j A_{ij}^{\text{sm}} + w$  with  $w$  measured and reported — never that the binding set is device-independent inside the window. *Reporting both numbers is the point*: a mechanism whose guarantee is checked in floating point owes the reader the size of the gap between the program and the arithmetic that runs it.

**How large must  $K$  be?** Prop. 11(i) makes hierarchical selection exact whenever  $K \geq k^*$ , so the practical question is how large  $k^*$  gets. Over 4000 columns per regime, with  $n_q$  drawn from  $\{64, \dots, 2048\}$  and scores  $\mathcal{N}(0, 2^2)$  scaled by  $\tau$ : when the column is *exclusive* ( $\tau \in [0.5, 6]$ ) the median is 1, the 99th percentile 5 and the maximum 8; in a *mixed* regime ( $\tau \in [0.1, 20]$ ) the maximum is 14; and on deliberately *broad* columns ( $\tau \in [0.02, 1]$ ) the median is 4, the 99th percentile 35 and the maximum 74. An earlier version of this work reported a single figure and concluded that a fixed  $K$  in the tens is universally safe; that was too strong. The corrected statement is more useful, because it makes the choice of  $K$  an empirical question about the deployed model rather than a constant asserted in advance. This is why E8 reports the realized  $k^*$  distribution jointly with  $|\mathcal{E}_j|$

Table 7: Verification of the analytical claims. Denominators are **evaluations**, not columns: a column swept at four temperatures contributes four. <sup>†</sup> **marks a row whose conclusion follows from how the generator builds its input**, so it can confirm an algebraic identity but cannot fail — see App. G, which says what each of these does and does not test. <sup>‡</sup> **marks a row that re-expresses evaluations already counted** in the Thm. 2 sharpness rows. No column is discarded for failing a claim; rows testing the regime above  $1/W_j$  count only the columns for which that regime exists ( $W_j > 0$ ), which is why three denominators fall short of the columns drawn. App. E is checked separately in Table 6.

| Claim                                                   | Test                                                                                    | Violations       |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------|
| Lem. 1, monotonicity of $G$                             | random vectors, $n \leq 30$                                                             | 0 / 5000         |
| Lem. 1, support characterization                        | sparsemax support vs. $\max\{k : G(k) < 1\}$                                            | 0 / 3000         |
| Monotone exclusivity, $\alpha = 2$                      | support size vs. $\tau$ sweep                                                           | 0 / 2000         |
| Monotone exclusivity, $\alpha \in \{1.2, 1.5, 1.8, 3\}$ | support size vs. $\tau$ sweep                                                           | 0 / 600 each     |
| Thm. 2, sharp recovery interval                         | $\text{supp} = T_j$ for $\tau_j \in \text{equation 10}$                                 | 0 / 24000        |
| Thm. 2, lower endpoint sharp                            | $\tau_j$ below it: $\text{supp} \supsetneq T_j$ , so $P_j < 1$ , $R_j = 1$              | 0 / 12000        |
| Thm. 2, upper endpoint sharp                            | $\tau_j$ at or above it: $\text{supp} \subsetneq T_j$ , so $R_j < 1$ , $P_j = 1$        | 0 / 10098        |
| Prop. 10, dilution bound                                | bound on random configurations                                                          | 0 / 4000         |
| Prop. 8(i), graded column recall                        | $k(\tau)$ vs. prediction, $\tau > 1/W_j$                                                | 0 / 3000         |
| Prop. 8(ii), row interval <sup>†</sup>                  | $P_i = R_i = 1$ on the row interval                                                     | 0 / 4000         |
| Prop. 8(ii), precision monotone <sup>†</sup>            | $P_i$ non-decreasing along a $\theta_i$ sweep                                           | 0 / 2000         |
| Prop. 8(ii), graded row recall <sup>†</sup>             | $R_i$ vs. prediction, weakest-first order                                               | 0 / 18000        |
| Prop. 11(i), exact hierarchy                            | hierarchical top- $K$ vs. global, $K \geq k^*$                                          | 0 / 2000         |
| Prop. 11(ii), retention fidelity                        | $P_j, R_j, \rho_j$ at $K < m_j$                                                         | 0 / 7200         |
| Prop. 6, ceiling <sup>†</sup>                           | softmax $P_j, P_i$ vs. $m_j/n_q,  K_i /n_k$                                             | 0 / 24           |
| Prop. 12(i)–(iii), causal form                          | threshold monotone; exclusion permanent; frozen over-admits                             | 0 / 4000 ea.     |
| Prop. 9(i), support shrinks <sup>†</sup>                | $S_j^{(2)} \subseteq S_j^{(1)}, R_j^{(2)} \leq R_j^{(1)}$                               | 0 / 17928        |
| Prop. 9(ii), precision preserved <sup>†</sup>           | $P_j^{(1)} = 1 \Rightarrow P_j^{(2)} = 1$ , over the swept $\theta_i$                   | 0 / 17928        |
| Prop. 9(iv), false negatives stick <sup>†</sup>         | Stage-1 zeros never resurrected                                                         | 0 / 17928        |
| <b>Cor. 2, joint attainment</b>                         | <b>full pipeline, all four quantities</b>                                               | <b>0 / 3000</b>  |
| Prop. 2(i), column total                                | $\sum_i \tilde{A}_{ij} = c_j$ at any $\mathcal{E}.j$                                    | 0 / 14937        |
| Prop. 2(ii), the anchor is derived                      | $\sum_j c_j = n_q$ without imposing it                                                  | 0 / 3000         |
| Prop. 2, fan-out quota conserved                        | step 2 returns exactly $\tilde{c}_j$                                                    | 0 / 14937        |
| Prop. 2(v), zeros permanent                             | no Stage-1 zero resurrects under equation 7                                             | 0 / 80741        |
| Prop. 2(v), fan-in quota a ceiling                      | $\sum_{\mathcal{E}.i} \tilde{A}_{ij} \leq \tilde{c}_i$                                  | 0 / 80741        |
| Prop. 2(iii), sub-unit rows                             | $\sum_j \tilde{A}_{ij} \leq \sum_j A_{ij}^{\text{sm}}$ ; 69.3% spend less               | 0 / 80741        |
| Prop. 2(iv), no relocation                              | entries off $\mathcal{E}$ fixed by step 1                                               | 0 / 80741        |
| Prop. 2(iv), column half                                | entries off $\mathcal{E}.j$ equal $A^{\text{sm}}$ exactly                               | 0 / 14937        |
| Prop. 2, cap is the identity                            | $\sum_j \tilde{A}_{ij} \leq 1 \Rightarrow a_i^{(1)} = \tilde{A}_i$ . exactly            | 0 / 80741        |
| Sec. 2.2, $\tau_i = 1$                                  | step 2 returns $a_i^{(1)}$ exactly, cap-binding rows included                           | 0 / 80741        |
| Prop. 2(vi), empty relation                             | output = $A^{\text{sm}}$ entrywise                                                      | 0 / 2000         |
| App. E (e), $\tau_j \rightarrow 0$ is uniform           | entries $\rightarrow \tilde{c}_j/ \mathcal{E}.j $ on the relation; $\neq A^{\text{sm}}$ | 0 / 1992 ea.     |
| Prop. 3(i), margin monotone                             | $\delta_j(\mathcal{E}') \geq \delta_j(\mathcal{E})$                                     | 0 / 6000         |
| Prop. 3(ii), width monotone                             | $m_j \delta_j / W_j$ non-decreasing                                                     | 0 / 6000         |
| Prop. 3(iii), containment                               | $I(\mathcal{E}) \subseteq I(\mathcal{E}')$                                              | 0 / 4506         |
| <b>Cor. 1, degree bound</b>                             | $ \text{supp}  = m_j,  \mathcal{E}.j $ growing                                          | <b>0 / 10000</b> |
| <b>Cor. 1, precision</b>                                | $P_j = 1,  \mathcal{E}.j $ growing                                                      | <b>0 / 10000</b> |
| Thm. 2 at $m_j = 1$ , argmax                            | $\tau_j \geq 1/\delta_j$ : whole quota on one member, zeros elsewhere                   | 0 / 9000         |
| Thm. 2 at $m_j = 1$ , continuity                        | $\ p(\tau') - p(\tau)\  \leq  \tau' - \tau  \ s.j\ $ through $1/\delta_j$               | 0 / 3000         |
| Composition, Sec. 2.1                                   | a key concentrating elsewhere is an exact 0 on the answer row                           | 0 / 12082        |
| Fig. 1, trace below the interval <sup>‡</sup>           | $R_j = 1, P_j = m_j/k^*$                                                                | 0 / 24000        |
| Fig. 1, trace on the interval <sup>‡</sup>              | $P_j = R_j = 1$                                                                         | 0 / 16000        |
| Fig. 1, trace above the interval <sup>‡</sup>           | $P_j = 1, R_j = k^*/m_j$                                                                | 0 / 20472        |
| Fig. 1, full support is a point <sup>†</sup>            | softmax at $(1, m_j/n_q)$ , every $\tau_j$                                              | 0 / 2000         |

rather than either alone: the relation has the information needed to keep  $k^*$  small (App. B), and that report is what tells us whether it used it.

**Why some rows need few trials, and why the counts are not comparable.** Prop. 6 is a deterministic identity rather than a statement that could fail on a hard draw, so it is checked on a small grid of  $(n_q, n_k, m_j, |K_i|)$  spanning four orders of magnitude rather than on a large sample; a larger count would add no risk to it. That reasoning generalizes, which is why the tables carry daggers rather than a single total: **a large denominator on a daggered row means the loop ran many times, not that the claim survived many chances to fail.** The counts are also in different units of work — Prop. 12 under a nested scope contributes 131,180 because it counts every prefix timestep of every sequence, where a column swept at four temperatures contributes four — so the rows should be read

individually and not summed. The blocks with large *informative* counts are those whose hypotheses interact.

**The row checks and  $\tau_i$ .** Every row-margin claim quantifies over the admission floor  $\theta_i$ , and the harness sweeps  $\theta_i$  to match. This is legitimate, but not because  $\theta_i$  is free: it is the dual of the unit cap, and  $\tau_i \mapsto \theta_i$  is a strictly increasing bijection onto  $[0, \max_j \tilde{A}_{ij})$ , so sweeping  $\theta_i$  over that range is *equivalent* to sweeping  $\tau_i$  over  $(0, \infty)$ . What the gain  $\tau_i > 0$  then contributes is a positive common factor, which moves no support and changes no precision or recall, so the support-level counts below are  $\tau_i$ -free and the shipped harness needs no  $\tau_i$ . Note what this does *not* cover: the harness tests the row claims at a given floor, never whether the head’s realized  $\tau_i$  puts the floor where the row needs it, which is E7.

**What the daggered rows do and do not test, and why we mark them.** A check can only fail if the generator can produce an input on which the conclusion is false, and for a *constructive* proposition it often cannot. Prop. 9’s three rows are the clearest case: the program forms  $A_{ij} = [\tilde{A}_{ij} - \theta_i]_+$  from a non-negative  $\tilde{A}$ , which makes  $S^{(2)} \subseteq S^{(1)}$ , and (i), (ii) and (iv) are corollaries of that inclusion evaluated on the same columns. Prop. 8(ii)’s rows plant the threshold strictly between the true and false sources, and its recall row sorts the row before naming the top entries as the true ones, so “weakest-first” is guaranteed by the sort. Prop. 6’s rows evaluate  $m_j/n_q = m_j/n_q$  on a full support. Prop. 1’s restricted row is  $\max \geq \text{mean}$  over  $n_k$  non-negative numbers. **These rows confirm that the algebra in the proof is the algebra in the statement, which is worth having and is how two sign errors were caught; they are not independent evidence that the propositions are true, because the generator cannot produce a counterexample.** Rebuilding them to sample inputs that could violate the conclusion — unsorted sources, thresholds that may land inside a group — is the natural next version of this harness and we have not done it.

The undaggered rows are the ones that could have failed and did not. Thm. 2’s three rows place  $\tau_j$  on both sides of two independently computed endpoints; Prop. 11(ii) reports fidelity at  $K < m_j$ , where the claim is an inequality that a bad bound would break; Cor. 2 runs the full pipeline and asks for four quantities at once; Prop. 12 sweeps a growing prefix. Those are where a genuine error would have surfaced, and one did.

**What none of it establishes.** These are not evidence about real data: every score configuration is synthetic and every ground-truth relation planted. How wide the recovery interval is on a trained network is E7, not anything in this appendix. The joint-attainment row draws the target sets  $\{T_j\}$  as a partition of the queries so that  $K_i = \{j : i \in T_j\}$ ; that is precisely the consistency hypothesis Cor. 2 now states, and without it “both margins at once” has no well-defined ground truth to check against.

## H EXPERIMENTAL PROTOCOL IN FULL

Table 8: Estimated single-GPU cost. H100-hours,  $\pm 50\%$  per item; the per-stage figures are run estimates and the  $1.5\times$  debugging allowance is applied once, at the bottom, so the column sums as printed. Runs within a stage parallelize across a node.

| Stage        | Contents; gate and pivot                                                                                                                                                                                                                    | H100-h                          |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| S0           | E1 on trained-backbone scores. <i>Gate</i> : a replication pair, $M > m_j$ at stated $a$ .                                                                                                                                                  | 1                               |
| S1           | Mechanism correctness: identities, hierarchical-vs-flat agreement, relation logging, end-to-end training. <i>Gate</i> : identities to tolerance; relation coverage non-zero.                                                                | 15                              |
| <b>S2</b>    | <b>E3 <math>n</math>-sweep with both controls; E2 and E4 on RULER.</b> <i>Gate</i> : precision flat in $n$ on $\delta_j > 0$ columns; hit rate not falling in $m$ . <i>If not</i> : report the training-effect claim and rescope the paper. | <b>60</b>                       |
| S3           | E5 MuSiQue/HotpotQA; E7 fidelity; E8 diagnostics.                                                                                                                                                                                           | 68                              |
| S4           | E6 IHEval depth stratification; E9 ablations.                                                                                                                                                                                               | 80                              |
| Subtotal     | (adaptation fine-tunes included in S2–S4)                                                                                                                                                                                                   | 224                             |
| <b>Total</b> | <b>with the <math>1.5\times</math> debugging allowance</b>                                                                                                                                                                                  | <b><math>\approx 336</math></b> |

**Staging.** Table 8 gives the five stages, their gates and the pivot taken if a gate fails. S0 and S1 together cost 16 H100-hours and can invalidate either the applicability of Thm. 1 (no replication pair on real score columns) or the correctness of the implementation, before any expensive stage is entered. Figures carry  $\pm 50\%$  per item; the stages sum to 224 and the  $1.5\times$  debugging allowance is applied once to that subtotal.

**E1, construction.** The probe measures degree, not mass, because under equation 2 the mass a key emits into its relation is inherited and a mass sweep would show MarSea tracking softmax by construction. Panel (a): the true relation size is held at  $m_j = 4$  and the candidate set  $|\mathcal{E}_{\cdot j}|$  takes the values  $4, 8, \dots, 256$ ; the  $m_j$  members score  $1 + U(0, 1)$  and the distractors  $0.9 \cdot U(0, 1)$ , so Assumption 2 holds with  $\delta_j \geq 0.1$  at every size — *by construction*, which is the point of a possibility result and is not what App. E says happens on an unrestricted column. MarSea’s  $\tau_j$  is placed inside Thm. 2’s interval computed from the planted set, at  $\text{lo} + \frac{1}{2}(\min(\text{hi}, 20 \text{lo}) - \text{lo})$ ; **it is an oracle temperature**, and MarSea’s curve is therefore Thm. 2’s *iff* drawn rather than a measurement of a head — what the figure shows is the shape of the guarantee, and the comparator’s side is Prop. 6, also an identity. The comparator’s support is  $|\mathcal{E}_{\cdot j}|$  and its precision  $m_j/|\mathcal{E}_{\cdot j}|$  at every temperature and scoring function, so it is drawn analytically rather than sampled; MESH and every dense scheme lie on the same line. Twelve seeds vary the scores and are inert on both curves, which is what an identity looks like. Panel (b): six keys at  $n_q = 63$  with true sizes  $1, 2, 4, 8, 16, 32$ ; three of the 72 columns drawn violated Assumption 2 on the first draw (their interval is not an interval and the midpoint temperature is negative) and were re-clipped rather than reported, and the script prints how many. Everything E1 establishes is already in Table 7’s rows; its value is the picture, and S0 is the measurement.

**E4, E5 and E6 in full.** E4 repeats E2 at  $m = 1$  on RULER S-NIAH, the regime Prop. 5(i) concedes to query-local schemes. *Prediction:* a gap near zero. A *large* gap here is the interesting negative — it would mean the benefit is not coming from multivariate structure, and E2’s reading would need revision. E5 externalises to MuSiQue ( $m \in \{2, 3, 4\}$  by hop) and HotpotQA-distractor; its  $m$ -stratification is confounded with difficulty by construction, so it corroborates E2 and never replaces it. E6 tests the ranked half of Sec. 1 on IHEval, stratifying by the number of conflicting instruction tiers and reporting accuracy *as a function of depth*, the slope rather than the pooled mean. *Prediction:* at depth one — two sources, one overriding — MarSea and a well-tuned query-local baseline should be close, a two-level decision being within a pointwise rule’s reach; as depth grows the query-local baselines should degrade markedly and MarSea gently. *Falsification:* a flat gap across depth withdraws the ranked framing for the weaker binary-exclusivity claim.

**Grids.** E2 sweeps  $m \in \{1, 2, 4, 8, 16\}$  at fixed  $n$  and fixed sequence length. E3 sweeps  $n$  over five levels up to a few hundred candidates at fixed  $m$ , each point run twice (real distractors versus inert padding of equal token count) and at a fixed gold position, with the position-marginal profile reported alongside. Three seeds on all headline configurations; mean and standard deviation; matched epochs and schedules across methods. Gains within seed variance are reported as such. Because the fan-out program couples queries, padding tokens consume budget unless masked out of the program, and we verify invariance of outputs to padding length as a correctness check.

**Baselines, in full.** Six: unmodified attention; softmax<sub>1</sub> (Miller, 2023); a learned-marginal transport in the MESH form; **an exclusivity head reading  $k_j$  alone**,  $\tau_\phi^K(k_j)$  with  $s_{\cdot j}$  and  $\nu_j$  withheld; row- $\alpha$ -entmax (Peters et al., 2019); and softmax sparsified to MarSea’s realized support size. Each isolates one design decision and is retuned under a matched schedule rather than quoted from a paper.

**The third and fourth are the two contribution (i) rests on, and the third is not an ablation we invented.** Zhang et al. (2023) learn *both* marginals, each as  $m \cdot \text{softmax}(h(\cdot))$  from a network reading one element at a time, and solve an entropic transport between them at a global regularization, so the output is dense: read in this paper’s orientation it is a learned  $c_j$  and a learned row total with no shape on either margin, the mirror image of Sec. 2 (Table 1). Prop. 6 pins its column precision at  $m_j/|\mathcal{E}_{\cdot j}|$  however well the marginals are learned, so it is the decisive test of whether *shape*, not capacity, is what moves precision. One limitation of that arm is the baseline’s, not ours, and we state it: under a causal mask a two-marginal transport is generically infeasible — the last key is visible to one query, whose row must then carry that key’s whole marginal — so Sinkhorn

cannot satisfy both marginals; we end the iteration on the row scaling, which the value mix needs, and report the column-marginal residual. The fourth withholds the field from  $\tau_\phi^K$  and isolates the field argument: Cor. 3 says a head that cannot see the column cannot serve two keys whose recovery intervals are disjoint, and Fig. 2(b) shows training-free that the statistic such a head is denied,  $\nu_j$ , is the one that carries  $m_j$ . *We predict the MESH-form arm is the strongest of the six on answer score and sits on the comparator’s line in E3, and that the key-alone arm loses on E7’s interval-hit rate.* If the key-alone arm matches MarSea’s hit rate the field argument is decoration; if the MESH-form arm matches MarSea on exact-set accuracy, shape is not what the task needs, and contribution (i)’s task-level claim is what fails.

The last two answer the objection that Prop. 6’s ceiling is an artifact of full support that any sparsifier clears. **The answer differs by margin, and an earlier draft got the row wrong.** On the *column*, neither row- $\alpha$ -entmax nor matched-sparsity softmax enforces a column constraint, so by Thm. 1 their support *membership* is uncontrolled even where its *size* matches MarSea’s, and reporting  $P_j$  against them compares mechanisms rather than sparsity levels. On the *row* that argument does not apply: a query-local sparsifier controls its own row membership completely, and what separates MarSea from it there is only *which keys Stage 1 lets reach the row*. We therefore measure that directly rather than assert it (next paragraph).

**The single-target and the empty column are row-side tests, and the split is reported.** With one answer query, every passage key has  $m_j \in \{0, 1\}$ . At  $m_j = 1$  Thm. 2 has  $W_j = 0$ : for  $\tau_j \geq 1/\delta_j$  the key’s whole quota goes to its largest-scoring query and the rest of the column is exactly zero, continuously in  $\tau_j$  (Sec. 2.1). At  $m_j = 0$  no temperature can empty a simplex, so the key reaches whichever query scores it highest,  $P_j = 0$  by definition, and the column is a test of nothing; **we declare it a row-side test** and exclude  $m_j = 0$  columns from every column average, reporting their count. What the row sees is the composition of the two stages, and E8 splits the distractor keys of each answer row three ways: *excluded by Stage 1* (their largest-scoring query is not the answer query, so their entry on the answer row is an exact zero before any row program runs), *rejected by the row* ( $\theta_i$  removed them), and *surviving*. The first share is the Stage-1-attributable part of MarSea’s row precision, and it is the quantity that separates MarSea from row- $\alpha$ -entmax and matched-sparsity softmax on the row margin, since a query-local sparsifier has no Stage 1 to do it; the harness reports the complementary share — keys that concentrate *onto* the answer row and are left to  $\theta_i$  — at 8.5% of keys under random scores (App. G). The column theory is exercised on columns with  $m_j \geq 2$ , keys that govern several annotated queries; where those come from is stated with E7.

**Reporting rules.** Fine-tuned and zero-shot numbers are never pooled: every comparison is against the same decoder under the same schedule, and a zero-shot figure quoted from another paper is not a control. Nor do we contest multi-stage retrieve-then-read pipelines with much larger readers — MarSea is a change to one normalization step, evaluated as a mechanism ablation, and a pipeline that beats it is not evidence about the mechanism.

**Metrics, and why the task criteria are not recall-only.** Evidence selection is scored as retrieval against the annotated gold set: support precision, support recall and, where a graded reading is wanted, target mass, each reported separately and never as a single aggregate (Prop. 9). **Task performance is reported on each benchmark’s own precision-sensitive criteria, not only its answer score**, because an answer score is blind to over-selection and a full-support mechanism’s recall is 1 for free. On HotpotQA we report the official supporting-fact EM and F1 and the *joint* metrics,  $P_{\text{joint}} = P_{\text{ans}}P_{\text{sup}}$  and likewise for recall, with joint EM demanding both exact (Yang et al., 2018); supporting-fact EM is exact set recovery, which is the task-level image of  $P_j = R_j = 1$  and the place the precision claim has to cash out. On MuSiQue-Answerable we report support F1 beside answer F1 (Trivedi et al., 2022); MuSiQue-Full’s paired sufficiency scores are a different instrument — a binary judgement of whether a context suffices, scored zero on a pair if either half is called wrongly — so we report them as an abstention diagnostic and not as a precision criterion. **One granularity caveat:** HotpotQA annotates supporting facts at the *sentence* level while  $m_j$  here is a count of *passages*, so the supporting-fact metrics are read at the granularity the benchmark annotates and the  $m$ -sweep at the granularity the theory is about; we say which is which wherever both appear. RULER is the exception and it is the one that matters most, since E2 runs there: it ships a *recall-based* score that checks only for the presence of the target values (Hsieh et al., 2024), so over-selection is free. **We therefore pre-commit E2 to exact-set accuracy** — the returned set

equals the gold value set, no distractor value admitted — reporting RULER’s own score alongside it; needle and distractor identity are configuration parameters of the generator, so the stricter criterion costs nothing and is the one the theory speaks to. IHEval is scored as task accuracy, and the depth stratification E6 reports is taken on its multi-turn rule-following task, which is where the number of conflicting components varies; elsewhere its conflicts are between two levels (Zhang et al., 2025). Efficiency is reported as parameters, FLOPs, peak memory and wall-clock latency *as a function of the measured relation coverage* rather than as a single figure, since by Sec. 2.3 the cost of materializing a relation’s columns is proportional to it.

**E7, E8 and E9 in full.** Wherever the gold evidence set is annotated,  $T_j$  is available directly and E7 reports all five quantities of Def. 2 — never a single aggregate — and recall *twice*, before and after Stage 2. By Prop. 9(i) recall cannot rise through Stage 2, so any recall deficit between the two readings is Stage 2’s; precision is *not* bracketed the same way, since a row-indexed floor can remove true edges from a column while leaving false ones in another row, so we report both precision readings and attribute nothing by subtraction. E7 also measures  $(m_j, \delta_j, W_j)$  from logged scores and reports the distribution of the *relative interval width*  $m_j \delta_j / W_j$  — Thm. 2 makes the interval non-empty whenever  $\delta_j > 0$ , so width, not feasibility, is what a hard column costs, and it is width that Assumption 2 turns into a measurement. Since equation 5 reads  $s_{.j}$ , E7 additionally reports the *interval-hit rate*: the fraction of active columns whose realized  $\tau_j$  lies in  $[1/(W_j + m_j \delta_j), 1/W_j)$ , together with the signed distance to the nearer endpoint when it does not. *The sharp endpoints matter here operationally*: scored against the sufficient interval  $[1/(m_j \delta_j), 1/W_j)$  a head landing in  $[1/(W_j + m_j \delta_j), 1/(m_j \delta_j))$  would be recorded as a miss while in fact recovering exactly. This is the most direct test of Thm. 2 available on real data, and it is only meaningful because the head sees the column the interval is defined by — a head reading  $k_j$  alone would be scored against a target it cannot observe. *Falsification*: a hit rate at chance, with the misses unbiased in sign, says the head is not using the column and the mechanism is running on a temperature that happens to work rather than one that was inferred. The same quantity is reported for the row against the interval of Prop. 8(ii). **E7 also reports the measured precision–recall trace against the predicted one.** On a column meeting Assumption 2, Thm. 2 with Lem. 1 and Prop. 8(i) fix the whole curve in advance, so sweeping  $\tau_j$  on held-out columns should reproduce Fig. 1 column by column:  $R_j = 1$  with  $P_j = m_j / k^*(\tau_j)$  below the interval,  $P_j = R_j = 1$  on it,  $P_j = 1$  with  $R_j = k^*(\tau_j) / m_j$  above, and no interior point. *Be clear about what this does and does not test.* Given the assumption and  $\alpha = 2$  the trace is an identity, so a departure indicts the assumption on that column or the implementation, never the theorem; what it earns is the recall side of the story, which is given exactly rather than bounded —  $k^*(\tau_j) / m_j$  on the column,  $K / m_j$  under retention (Prop. 11(ii)),  $k(\theta_i) / |K_i|$  on the row — and a like-for-like comparison, since the same sweep run on each baseline shows a full-support mechanism fixed at  $(1, m_j / n_q)$  and a matched-sparsity one moving along no curve the theory predicts, its support *membership* being uncontrolled by Thm. 1. The genuinely empirical question remains the interval-hit rate: whether the learned  $\tau_\phi^K$  lands where the column says it must. E8 logs the relation’s coverage on both margins from epoch one — the fraction of keys with  $\mathcal{E}_{.j} \neq \emptyset$ , of queries with  $\mathcal{E}_{.i} \neq \emptyset$ , and the full distributions of  $|\mathcal{E}_{.j}|$  and  $|\mathcal{E}_{.i}|$  rather than their means — because the degenerate optimum is an empty relation everywhere, which is standard attention, and the subtler one is a relation of singleton columns, where Thm. 1 says nothing (App. E, cost (a)). It reports  $k^*$  stratified by  $|\mathcal{E}_{.j}|$ . There is one fan-in trigger: a row enters equation 7 iff its slice of the relation is non-empty, and because the relation is one object every query a key re-shapes lies in that key’s column slice and so has a non-empty row slice; the row rate is bounded below by the column-induced rate *by construction*, and a measured violation is an implementation error rather than a finding. It reports the distribution of  $\tau_i \tilde{R}_i / \bar{c}_i$  about one, the regime threshold of Sec. 2.2, and the correlation between each membership decision and the realized separation  $\delta_j$  of the column it entered, which is how we learn whether the relation keys on the structure the theory says matters or on something else. Note that  $\tau_\phi^Q$  collapsing to 1 is *not* the fan-in analogue of an empty relation, since covered rows still run there and are still capped at one unit; it is the safe default. The failure to watch for is the opposite,  $\tau_i$  large enough that rows are routinely truncated to one or two keys, which would show up as a row-recall deficit in E7. *Falsification*: if  $k^*$  on covered columns approaches the deployed  $K$ , the hierarchy is silently truncating and no E7 number is meaningful until  $K$  is raised. **E9** ablates every decision in turn — the relation’s parameterization (its rank  $r$ , pairwise against key-alone, and which layers carry it),  $K$ , block size and emission policy,  $\tau_i$  learned against pinned at 1 (which reduces Stage 2 to a plain unit cap),  $\tau_j$  learned against fixed or uniform, and the

inherited capacity of equation 2 against a uniform one (which is the ablation that puts a predicted budget back and is reported as such). **The load-bearing ablation of this group strips the field argument from each head** —  $\tau_\phi^K(k_j)$  against  $\tau_\phi^K(k_j, s_{\cdot j}, \nu_j)$ ,  $e_{ij}$  from  $u_\phi(k_j)$  alone against the pair, and  $\tau_\phi^K$  with and without  $\nu_j$  — since a mechanism that claims to model a multivariate relation should lose measurably when its parameters are made blind to the field, and if it does not, the claim is decoration.

**E7, the column ground truth.** No benchmark annotates  $T_j$ , the set of queries that *should* attend to a key, so E7 derives it by exact string match from annotation the benchmarks do ship, on both instruments alike, and it uses two kinds of column because the backbone routes them through different heads. The *coreference column*: a key is the first token of the *first mention* of a bridge string, and  $T_j$  is the first token of every *later mention* of the same string. On RULER’s multi-value task the  $m$  needles share one key phrase, so its first mention has  $m + 1$  later mentions (the other  $m - 1$  needles, the question, and the answer prefix) and  $m_j = m + 1$  moves with the sweep; on MuSiQue the bridge strings are the decomposition’s intermediate answers, giving a leaf passage  $m_j = 2$  and a bridging one 3–5, and because we assemble the prompt the gold passages are placed in hop order so that every later mention is visible to the earlier one. Reversing that order changes  $m_j$  deterministically, which is a free check that the annotation measures what we say. The *value column*: a key is a needle’s value span and  $T_j$  is the answer positions that emit that value, so  $m_j$  is the number of emitting tokens, typically one; this is the column the answer rows are scored on, and it is the  $m_j \leq 1$  regime of Sec. 2.1. The two are not merged into one target set: S0 found that answer positions attend to values and not to the key phrase’s first mention, so a merged  $T_j$  would score the coreference column against a target the backbone never routes to it. Two things are kept separate from both. First, a target the relation excluded is a *relation recall* miss, logged on its own and never charged to  $\tau_j$ ; the interval is computed on  $T_j \cap \mathcal{E}_{\cdot j}$  against the relation’s complement, and a column whose complement is empty has  $\delta_j = +\infty$  and a lower endpoint of 0. Second, the construction is ground truth about the *task* and a hypothesis about *routing*, so S0 first checks on the unmodified decoder, sweeping every head, that some head sends the later mentions to the earlier one — the mass those queries put on the key span exceeds half the row’s maximum over candidate keys, the sink excluded, on at least 80% of examples — and a construction that fails the check is reported as not measurable on this backbone rather than used; the column test runs at the head that routes it, which must be in a patched layer. Dense arms have no relation, so their column precision is reported on two domains, the full visible column ( $m_j/n$ ) and MarSea’s paired relation ( $m_j/|\mathcal{E}_{\cdot j}|$ , the identity of Prop. 6); the second is the headline.

**The coverage residual.** Where  $T_j$  is annotated, E7 also logs the signed residual  $n_q - \sum_j m_j$ : positive when queries go ungoverned, negative when they are claimed by more than one key. It is a dataset statistic rather than a property of any head, and it is reported because Cor. 2’s consistency hypothesis is exactly the case where it is zero.

**Why these three datasets and not others.** Benchmarks in which one correct answer is chosen among distractors with no coupling between targets exercise only Prop. 5(i), the half we concede, and a gain there is evidence for a better scoring function rather than for Thm. 1; we therefore do not use multiple-choice reasoning suites as evidence. RULER, MuSiQue and IHEval were selected because each supplies something the others cannot: a controlled  $m$  at constant difficulty, a real task with annotated gold evidence, and a ranked order over conflicting sources respectively. In real multi-hop data  $m$  is not controlled — MuSiQue stratifies it by hop count, varying gold-paragraph count and reasoning depth together, so a gap growing in  $m$  there is confounded with “MarSea helps more on harder questions” (Min et al., 2019). Neither instrument is reported alone, synthetic recall being an unreliable predictor of downstream performance (Yen et al., 2025).